Springer Tracts in Nature-Inspired Computing

Nilanjan Dey *Editor*

Applied Multi-objective Optimization

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Preface

Multi-objective optimization (MOO) refers to an algorithm that aims to solve multiple competing objectives that need to be either minimized or maximized. Applied multi-objective optimization is a branch of multiple-criteria decision-making that deals with mathematical optimization problems involving multiple objective functions that must be optimized simultaneously. It is also referred to as multi-objective programming, vector optimization, multicriteria optimization, multiattribute optimization, or Pareto optimization. *This book highlights* the basic concept of multi-objective optimization and its application in numerous scientific domains. *The book* includes eight chapters in which *Banerjee* et al. in Chap. 1 provide information for the researchers and scientists working in this domain regarding the basic concepts and different approaches of multi-objective optimization. *Hasan* et al. highlighted counterfactual explanations and discussed federated learning for enhanced data analytics optimization in Chap. 2. In Chap. 3, *Farahmand-Tabar* et al. used multi-objective adaptive guided differential evolution on passively controlled structures with a tuned mass damper. In Chap. 4, *Harkare* et al. analysed different approaches for evolutionary multi-objective optimization for optimal solution selection in data analytics.

In Chap. 5, *Farahmand-Tabar* discussed the use of the multi-objective Lichtenberg algorithm for the optimal design of truss structures. In this chapter, they addressed the multi-objective optimization of truss structures through the Lichtenberg algorithm (MOLA), which can handle continuous variables. A performance analysis of a multi-objective function-based PID controller for system frequency regulation was introduced by *Boopathi* et al. in Chap. 6. In Chap. 7, *Farahmand-Tabar* and*Afrasyabi* discussed multimodal routing in urban transportation networks using multi-objective quantum particle swarm optimization. In this chapter, they focused on addressing the routing challenges within urban transportation networks using a multi-objective quantum particle swarm optimization (MOQPSO) approach. In the final chapter, *Dowlatshahi* and*Hashemi* provided an analytical review of multi-objective optimization for feature selection. This chapter reviews the presented feature selection methods focused on the MOP approach.

The editor expresses gratitude to the exceptional authors and referees for their valuable contributions to the book. Their hard work and cooperation have resulted in outstanding publication. Additionally, thanks are extended to the members of the Springer team for their support.

Kolkata, India Nilanjan Dey

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About the Editor

Nilanjan Dey is an Associate Professor, Department of Computer Science and Engineering, Techno International New Town, Kolkata, India. He is a visiting fellow of the University of Reading, UK. He was an honorary Visiting Scientist at Global Biomedical Technologies Inc., CA, USA (2012–2015). He was awarded his Ph.D. from Jadavpur Univeristy in 2015. He has authored/ edited more than 70 books with Elsevier, Wiley, CRC Press and Springer, and published more than 300 papers. He is the Editor-in-Chief of *International Journal of Ambient Computing and Intelligence*, IGI Global, Associated Editor of IEEE Access and *International Journal of Information Technology*, Springer. He is the Series Co-editor of *Springer Tracts in Nature-Inspired Computing*, Springer, Series Co-editor of *Advances in Ubiquitous Sensing Applications for Healthcare*, Elsevier, Series Editor of Computational Intelligence in Engineering Problem Solving and Intelligent Signal processing and Data Analysis, CRC. His main research interests include Medical Imaging, Machine Learning, Computer Aided Diagnosis, Data Mining etc. He is the Fellow of IETE and Senior member of IEEE.

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Chapter 1
A Guide to Meta-Heuristic Algorithms for Multi-objective Optimization: Concepts and Approaches

Archisman Banerjee, Sankarshan Pradhan, Bitan Misra, and Sayan Chakraborty

1 Introduction

Various kinds of problems or difficulties in the domain of technology and industries require concurrent optimization of a variety of objectives or targets [1]. It is possible to break down some practical problems into a single objective problem, but breaking down all problems like this is quite a difficult task. Therefore, it is necessary to define multiple objectives to obtain the best solution for a problem or task at hand. The multi-objective optimization technique (see Fig. 1) has been truly helpful in this aspect. The application areas of these techniques have become popular as far as real-life problems are concerned [2]. Single objective optimization is an approach that provides the concept of the characteristics of the problem, but generally, it cannot yield different possible solutions that deal with a variety of objectives against each other. The advantages of taking multiple tasks in the process of solving a problem [3] is that a large set of alternate solutions can be observed. Additionally, modeling the task will be more practical. Multi-objective optimization

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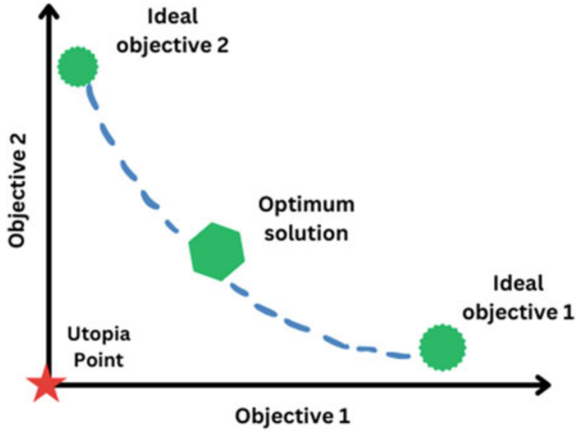

Fig. 1 Multi-objective optimization technique

promotes decision making and organization of the task. Multi-objective optimization yields several optimal solutions, often referred to as nondominated solutions or Pareto-optimal fronts, particularly when the objectives are not in conflict and there is only one unique solution. By applying multi-objective optimization and generating nondominated solutions, decision-makers can obtain a greater knowledge of the quantitative trade-off between objectives. Based on the results and their personal assessment, they may select one of the best options.

Multi-objective optimization is carried out by two basic approaches. The first is to accumulate all the single objectives into one complex objective function. To do this, the utility theory or weighted sum technique is generally applied. The second approach places every possible objective except one in an objective set. In recent years, multi-objective optimization has become very popular for solving real-world problems [8], such as reducing pollutant emissions and efficiently optimizing any power plant.

Meta-heuristic algorithms are algorithms that are used to find fair solutions to complex and tricky optimization problems. These types of algorithms are developed by observing some natural phenomena, such as genetics, the behavior of swarms, and the survival of the fittest species in the process of evolution. Some notable examples of meta-heuristic algorithms are particle swarm optimization (PSO), differential evolution (DE), genetic algorithm (GA), and ant colony optimization (ACO). These algorithms are quite popular for solving difficult problems in the fields of engineering, commerce, and agriculture [4]. Meta-heuristics algorithms compute and optimize a problem through repetitive iterations. It looks out for the best solution in a specific search space of solution sets and obtains the best possible solution. A primary advantage of meta-heuristic algorithms is that they do not depend upon the initialization of the problem to be optimized. As meta-heuristic algorithms make almost no assumptions, they may be implemented to generate solution of real-time problems in various areas [5, 6].

The application of meta-heuristic algorithms in multi-objective optimization has increased in the past few years. It can be noticed that when the normal or mainstream concepts of multi-objective optimization fail to solve complex nonlinear modeling problems, meta-heuristic algorithms can provide an effective solution. Meta-heuristic algorithm offers a new way of applying multi-objective optimization in the fields of wireless communication, health crisis management, agriculture, image segmentation, and many more. Another reason behind the popularity of meta-heuristic algorithms in solving problems having multiple objectives is that they provide the best optimal solution out of the large solution set. GA (evolution-based algorithm) is a popular and efficient meta-heuristic algorithm that helps to optimize complex searching processes. Bioinspired algorithm bacterial foraging optimization is useful in neural networks and machine learning-related optimization. Data clustering, data mining, and time series prediction can be successfully performed with the help of PSO algorithm. Grey wolf optimization can be applied in the process of satellite image segmentation in the field of space sciences and research [7]. Meta-heuristic algorithms are also very helpful in optimizing the deep structure of neural networks. The deep structures of neural networks have recently been used in the image classification process for chest infection detection during the COVID-19 pandemic [9]. In truss structures, vibration frequency constraint-related issues may also be resolved by utilizing meta-heuristic algorithms [10]. In the transportation of goods, warehousing, and cutting industries, the bin packing problem is a regular concern. To solve this type of multi-objective problem, a meta-heuristic algorithm can essentially be applied [11]. The meta-heuristic approach helps to comprehend the detailed analysis of the operation of hydraulic structures. Furthermore, it helps to study reservoir operation and flow control of fluid in hydropower industries [12]. Different types of antenna array geometries can be studied, and their radiation pattern can be improved efficiently using a meta-heuristic algorithm in case of wireless communication systems, which have large application domain in the military and space research fields.

In the engineering field, most of the problems are nonlinear problems. Additionally, problems related to antenna arrays, neural networks, and artificial intelligence consist of large datasets. Therefore, it becomes difficult to solve these kinds of complex problems with conventional algorithms. Again, meta-heuristic algorithms are preferred over other algorithms because they take less computing time and produce almost accurate results [13]. Numerous applications of multi-objective optimization are found in management and engineering. Most optimization issues that arise in practical applications contain many goals and thus can be categorized as multi-objective issues. Multi-objective optimization issues may be solved numerically, although it is difficult to do so because of theoretical and computational difficulties. As a result, multi-objective optimization has acquired a lot of attention in the last couple of years, and meta-heuristic algorithms have proven effective in solving problems of this nature.

2 Meta-Heuristic Algorithms in Solving Multi-objective Problems

Different variants of meta-heuristic algorithms, such as the genetic algorithm (GA), particle swarm optimization (PSO), ant colony optimization (ACO), and differential evolution (DE), can be implemented in real-life problems that require multi-objective optimization. The section below addresses the implementation of these algorithms in different multi-objective optimization problems.

2.1 Genetic Algorithm in Multi-objective Problems

Genetic algorithm (GA) refers to a technique for tackling confined as well as unconstrained optimization issues. It depends on natural selection, a method that drives biological evolution. The multi-objective genetic algorithm is a direct search method for multi-objective optimization problems. Genetic algorithm (GA) is a popular approach for solving multi-objective optimization problems. This algorithm seeks to locate a number of Pareto solutions as opposed to the typical multi-objective algorithm, which seeks to find a single Pareto answer. A collection of solutions known as the Pareto front or Pareto set is the outcome of a multi-objective genetic algorithm. These solutions show how competing goals are traded off, with one goal being improved at the expense of another. One should consider the diversity and elitism of solutions while utilizing genetic algorithms to solve multi-objective optimization problems.

Using genetic algorithms to solve multi-objective optimization issues has various advantages:

Population-based approach: GAs keep track of a population of potential solutions, which makes it possible to explore the solution space and find several Pareto-optimal answers.

Trade-off between diversity and elitism: To prevent convergence to a single solution, GAs balance fostering diversity among the options with protecting the best solutions (elitism).

Effective solution space exploration: It is made possible by GAs, which employ crossover and mutation strategies to explore various areas of the solution space and obtain a variety of Pareto-optimal solutions.

Flexibility: GA is appropriate for a variety of optimization issues since they can handle both continuous and discrete decision variables.

Multi-objective optimization using GA has been successfully applied in solving pavement maintenance problems in article [14]. The primary goal of the pavement maintenance system is to identify the most cost-effective ways to keep pavements in usable condition for a predetermined amount of time. A multi-objective optimization with minimal cost and maximum condition for the utilized road network was developed in this study. The model has been adopted for road networks, and an optimal solution has been obtained. It can be said from this study that pavement management has certain budget constraints, and pavement managers can decide the amount of funds that should be invested in the time based on desired road network conditions, hence can be identified as multi-objective problem and is successfully addressed using GA. In [15], GA was employed along with an artificial neural network (ANN) to assess building retrofit projects. In this case study, a school building is utilized to demonstrate the value of optimization and to identify any potential issues. It essentially entails the optimization of objective functions with a focus on the properties and functionality of buildings, such as retrofit cost, consumption of energy, and heat retention. GA has been applied in the thermodynamics cycle of an ideal turbojet engine [16] which may be identified as a multi-objective problem. The technique employed is the Pareto approach optimization. Specific thrust (ST), thrust-specific fuel consumption (TSFC), propulsion efficiency, and thermal efficiency are the aim factors considered in this study. For bi-objective optimization, different pairs of objective functions have been utilized in this case. The outcome achieved unequivocally demonstrates that the algorithm used is superior at upholding the Pareto approach's quality in both bi-objective and multi-objective optimization processes.

A variant of the GA, known as the nondominated sorting genetic algorithm (NSGA-II), was applied in [17] to find out the heat exchanger pressure drop, and an effective solution is obtained using two objective functions: maximum effectiveness and minimum total annual cost. GA embedded in multi-objective optimization can also be employed in the topology optimization of elastic structures [18]. The primary purpose of the work is to obtain evenly grouped solutions or populations to obtain the Pareto set for the given case. Moreover, this type of work can be extended to specific types of chromosomes that need new crossover and mutation operators in GA. Analyzing various multi-objective problems that apply GA, it is noticed that GA using multi-objective optimization yields good results in various real-life problems of different engineering and medical domain.

A potent method for resolving multi-objective optimization issues is the use of genetic algorithms. They have advantages including flexibility in managing various kinds of decision variables, trade-off between elitism and diversity, and effective exploration of the solution space. The trade-off between competing goals can be represented by a group of Pareto-optimal solutions that are found by applying genetic algorithm.

2.2 Particle Swarm Optimization Algorithm in Multi-objective Problems

A popular population-based optimization algorithm for resolving single-objective optimization issues is particle swarm optimization (PSO). Particle swarm optimization (PSO) is an efficient meta-heuristic optimization technique motivated by swarm behavior observed in an environment, such as fish and bird schools. To discover the optimal solution, PSO employs swarms that roam the search area. Each agent retains the best solution offered by both itself and its neighbors. Together, they can obtain the optimal solution more quickly because of this. Single-objective optimization issues are frequently resolved using this population-based stochastic optimization approach. However, PSO may be expanded to address multi-objective optimization issues, where several competing objectives must be optimized concurrently.

To modify PSO to be applicable to multi-objective optimization, a number of changes are suggested. With these changes, PSO should be better equipped to manage several competing goals and identify a group of solutions that present the trade-off between them. Some of these techniques are Pareto dominance, competitive strategy, dynamic neighborhood, crowding distance, etc. Pareto dominance is a concept in multi-objective optimization that compares solutions based on their performance with respect to multiple objectives. The amount of crowding a solution has in the goal space is indicated by its crowding distance. By giving preference to solutions situated in slightly populated locations of the target area, it contributes to preserving population variety. Crowding distance can be used to improve PSO and encourage Pareto front exploration. In PSO, each particle interacts with a fixed set of neighboring particles inside a predetermined neighborhood structure. The neighborhood structure in multi-objective PSO can be dynamically modified to enable particle interactions with various neighbor sets at various optimization stages. This promotes diversity preservation and the exploration of various Pareto front regions. A competitive strategy can be incorporated into the Pareto external archive generation process to expedite the search for nondominated solutions. This tactic improves PSO's capacity to identify an extensive range of Pareto-optimal solutions. These modifications and enhancements to PSO enable it to handle multi-objective optimization problems effectively.

PSO can be studied for multi-objective building energy performance in article [19]. In this study, a multi-objective optimization algorithm and building energy simulation software are integrated to find nondominated solutions and improve building energy efficiency. The target is to reduce the total energy requirement of the building and to understand the interrelation between the cost functions. Thus, this method provides a powerful tool for searching for an optimal solution while saving time, and therefore, decisions can be made in the early stage to enhance the energy production of the building. The PSO algorithm along with 3D simulation can be used in the case of natural convection in a cylindrical annulus filled with molten magnesium in the existence of a magnetic field [20]. The other walls are thermally separated, while the inner and outer walls are perpetuated at a fixed temperature. Here, the magnetic field strength and the rate of natural convection heat exchange are considered as target functions. The outcome demonstrates the efficiency of PSO algorithm in obtaining desired results. The reliability and redundancy of the multi-objective optimization problem may be addressed using PSO algorithm [21]. The target functions in this problem are the reliability of the system and the associated design cost. The multi-objective algorithm gives a Pareto optimal solution instead of a single solution. The model can be made more efficient by introducing fuzzy network logic, and it results in a fuzzy multi-objective optimization problem taking into consideration the reliability and costs.

An improved particle swarm optimization (IPSO), another variant of PSO, is used [22] for the multi-objective power transfer problem. The fitness functions that are addressed in this problem includes cost, voltage stability, loss, and emission effects. The algorithm uses a fuzzy mechanism to obtain the best solution from the Pareto set. The results imply that more than one nondominated solution is obtained, allowing the choice to select a good solution. The fuzzy-based PSO finds uses in environmental or economic power dispatch [22]. The economic and emission effects are treated as the objective. Here, a fuzzy-based PSO is devised to dispatch electric power, taking into consideration both environmental and economic matters. The bi-objective power dispatch basically minimizes pollution and fuel costs. Thus, the PSO algorithm is proven to yield better results in the discussed cases when implemented to obtain optimal solutions in multi-objective problems. The use of methods like multiple swarms, coevolution, fitness assignment, and diversity maintenance allow PSO to be modified to handle multi-objective optimization issues in general. Due to these additions, PSO can explore and approximate the Pareto front and generate a group of non-dominated solutions that show the trade-off between various goals.

2.3 Ant Colony Optimization in Multi-objective Problems

The fundamental idea behind ant colony optimization (ACO) is observing how ants leave their nests to travel the shortest distance feasible while looking for food. To choose the quickest path, the ants searching for food secrete a pheromone, which other ants follow. The two main criteria that obtain the minimized path from the food supply to the nest are the amount of pheromone and the rate of pheromone evaporation to other paths. As more ants use the shortest route, these two factors will rise. This well-known Meta-heuristic approach may also be employed to handle multi-objective optimization issues in ACO.

ACO has been modified for multi-objective optimization using a number of different methods. Pareto dominance can be used by ACO to evaluate and choose solutions. ACO can include user-defined weights or preferences to direct the search to particular Pareto front regions. As a result, the algorithm may concentrate on identifying solutions that suit the user's preferences. Combining ACO with multi-objective optimization methods depending on decomposition is possible. This combination breaks down the multi-objective problem into a number of single-objective subproblems, enabling ACO to investigate distinct areas of the Pareto front. Interval outranking can be added to ACO to create a novel multi-objective ACO optimizer. When there is ambiguity or inaccurate information in the objectives, interval outranking can help.

The exploration strategy of ants served as the inspiration for ACO, which is successfully utilized to address a few combinatorial optimization issues. ACO can be used to optimize the flow shop scheduling problem [23]. The multiple objectives for this model are makespan, total flow time, and total machine idle time. However, the important objective is to minimize scheduling costs. The efficiency of ACO is measured with respect to other algorithms, and the result obtained clearly shows that it is better and more effective than other meta-heuristic algorithms. The multi-objective resource allocation problem [24] can be optimized using modified ACO. The fitness function is to decrease cost and maximize efficiency. The algorithm is found to be efficient by comparing the solutions obtained by ACO with hybrid GA. The ACO algorithm outperforms the other considered algorithms with regard to the nondominated solutions by 50%.

Multi-objective optimization using ACO is employed for task scheduling in cloud computation [23]. Here, the makespan and user cost are taken as constraints in optimizing both the performance and cost. The constraints are defined to calculate and provide feedback related to the fitness function—performance and cost. Due to the constraint function, the solution changes over time so that optimal solutions can be achieved. As per the algorithm, based on the metrices, it can be said that this multi-objective optimization model using ACO outperforms other algorithms, and it increases up to 56.6%, in the best possible case considered here. Mixed model assembly line balancing [25] can be achieved using the ACO. For the model to be efficient, the design should be done by satisfying constraints such as space, time, and location. The cost function for this technique is to decrease the number of stations for a given time cycle. The main problems with this model are problems of capacity utilization and differences in station timings. In this model, the requirement is to minimize the delay and smoothness between the stations. According to the results, ACO performs better than the rank positional weighted method, GA and association rule algorithm with good computational time.

Moreover, ACO can be used in a secured routing protocol [26]. This is used with the motive of obtaining high network security with minimized energy utilization in a wireless sensor network. The objectives considered are the residual energy of nodes and the routing path. The results indicate that the algorithm can perform well against black hole attacks in WSN routing. In this study, Pareto optimization is used to address the limited resources and WSN routing issues. It is essential to understand that ACO wasn't developed with multi-objective optimization in mind, hence implementing it towards problems with multiple objectives necessitates changes and adaptations. ACO has been integrated with various meta-heuristic algorithms or local search techniques in a number of mixed strategies that have been put out to improve its performance in resolving multi-objective optimization issues.

2.4 Multi-objective Differential Evolution Algorithm

Differential evolution (DE) is an evolutionary algorithm that utilizes an evolutionary process to iteratively improve a candidate solution to solve global optimization challenges. Multi-objective optimization problems can be resolved by employing another popular evolutionary technique referred to as DE. This is a population-based algorithm that searches the search space for the best solutions by combining mutation, crossover, and selection. To take into account the trade-offs between various objectives, these operators might be altered or expanded. DE may be expanded to handle constraints in multi-objective optimization problems. DE can incorporate techniques to sustain diversity in the population. This helps in exploring different regions of the Pareto front and avoiding convergence to a single solution. Various techniques, such as penalty functions or repair operators, can be used to make sure that the generated solutions persuade the problem's constraints.

DE algorithm can be applied to the economic environmental dispatch problem [27]. The objective of this algorithm lies in optimizing the fuel cost and emission, reassuring all the equality and inequality constraints. It was found that it provides good quality solutions and good computational time. DE is very efficient in solving multi-objective problems and is extended to a greater number of objectives to yield good solutions. During the peak time of COVID-19 spread, there was a need for automated chest CT image analysis to cope with the limitations of medical professionals [28]. A convolutional neural network (CNN) was used along with DE and certain machine learning techniques to design this machine. The experimental outcomes reflect that the proposed model can categorize chest images with greater accuracy. The DE has also been adopted in optimizing the size of a PV/wind/diesel hybrid microgrid system (HMS) [29]. The need for optimizing the microgrid system size and its storage is to obtain reliability and system cost. The computational time is reduced as the objective functions are treated separately. The main requirement for the design of HMS is the management of energy. Then, it is used to analyze the cost of electricity, loss of power supply probability and renewable factor and the relation of these factors with HMS cost and reliability. The main motto of this is to decrease the price of renewable energy and thereby thrust renewable energy resources and their diversification.

The nondominated sorting differential evolution algorithm can be employed in optimizing the phasor management unit placement model, which has a significant role in wide-area measurement systems in power systems. [30]. The multi-objective model aims to achieve minimized phasor management unit number and maximized reliability in measuring. The nondominant sorting differential evolution efficiently improves the crowding mechanism and mutation strategy. This can easily optimize the multi-objective optimization model, produce many optimal solutions, and achieve an accurate Pareto front. Evolution methods for crop planning and the operation of irrigation reservoirs also make use of DE [31]. The objective is to optimize cropping patterns and working policies for multi-crop irrigation systems.

2.5 Other Multi-objective Meta-Heuristic Algorithms

There are many meta-heuristic algorithms other than these four classes of meta-heuristic algorithms which also show promising results in solving multiple kinds of multi-objective problems. One of such efficient algorithm is the teaching learning-based optimization (TLBO) algorithm that can be implemented in optimizing the power flow problem, taking into account the total fuel cost and total emission of the units [32]. The multi-verse optimization (MVO) algorithm is utilized in optimizing the distribution and sizing of the energy resources and working of the energy storage system to minimize the annual costs [33]. The obtained optimal solution is reduced under economic matters and cost issues to offer decision support for the utility. Moreover, this method turns out to be better than the nondominant sorting GA and PSO as per the multi-objective optimization metrics. In addition, there are other efficient meta-heuristic algorithms, like the grey wolf optimization algorithm and artificial bee colony optimization, that are also found to be effective and robust in solving optimal problems. Due to being less prone to the shape or continuity of the Pareto front, i.e., they can be effortlessly applied for discontinuous or concave Pareto fronts, this class of algorithms is generally preferred for optimization problems where there are multiple objective functions that required to be optimized without compromising the best solution. These can be considered merely a few illustrations of typical meta-heuristic methods for multi-objective optimization problems. Every algorithm has advantages and limitations of their own, and the best method depends on the particular issue at hand. To determine the best algorithm for a specific issue, it can be helpful to compare and assess multiple approaches.

3 Various Multi-objective Optimization Techniques

3.1 Weighted Sum Method

A highly popular and simple approach for multi-objective optimization is the weighted sum approach. This technique modifies the weight among the objective functions and obtains Pareto optimal solutions [34]. In this method, the overall performance rating of an alternative is calculated as the sum of the normalized performance rating and weights of the criteria. The mathematical expression of this method is given as

$$S_i = \sum_{j=1}^{n} w_j \cdot r_{ij} \tag{1}$$

where S_i represents the overall performance rating of alternative i, w_j denotes the weight of the objective, and r_{ij} denotes the normalized performance rating of alternative i with respect to criterion j. Vector normalization or max–min linear normalization is the normalization procedure where the weighted sum technique is used [35]. This technique is applicable in robotics, data processing, etc. [36]. In the case of robot selection, it is used to eliminate the subjective factors and the maximum and minimum expert values [37]. Various objectives in a mobile cloud database environment, such as the time required to execute a query, monetary cost, and energy consumed by a mobile device, are incorporated with the help of a decision process based on a weighted sum technique. The advantages of using weighted sum methods include its intelligibility and simple execution. This technique is helpful to find the optimal solution for convex problems in the entire Pareto optimal set. The weighted sum method has also proven to be efficient in biomedical applications.

Along with advantages, there are also some drawbacks to these methods. First, all the objectives need to be converted into similar types for the mixed optimization problems to be solved. Second, if there are uniformly distributed sets of weights, they do not ensure that we will obtain a uniformly distributed Pareto optimal set as the solution. Third, in the case of two different sets of weight vectors, they may or may not produce two different optimal solutions. Finally, various minimum solutions of a particular vector that represent various solutions in the Pareto optimal range can exist [38].

3.2 Epsilon Constraint Method

The epsilon constraint technique is a way to transform algorithms. It converts unconstrained problem algorithms to constrained problem algorithms. To perform this conversion, we use epsilon level comparison. Search points are compared based on objective value pairs and violations of their constraints [39]. This technique modifies multi-objective problems into various single objective tasks. This transformation is performed by using the following process to minimize $f_z(x)$ and can be given as

$$f_z(x) \leqslant \varepsilon_j \tag{2}$$

where $j = 1, 2, 3, \ldots\ldots m$, and $x \in S$, where S is the feasible domain that is described by the constraint of equality or inequality. The vector of the upper bound is ε, which sets the maximum value of each objective function. To obtain the total set or the Pareto optimal set, the ε value has to be varied along the Pareto front for each objective [40]. In energy management systems, optimal power flow is optimized by this method. Using this method, the generation cost of power is optimized and minimized in such a manner that the total power loss can be limited to some predefined limits [41]. The epsilon constraint method can also be employed to optimize the average waiting time in the scheduling problem of cloud computing [42]. The predator–prey model is another multi-objective optimization technique that utilizes epsilon constraint method to activate the adaptation of selection behavior [43]. In research related to multitarget urban unmanned aerial vehicles, the shortest total flight distance and the maximum average satisfaction of the clients are taken as objective functions. To optimize these tasks, one can take help of the epsilon constraint method. Studies show that the results produced by the epsilon method are more effective and efficient than those of other multi-objective algorithms [44].

The advantage of using the epsilon constraint method in various real-life optimization tasks is that it can be implemented in both convex and nonconvex optimization problems. The main disadvantage of this method is that we have to carefully choose the upper bound vector ε in such a manner that it lies between the lowest and the highest value of every.

3.3 Pareto Dominance Method

In a multi-objective practical problem, the Pareto dominance method is used to sort out the best optimal solution out of a set of solutions. When it is observed that a particular solution in the solution set is not inferior to the other solution in any characteristic, then we can state that this solution strictly dominates the other one. According to this method [45], if in a minimization problem a solution x is better than the other solution y, then it is said that f(y) is dominated by f(x) if $f_k(x) \leq f_k(y)$, where k = 1, 2, 3, 4……, N. The Pareto dominance method is useful in the multi-objective problem related to energy saving, cost minimization, and emission reduction in power plants [8, 46]. A Pareto-based method typically consists of two components: a diversity estimator and a Pareto dominance-based decision. While the latter encourages population diversity, the former directs selection toward the best option. However, when trying to solve optimization issues with numerous targets, the Pareto dominance-based approach fails.

3.4 Other Multi-objective Optimization Techniques

In addition to these multi-objective optimization techniques, there are several more methods that are widely used in the fields of engineering, health care, commerce, and agriculture. The global criterion approach is a multi-objective optimization method in which the decision maker assumes an approximate solution to point out the optimal solution. The goal programming method is introduced when it becomes difficult to approach some constraints. This method is applied by solving linear programming. Using this linear programming, the minimization of the deviation of the objective functions from the prespecified goals is performed. However, it cannot be said that the solution achieved by this method is the most efficient solution.

The multi-objective fuzzy concept is based on a multivalued approach. It is formulated by the expression

$$F_{fuzzy}(x) = \prod_{i=1}^{k} \mu_i(f_i(x)) \tag{3}$$

Here, μ is known as the degree of truthfulness. If μ is 0, it implies that the statement is false, and if its value is 1, it indicates that the statement is true. The entire fuzzy optimization concept can be formulated as

$$\begin{gathered} \max\ F_{fuzzy}(x) \\ \text{s.t.x} \in X \end{gathered} \tag{4}$$

The complex method is used in multi-objective optimization to substitute the worst solution point with a new better solution point. This new point is considered the reflection of the previous point. The complex technique has several possible solutions. When all the points in the complex converge, the best solution is found [47, 48].

4 Applications

Multi-objective optimization algorithms find applications in numerous domains. It is not limited to a specific field and can be used to deal with some classic problems. Recently, there has been a trend of multi-objective optimization in the field of structural design. Multi-objective optimization is employed in optimizing concrete mixture proportions, along with some machine learning algorithms [49]. It is the case in which multiple objective functions, such as strength and cost, along with many design variables, such as concrete components, are optimized under nonlinear constraints. The selected machine learning algorithm from the literature survey gives the objective function for this optimization problem. This employs a PSO algorithm to obtain the desired objective. The multi-objective optimization model serves as a guide to make decisions before the construction phase. Multi-objective optimization has been applied to design finance-based scheduling for multiple projects by using a variant of GA [50]. This study basically checks the performance of the multi-objective optimization model and helps in developing parity between project financing and project scheduling. Multi-objective optimization has an application in IoT-based cognitive radio networks. Multi-objective optimization proposed with the ant colony optimization algorithm uses the data and saves energy by the process of efficient clustering and data aggregation. The energy is enhanced using multi-objective ant colony optimization compared to artificial bee colony optimization or GA [51].

Multi-objective optimization is used to design a sparse span array [52], which can be used to detect microwave signals. The two main objectives of this multi-objective optimization problem are to minimize the number of antenna elements and to maintain the side lobe level to search for the optimal antenna array distribution from the entire array framework. The problem involves a PSO algorithm. Multi-objective optimization is also applied for optimizing the maximum area with wind and thermal comfort in building design [53]. Algorithms such as PSO and GA have been applied in multi-objective optimization techniques such as the Pareto dominance method, and it is observed that different algorithms perform differently depending upon the technique applied to the problem in hand. Building performance optimization method is created based on GA and PSO to maintain the comfort and energy content of a residential building [54].

The scope of multi-objective optimization can be extended to home care services [55] also by using a two-stage meta-heuristic algorithm. It aims to minimize the costs of routing and other related services and maximizes nurse–patient compatibility. However, there is enough room for further implementation of this model on a large scale. Another benchmark application of multi-objective optimization is project-based financing and scheduling [50]. A finance-based multi-objective optimization model is developed using the nondominated sorting GA as the basis. Multi-objective optimization has also been extended in devising a model for sustainable harvest decision problems, which is currently a major issue [56]. The main target is to achieve sustainable harvest planning by balancing constraints such as weather and climatic conditions, the price of commodities, changes in production, and crop ripening patterns. It maximizes the profit while minimizing the harmful gaseous discharge and release of wastes, and this makes use of the nondominated sorting GA based on a concept called the nondominated sorting genetic engineering algorithm.

The efficient utilization of wind energy also utilizes multi-objective optimization techniques [57]. In this case, the meta-heuristic algorithm has improved wind energy utilization in terms of stability and accuracy. The MOO is also associated with the distribution of available resources so that multiple objective functions can be satisfied without violating the constraints. This type of real problem involves hydrothermal scheduling and transportation scheduling. Natural gas transportation through pipeline networks employs multi-objective optimization of the nondominated sorting GA. The main objective function is to maximize the gas flow rate and minimize the power, and the constraints are the pressure at the nodes and the rotational speed of the compressor.

A new multi-objective optimization model on the basis of ratio analysis is given in [58]. In this technique, the denominator is the square root of the sum of squared responses. These dimensionless ratios between zero and one should be added in the case of maximization and subtracted in the case of minimization. This model is implemented in privatization in a transition economy [59]. Due to the optimization of more than one fitness function at a time, multi-objective optimization has become prominent in recent times. In all cases, the multi-objective optimization technique yields better results than the single-objective optimization technique for the same problem statement. The difficulty of problems enhances with the increase in the number of objective functions, making the multi-objective optimization algorithm difficult to deal with.

5 Challenges and Future Scope

For the design, operation, control, and optimization of industrial processes, multi-objective optimization is an essential tool. There are two types of difficulties in this area. The primary difficulty is in the handling of large-scale nonlinear problems and understanding the application of unpredictability to mathematical models to use novel approaches to resolve conflicts between various objectives. Additionally, enhancing the level of specifics in mathematical designs [6]. This entails applying demanding thermodynamic models to energy systems, taking individual mass balances or property balances into account when designing water networks, and concurrently working on the optimization of ecological and safety indices along with financial and operational goals. It also involves including the comprehensive layout [8–10] of processing equipment and processing facilities.

The future scope of multi-objective optimization is vast and holds great potential for various fields and applications. Multi-objective optimization can enhance the capabilities of artificial intelligence and machine learning algorithms [12–14]. By optimizing multiple objectives simultaneously, these algorithms can exhibit improved performance, robustness, and adaptability. Multi-objective optimization can play a crucial role in achieving sustainable development goals by maintaining balance among economic, social, and environmental objectives. It can help optimize resource allocation, energy systems, transportation networks, and land use planning to minimize negative impacts on the environment while maximizing overall social welfare. Multi-objective optimization can revolutionize engineering design processes by considering multiple conflicting objectives simultaneously. It can optimize the placement and capacity of renewable energy sources, energy storage systems, and grid integration strategies to maximize renewable energy generation, minimize costs, and reduce carbon emissions. Multi-objective optimization techniques can aid financial institutions and investors in portfolio optimization, risk management, and asset allocation.

6 Conclusions

In conclusion, metaheuristic algorithms have shown to be strong and useful resources for resolving issues involving multi-objective optimization. By examining the solution space and identifying a set of Pareto-optimal solutions that illustrate the trade-off between opposing goals, these algorithms provide effective answers. In a number of fields, the use of metaheuristic algorithms, including Differential Evolution (DE), Ant Colony Optimization (ACO), Particle Swarm Optimization (PSO), and Genetic Algorithms (GA), has produced encouraging results. These algorithms offer adaptable and flexible search techniques that are capable of solving challenging non-linear situations. The capacity of metaheuristic algorithms to manage several goals at once is one of its main features. Metaheuristic algorithms can identify a multiple range of Pareto-optimal solutions by incorporating diversity maintenance and Pareto dominance methodologies. This enables decision-makers to consider various trade-offs and arrive at well-informed conclusions. In addition to their scalability and robustness, metaheuristic algorithms can be used to solve real-world issues involving numerous choice variables and vast solution spaces. Additionally, by adding penalty functions or repair operators, they may deal with limits and uncertainties and guarantee that the created solutions meet the criteria of the problem. It is crucial to remember that the particular issue domain, the characteristics of the challenge, and algorithmic variants can all affect the efficiency and usefulness of metaheuristic algorithms in multi-objective optimization. The needs and constraints of the task should be properly taken into account before selecting an algorithm. All things considered, metaheuristic algorithms have proven to be successful in resolving multi-objective optimization issues and offering decision-makers a range of varied Pareto-optimal solutions. Their adaptability, resilience, and expandability render them advantageous instruments for addressing intricate decision-making quandaries across diverse fields.

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Chapter 2
Counterfactual Explanations and Federated Learning for Enhanced Data Analytics Optimisation

Syed Irtija Hasan, Sonia Farhana Nimmy, and Md. Sarwar Kamal

1 Introduction

The landscape of machine learning and artificial intelligence (AI) is constantly evolving to address the complicated challenges posed by real-world data while upholding the principles of privacy and transparency. As the digital universe grows, it is becoming increasingly clear that traditional centralised models, while efficient, can falter in the face of the twin challenges of privacy and heterogeneity [1–3].

This is where Federated Learning (FL) comes in, a groundbreaking paradigm shift with the potential to reshape our understanding of machine learning [4–6]. FL the genius of the system is that it allows decentralised devices to collectively contribute to a unified model. Crucially, the raw data is restricted to its original source, mitigating privacy risks [7, 8]. However, like any innovation, FL is not without its challenges. One major problem is achieving consistent model performance when data distributed across different devices is not uniform, or what experts call non-ID.

Simultaneously, as the gap between complex model predictions and human interpretability narrows, the demand for transparent mechanisms for model decision-making increases. This is where counterfactual explanations stand out [9, 10]. By illuminating alternative scenarios, they have become an indispensable tool for explainable AI. However, as these models continue to advance, we are faced with a pressing question: How can we maintain the consistency and relevance of these explanations over time?

At a time when data protection is paramount, the health sector is at a unique crossroads. Faced with strict regulatory requirements and a duty to protect patient data, the

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sector is perfectly positioned to benefit from advances in federated learning (FL) [11–13]. The potential to gain actionable insights without breaching the confidentiality of individual datasets could revolutionise personalised medicine, epidemiology and the broader spectrum of health analytics.

But as tempting as the promise is, the road to realisation is paved with obstacles. First and foremost, the vulnerability of the system to malicious actors becomes apparent when they inject poisoned data, compromising the integrity of the entire federated architecture. In addition, there is the latency caused by the decentralised structure of FL and the resulting communication costs, which are still the subject of intensive research and improvement.

In this endeavour, our work attempts to weave these multi-layered strands into a coherent tapestry, taking an integrated approach that combines the power of FL with the transparency of counterfactual explanations [14–17]. Building on the pioneering efforts of forerunners in the field, we seek to chart a course towards a future of federated machine learning that is not only efficient and transparent, but also inherently secure.

In this book chapter, our explorations in a structured way. We begin with Sect. 2, which deals with a comprehensive literature review. We then present our methodology in Sect. 3. Section 3 is divided into various subsections: Sect. 3.1 deals with federated learning, Sect. 3.2 with graph theory, Sect. 3.3 with hierarchical Bayesian models for gene expression in cancer research, Sect. 3.4 with counterfactual explanations, and Sect. 3.5 with structural causal models for gene expression data. Our findings and results are the focus of Sect. 4, followed by a thoughtful discussion. Finally, we summarise our findings in Sect. 6 and draw a comprehensive conclusion.

2 Literature Review

Counterfactual explanations, a key tool of explainable AI, provide opportunities to understand and potentially challenge unfavourable decisions made by machine learning models. However, this research [18] highlights a limitation: When machine learning models are retrained, their previous counterfactual explanations may become irrelevant or incorrect. The authors provide a structured understanding of this problem and propose a solution, 'counterfactual data augmentation'. This technique improves the stability of counterfactual explanations as models evolve [19–21]. Using simulated model retraining experiments, the method is shown to improve the consistency of the explanations and make them more reliable for real-world applications.

Zeou et al. [22] address the challenges of federated learning, a method for training shared models on numerous devices while preserving user privacy. To achieve both fairness among participants and resilience against potential threats, they reinterpret federated learning as a multi-objective optimisation task. By introducing a new algorithm, FedMGDA+, they ensure that it reliably converges to balanced solutions and guarantees that each participant's performance is not overlooked. FedMGDA+ features a user-friendly implementation and requires minimal adjustments to the hyperparameters. Compared to other methods on different datasets, FedMGDA+ consistently outperforms, cementing its position as the top solution for federated learning.

Budrionis et al. [23] highlight the challenges faced by sectors such as health care, which are constrained by strict regulations on data reuse, hindering the adoption of sophisticated data analytics methods. This limitation has led to significant delays in healthcare research and development, over-allocation of resources and reliance on insufficiently large datasets. One promising solution is federated machine learning, which meets regulatory requirements without compromising privacy. However, current computing frameworks that support federated learning are still under development and have not been sufficiently evaluated in real-world scenarios. The authors conducted three experiments on the dataset MIMIC-III to evaluate the performance of neural network models trained with federated learning compared to traditional centralised methods. The objective was a binary classification task for predicting hospital mortality. The experiments found that the prediction performance of the federated models closely matched the centralised baseline in terms of metrics such as the area of the ROC curve and the F1 score, but the training and inference times were about 9 and 40 times longer, respectively. Interestingly, the distribution of data across nodes had a negligible effect on performance or training duration. In summary, federated learning appears to be a viable approach for sophisticated data analysis in domains with strict privacy constraints.

Spooner et al. [24] present a novel method for counterfactual explanations (CFEs) that uses Bayesian optimisation and is suitable for both classification and regression models. This globally convergent algorithm can handle various regression models and constraints, such as sparsity of features. Impressively, it can handle multiple counterfactual queries simultaneously and continuously improves with each query. The team has carefully structured CFE search for regression models and uses differentiable potentials to solve stability problems related to threshold-driven targets. Within this structure, they demonstrate two significant complexities: (a) the act of confirming counterfactual existence is NP-complete and (b) the use of such potentials for instance identification is CLS-complete. The research presents a comprehensive algorithm for CFEs that integrates both expected improvement and an exponential-polynomial (EP) family into a unique capture function. When evaluated against real benchmarks, their approach shows remarkable efficiency and accuracy.

Sun et al. [25] are looking into the area of federated machine learning, a technique that allows decentralised node devices such as IoT devices and smartphones to collaboratively build a model without transmitting raw data. This approach has potential advantages in terms of privacy and economic efficiency, especially when combined with a well-crafted communication protocol. However, the same protocol could be exploited by malicious entities to perform data poisoning attacks on different nodes, which poses a significant risk to most machine learning models. In this paper, the authors investigate the vulnerability of federated machine learning models, especially in IoT systems. They focus on attacks on a well-known federated multitasking system designed to overcome the statistical challenges typical of federated systems. The task of planning optimal poisoning attacks on this system is formulated as a two-stage programme that can be adapted to any choice of target or attack nodes. A pioneering system-aware optimisation method called 'Attack on Federated Learning' (AT2FL) is introduced to efficiently compute the implicit gradients for poisoned data and find the best attack strategies in the federated context. To the authors' knowledge, this is a pioneering attempt to study federated machine learning attacks based on data poisoning. Experiments conducted on various real-world datasets confirm the sensitivity of the federated multitasking model to direct and indirect poisoning attacks.

Wang et al. [26] address the challenges in federated learning where high concurrent access to cloud servers leads to increased transmission delays. Existing solutions do not effectively deal with the latency issues caused by server concurrency. The authors introduce an edge-based communication framework that groups devices by network location and uses mobile edge nodes as communication nodes, reducing direct connections to the main server and the associated latency. They also present a cosine similarity-based method to eliminate redundant local model updates, minimising unnecessary communication. Tests show that this method reduces local updates by 60% and increases the convergence speed of the model by 10.3% compared to traditional federated learning.

Zhang et al. [27] address the challenges of Federated Learning (FL) with non-IID client data that affect optimisation and feature consistency. Their proposed solution, FedUFO, uses an adversarial module and consensus loss to improve feature matching and reduce objective inconsistencies. In this way, they demonstrate superior performance and balance between clients.

Xu et al. [28] discuss the potential privacy and security vulnerabilities in federated learning, where shared gradients can be exploited by attackers to decrypt user privacy. They cite two challenges: protecting user privacy during training and verifying the integrity of the results aggregated on the server. While some solutions aim at secure federated learning, ensuring correct server operations without compromising user privacy remains unsolved. To solve this problem, they introduce VerifyNet, a pioneering federated learning system that protects privacy and verifies the training process. It uses a double-masking protocol to protect user data and requires cloud servers to prove aggregation is correct. VerifyNet ensures that attackers cannot fake this proof without solving an associated NP-hard problem and accepts user aborts during training. Tests with real data confirm the effectiveness of VerifyNet.

There have been many innovations in the field of federated learning, especially with regard to privacy and security. However, there are still gaps, especially in reconciling accurate model training and privacy. Ferrario et al. found that counterfactual explanations are unstable during model training, which can cause previous explanations to become outdated. While Zeou et al. proposed the FedMGDA+ algorithm for balanced solutions in federated learning, it does not directly address the difficulties of simultaneously ensuring model accuracy and privacy. Budrionis et al. and Wang et al. pointed out latency issues in federated learning, but their solutions rely heavily on technical adaptations rather than holistic model improvements. Bayesian optimisation proposed by Spooner et al. for counterfactual explanations and FedUFO proposed by Zhang et al. both offer methodological advances but do not holistically close the gap between data reliability and accurate, efficient model training. Sun et al. focussed on security threats and pointed out the vulnerability of federated learning to data poisoning attacks. VerifyNet proposed by Xu et al. , while emphasising both user privacy and validation of results aggregated from the server, does not provide a unified solution to the overarching challenge.

In contrast, our proposed approach combines the strengths of counterfactual explanations and federated learning to improve data analysis optimisation. This fusion not only addresses the gaps that arise when retraining models and inconsistent counterfactual explanations, but also covers a broader range of concerns, from model accuracy to privacy. Through this unified approach, our model aims to fill the existing research gaps and provides a more robust, efficient and privacy-friendly solution for federated machine learning.

3 Methodology

The methodology is divided into two sections. Federated learning and counterfactual explanations. First this research introduces federated learning and then counterfactual explanations. Figure 1 describes the steps of federated learning with counterfactual explanations for optimal biological outcomes for cancer data.

3.1 Federated Learning

Federated learning (FL) is a breakthrough approach in machine learning, notable for its usefulness in training models on decentralised devices or servers with local data samples without sharing them. The method introduces a sophisticated privacy-preserving mechanism where the model is trained on multiple decentralised devices or servers that keep their local data samples instead of transmitting the raw data to a central location. This paradigm not only facilitates the incorporation of insights from multiple and dispersed data sources, but also significantly mitigates privacy risks as the raw data never leaves its original location. The collected insights or model updates from these scattered sources are then collaboratively combined into a global model that reflects the learned patterns and insights of all participating entities. In this study, the applications and challenges of FL are explored in depth. The focus is on the effectiveness of FL in orchestrating efficient model training without compromising data privacy and security.

3.1.1 Graph Theory

- **Graph Communication Models**: Use graph algorithms to model and optimise communication patterns between devices or nodes in federated networks. This can

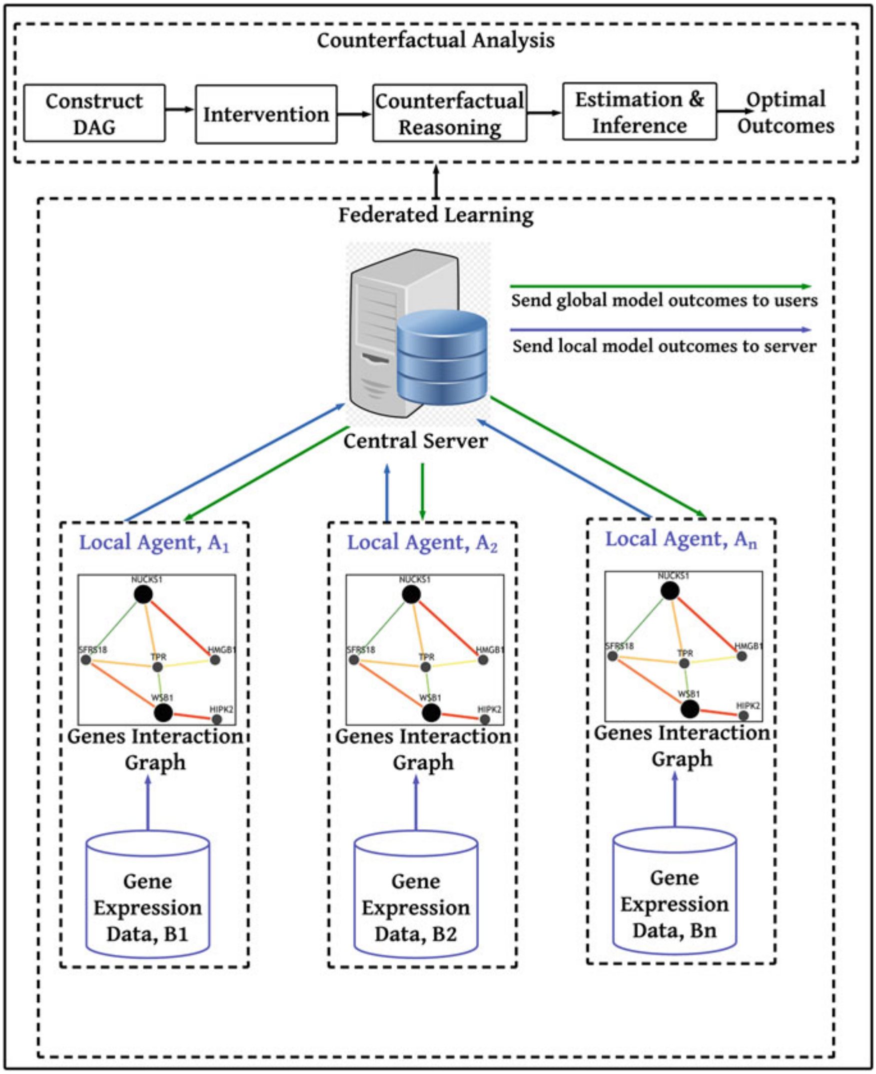

Fig. 1 A structural overview of federated learning for gene expression data with counterfactual explanations

be especially helpful in environments where bandwidth is at a premium or network topology plays a crucial role.

- **Distributed Graph Algorithms**: Utilise distributed graph algorithms to analyse and improve data distribution and computation across nodes.

3.1.2 Statistics

- **Hierarchical Bayesian Models**: In federated settings, consider each device or node as a different 'group' and model the global model as a hierarchical model. This can be used to efficiently combine local updates.
- **Differential Privacy**: Integrate differential privacy to ensure that updates shared by individual nodes do not reveal sensitive information. Noise addition mechanisms like the Laplace or Gaussian mechanism can be applied.

3.1.3 Mathematics

- **Convex Optimization**: Federated Learning can be viewed as a form of distributed optimization. Techniques from convex optimization can provide insights into convergence rates, robustness, and efficiency of the global model updates.
- **Distributed Matrix Factorization**: For models like collaborative filtering in Federated Learning setups, one could use distributed matrix factorization techniques.

In this research, we present two different approaches, both focussing on graph-based federated learning, but applied in different contexts of gene expression analysis for cancer research. Let us explore them in a simple way. While both approaches focus on gene expression in cancer research, the first uses a network (graph theory) to explore and visualise the interactions between genes, and the second uses a multi-level statistical model (hierarchical Bayesian models) to analyse gene expression considering different levels of influence and dependencies. Both methods aim to better understand the complexity of gene interactions and gene expression in the field of cancer research, which could help scientists make breakthrough discoveries in cancer treatment and control.

3.2 Graph Theory in Gene Expression Analysis for Cancer Research

Imagine a giant family tree in which each person (or node) is connected to others by lines (or edges) that show relationships. Now imagine that these are genes and not people. Graph theory uses a similar concept with nodes and edges, but in the context of genes and their interactions. In this approach, genes (nodes) and their interactions (edges) are mapped to help us visualise and understand how different genes interact with each other in cancer. This 'mapping' helps researchers understand which genes talk or interact with each other and may play a key role in the development of cancer. By studying the interactions and connections between genes, we can identify crucial players and pathways involved in the disease and potentially find new targets for cancer treatment.

3.2.1 Gene Interaction Graph Model

Gene interaction graphs can model relationships between genes based on their expression patterns in cancer cells.

Model Definition

- Let $G(V, E)$ be a graph where:
  - V is the set of nodes, representing genes.
  - E is the set of edges, representing interactions or correlations between genes based on expression patterns.
- $w(e)$ represents the strength of the correlation between two genes connected by edge $e \in E$.
- x_v represents the expression level of gene $v \in V$ in a particular sample or group of samples.

A typical objective might be to identify subgraphs or clusters where the average correlation $w(e)$ is above a certain threshold, suggesting a group of genes that act in concert in the context of cancer.

3.2.2 Distributed Graph Algorithms for Large-Scale Genomic Data

Given the vast amount of genomic data, distributed algorithms can be beneficial.

Model Definition

- Let $f(v)$ be a function computed at node v, indicating a specific metric of interest (e.g. differential expression in cancer versus healthy cells).
- Communication cost $C(v, u)$ is defined as the cost (in terms of computational time or resources) to compare or relate the expression data of genes v and u.

The objective in distributed algorithms would be to compute $f(v)$ for all $v \in V$ while minimising:

$$\text{Total Cost} = \sum_{\{v,u\} \in E} C(v, u). \tag{1}$$

This approach allows for the scalable analysis of gene expression data, potentially identifying key genes or pathways implicated in cancer.

3.3 Hierarchical Bayesian Models for Gene Expression in Cancer Research

Think of hierarchical Bayesian models as a multi-level, detailed team of investigators. When detectives try to solve a mystery, they gather different clues and information from different sources. Each piece of information (such as a gene expression level) doesn't tell the whole story on its own, but when you start to put them together and consider how one clue might influence the others, a clearer picture emerges. Hierarchical Bayesian models work in a similar way when analysing gene expression in cancer research. This method considers different levels of information and how they influence each other to help researchers understand the complex biological processes involved in cancer development. By carefully analysing the data and considering the hierarchical nature of biological information, researchers can learn more about how genes are expressed in cancer, which can be crucial for developing new therapeutic approaches.

Problem Definition
Given gene expression data from 105 patients (some healthy, some with cancer), we aim to determine genes whose expression patterns are significantly different between the two groups. This could help in identifying potential biomarkers or therapeutic targets for cancer.

Data Structure
Let's denote:

- y_{ij}: expression level of gene i in patient j.
- G: total number of genes under study.
- $P = 105$: total number of patients.
- z_j: a binary indicator, with $z_j = 1$ if patient j has cancer, and $z_j = 0$ otherwise.

Hierarchical Model

1. **Likelihood**:

   The expression level of gene i in patient j is modelled as

   $$y_{ij} \sim \mathcal{N}(\mu_{i,z_j}, \sigma^2_{i,z_j}),$$

   where $\mathcal{N}$ is the normal distribution, μ_{i,z_j} is the mean expression level of gene i for the group (cancer or healthy) of patient j, and σ^2_{i,z_j} is the variance.

2. **Priors**:

   To capture the hierarchical structure, we place priors on the parameters:

$$\mu_{i,0} \sim \mathcal{N}(M_{i,0}, S^2)$$
$$\mu_{i,1} \sim \mathcal{N}(M_{i,1}, S^2)$$
$$\sigma_{i,0}^2, \sigma_{i,1}^2 \sim \text{Inverse-Gamma}(\alpha, \beta),$$

where $M_{i,0}$ and $M_{i,1}$ are hyperparameters representing the overall mean expression levels of gene i in the healthy and cancer groups, respectively.

3. **Posterior Inference**:

   Using Bayes' theorem, we can compute the posterior distribution of the parameters given the data. With this, we can determine if the expression level of each gene is significantly different between the healthy and cancer groups by analysing the posterior distributions of $\mu_{i,0}$ and $\mu_{i,1}$.

Analysis

Upon obtaining the posterior distributions, we can

- Identify genes whose 95% credible intervals for $\mu_{i,0}$ and $\mu_{i,1}$ do not overlap, indicating a significant difference in expression levels between the two groups.
- Rank genes based on the magnitude of difference, which might be important for further biological validation or therapeutic interventions.

This hierarchical Bayesian model provides a principled way to account for the inherent variability in gene expression data and group-level differences. It can be a powerful tool in the context of cancer research, aiding in the discovery of potential biomarkers or therapeutic targets.

3.4 Counterfactual Explanations

Counterfactual explanations (CE) navigate the complex corridors of interpretability in machine learning and provide insightful narratives about model predictions. Primarily, these explanations seek to set out what minimal changes to input features would alter the model's prediction, providing a 'what-if' analysis that is both insightful and easy for the end user to interpret. CE serves not only as a tool for explaining model predictions, but also for potentially revealing inherent biases or inconsistencies in those prediction results, ensuring that the model's decision-making mechanisms are questioned and understood. Diving into the vast ocean of counterfactual explanations, this research aims to shed light on their crucial role in improving transparency, accountability and trust in machine learning models by providing understandable and actionable insights into their predictive behaviour and ensuring compliance with ethical and legal standards. Consequently, CE paves the way for the creation of more robust, fair and accountable artificial intelligence systems.

3.4.1 Counterfactual Explanations Graph Theory

Counterfactual explanations can be obtained by identifying the smallest change to an input vector that positively affects a prediction from the user's perspective. However, the counterfactuals generated must be close and associated with regions of high data density, and our experiments suggest that faithful counterfactuals come at the price of a higher degree of difficulty.

In the field of explainable artificial intelligence (XAI), counterfactual explanations within the framework of graph theory illuminate the complicated network of possibilities, alternatives and explanatory paths within complex models. Graph theory, with its basic principles built on nodes and edges, provides a structured yet flexible canvas to traverse the myriad potential scenarios represented by counterfactual explanations. Nodes, representing conceivable states or instances, and edges, symbolising transitional possibilities, weave a coherent narrative that can be mapped to visualise alternative realities and show how minimal changes in certain features can lead to altered predictions. Thus, counterfactual explanations using graph theory pave the way not only to illuminate the often opaque decision paths of machine learning models, but also to show how complicated, interconnected networks of data points and features interact to influence these decisions. This integrative methodology improves interpretability by providing a visual, networked map that simplifies understanding of complex predictive mechanisms, increasing trust and transparency in artificial intelligence systems.

- **Causal Graphs**: Use directed acyclic graphs (DAGs) to model causal relationships between variables. It can help in creating interpretable counterfactual explanations by traversing the causal paths.
- **Shortest Path Algorithms**: Identify the most direct paths to reach a counterfactual state, which can be visualised as a node in a graph.

3.4.2 Counterfactual Explanations Statistics

- **Potential Outcomes Framework**: A foundational statistical approach to causality, this can be used to frame counterfactual queries and reason about interventions.
- **Propensity Score Matching**: Match treated and untreated units in observational data to estimate causal effects, aiding in the generation of plausible counterfactual instances.

3.4.3 Counterfactual Explanations Mathematics

- **Optimization on Manifolds**: Find the closest counterfactual instance subject to certain constraints, for instance, by ensuring changes lie within a certain manifold defined by the data.

- **Distance Metrics**: Use mathematical distance metrics like Euclidean, Mahalanobis, or even more domain-specific metrics to identify and score the closeness of counterfactual examples.

3.5 Structural Causal Models for Gene Expression Data

A structural equation model-based analysis found strong support, in particular, for a traditional causal model in which gene expression is regulated by genetic variation via DNA methylation instead of gene expression affecting DNA methylation levels.

Structural causal models (SCMs) are proving to be a key method when it comes to studying gene expression data. Using these models in a biological context, especially for gene expression data, facilitates a structured analysis of how different genes may influence others and how they may contribute to certain phenomena such as the development or progression of diseases like cancer.

For example, let's think of SCMs as a way of tracing the 'cause-and-effect' relationships between genes. Think of a network of dominoes where flipping one domino (changing one gene) can trigger a cascading effect (changes in the expression of other genes or the development of a disease). In the context of gene expression data, SCMs help to create a systematic map that illustrates how different genes interact and influence each other, guiding us through a web of biological information and relationships. This structured model allows researchers to visualise and understand how the change in one element (gene) propagates through the network and can lead to various observable effects.

Of particular note is the ability of SCMs to identify causal relationships and not just correlations in the gene network. This means that SCMs not only identify which genes tend to express together or under certain conditions, but also how one gene can directly influence the activity of another. In the field of gene expression data, this is crucial as it allows researchers to identify potential targets for therapeutic intervention and predict the likely outcomes of manipulating specific genes.

In cancer research, for example, the use of SCMs could illuminate pathways of genes that contribute to tumour development, progression or resistance to therapy. By deciphering these causal pathways, researchers could potentially identify new therapeutic targets and develop strategies to more effectively interrupt the disease process. SCMs are thus important tools in biomedical research. They help scientists navigate the dense and complicated networks of gene interactions and effects and open up avenues for breakthrough discoveries in disease treatment and therapy development.

3.5.1 Setting the Context

Given a matrix E where each row i represents a gene and each column j represents a sample, the entry E_{ij} denotes the expression level of gene i in sample j.

3.6 Components

- **Variables**: Each gene G_i is treated as a variable. It's possible that some genes are latent or unmeasured.
- **Structural Equations**: For each gene G_i, there's a structural equation that defines it in terms of its direct causes or regulator genes:

$$G_i = f_i(G_{pa(i)}, U_i),$$

where $G_{pa(i)}$ represents the parent genes that directly regulate G_i and U_i is an error term capturing unobserved factors or noise.

3.6.1 Constructing the Directed Acyclic Graph (DAG)

The causal relationships can be represented as a DAG. Here, an edge from G_x to G_y indicates a direct causal effect of G_x on G_y, possibly implying a regulatory relationship.

3.7 Interventions

In the context of gene expression data, interventions can be thought of as gene knockouts or overexpressions. Mathematically:

- A knockout of gene G_k can be represented as $do(G_k = 0)$.
- Overexpression might be represented by setting: $do(G_k = \max(E_{k.}))$, assuming $\max(E_{k.})$ is the maximum observed expression of gene G_k.

3.7.1 Counterfactual Reasoning

- Given observed outcomes after a certain treatment (e.g. a drug), one might want to reason what the gene expression would have been in the absence of the treatment.
- Mathematically, if G_i^{obs} is the observed expression of gene G_i under treatment and one wishes to know its expression had another gene G_j not been knocked out, it

would be computed as

$$G_i^{\mathrm{cf}} = f_i(G_{pa(i)}^{\mathrm{obs}}, U_i^{\mathrm{obs}} | do(G_j = G_j^{\text{pre-knockout}})),$$

where $G_j^{\text{pre-knockout}}$ is the expression level of G_j before the knockout.

3.7.2 Estimation and Inference

For gene expression data, one can use various statistical and machine learning techniques to estimate the functions f_i. Common methods might include

- Linear regression if the relationship is assumed to be linear.
- Non-linear methods like neural networks or Gaussian processes if complex relationships are expected.

Challenges

- **High Dimensionality**: Gene expression data typically has thousands of genes but far fewer samples, leading to challenges in estimating accurate models.
- **Unobserved Confounding**: There might be latent factors, like unmeasured genes or pathways, affecting the observed expressions.

SCMs can provide a powerful framework to understand gene regulatory networks from expression data. They can help deduce causal relationships, predict outcomes of interventions, and lead to insights into the underlying biological processes. However, due care must be taken considering the complexity and challenges inherent in gene expression data.

4 Experimental Results

In this section, the outcomes of the proposed federated learning method describe the counterfactual paradigm of gene expression dataset. This approach enhances security with optimisation of the data analytics. In this section, our primary emphasis on evaluating the federated learning approaches with the optimal outcomes, employing breast cancer gene expression datasets for analysis. This section has several subsections; at first, we describe the collected data, next subsection focuses on counterfactual analysis to identify the optimal biological function.

4.1 Description of Data

In this study, microarray data related to breast cancer (specifically, breast carcinomas) were employed to assess histologic grade. Histologic grade refers to the characterisation of tumours to identify abnormal cancer tissues or cells. The classification of histologic grade is based on the growth and spread of cancer cells and is categorised into three grades: grade 1, grade 2 and grade 3. These gene expression patterns associated with breast cancer in order to enhance the accuracy of histologic grading. These microarray datasets (accession no: GSE2990) of breast carcinomas and identified genes that were differentially expressed by comparing the expression profiles between grade 3 and grade 1 tumours. Notably, among all the histologic-grade categories, grade 2 tumours exhibited a high number of genes associated with recurrence.

4.2 Optimal Gene Analysis

We identify genes that are connected to cancer patients. We will compare the average expression levels of each gene for cancer patients and non-cancer patients. Significant differences in expression levels can suggest a gene's association with the disease. We can achieve the gene connectivity among cancer patients by computing the correlation matrix of gene expressions for cancer patients. If two genes are highly correlated, it may suggest some form of interaction or common regulatory mechanisms between them.

Table 1 describes the top ten genes with the highest differences in mean expression between cancer and non-cancer patients. A larger absolute difference in mean expression suggests a stronger association of the gene with cancer.

Table 1 Top ten genes with the highest differences in mean expression between cancer and non-cancer patients

Gene name	Mean gene expression value
NAP1L1.3	1.561369
ZNF385D	1.367639
NAP1L1	1.332179
FAM181A	1.325632
EXOSC2	1.248139
WBP2	1.154432
TIFA	1.15096
PPOX	1.138391
RBX1	1.135169
DHRS13	1.084373

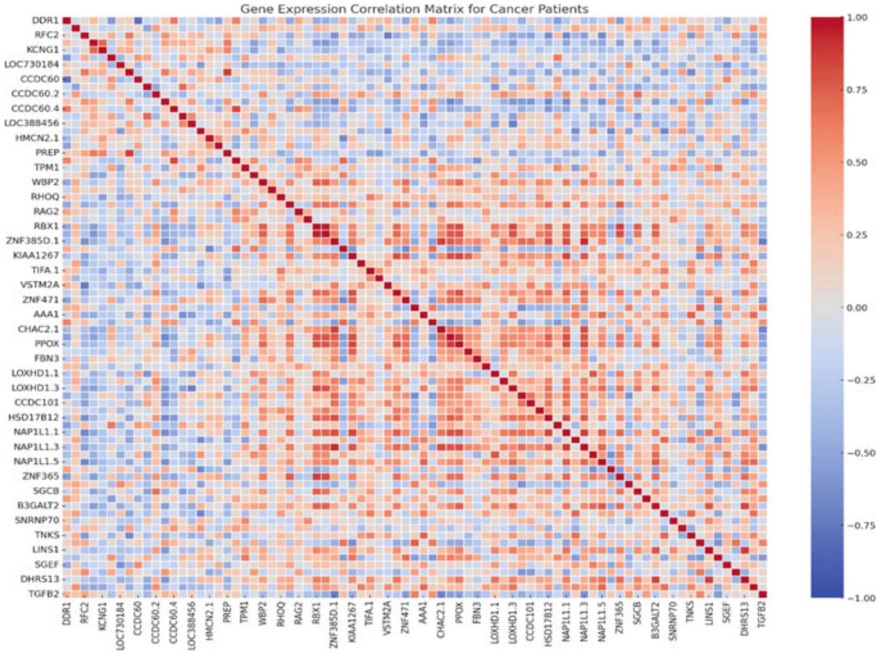

Fig. 2 Correlation between the gene expression of genes

We'll compute the correlation matrix for gene expressions among cancer patients. This will allow us to see which genes have expression patterns that are strongly correlated with each other, potentially suggesting some form of interaction or common regulatory mechanisms between them.

Figure 2 illustrates the correlation among the genes. Here, red shades represent positive correlations: when the expression of one gene increases, the expression of the other gene also tends to increase. Blue shades represent negative correlations: when the expression of one gene increases, the expression of the other gene tends to decrease.

4.3 Network Graph for the Optimal Genes

In this graph, genes will be represented as nodes, and their correlations will be represented as edges. Stronger correlations will be shown with thicker or more pronounced edges.

In Fig. 3, the nodes represent genes, and edges (lines connecting nodes) represent the strength of the association between genes, with thicker edges indicating stronger associations. The edges connecting the nodes represent a significant asso-

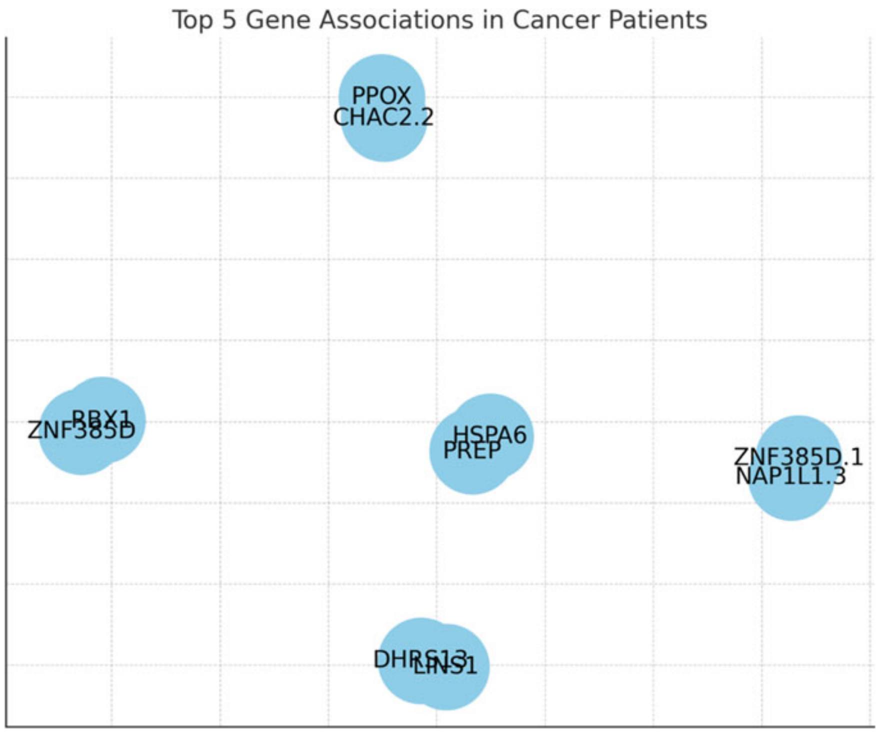

Fig. 3 Network graph for top genes

ciation between the expressions of those genes in cancer patients. The thickness of the edge is proportional to the strength of the association (correlation).

The edge labels show the correlation coefficient (ranging from −1 to 1) between the connected genes. A value closer to 1 indicates a strong positive association, while a value closer to −1 indicates a strong negative association. In this network graph, genes that are connected have expressions that are strongly associated with each other in cancer patients. When the expression of one gene changes, there's a likelihood that the expression of the other gene it's connected to will also change. This can be a basis for further investigation to understand their role in cancer progression or their potential as therapeutic targets.

4.4 Genes Analysis for Individual Patient

To find genes that are associated with an individual cancer patient, we first need to define what 'associated' means in this context. Here are a few potential interpretations and the corresponding analyses we could perform:

- **Differentially Expressed Genes**: Identify genes that are expressed at significantly different levels in a given cancer patient compared to the average expression levels of healthy (or non-cancer) patients. This would involve comparing the gene expression values for the chosen patient against the average expression values for all non-cancer patients.
- **Patient-Specific Network Analysis**: Create a correlation matrix for the gene expression values of the individual patient. This will highlight which genes have expression patterns that are related to each other specifically for that patient.
- **Comparison to the Group**: Identify genes in the individual patient that have expression values that are outliers when compared to the distribution of expression values in the entire dataset (or in all cancer patients).

Figure 4 illustrates top 50 genes with the highest absolute Z-scores for patient ID:17. The green dashed line represents the Z-score threshold we have chosen for outliers, which is 2. Genes with Z-scores beyond this threshold (either positively or negatively) can be considered as having expression levels that are notably different from the average expression levels in cancer patients.

We have ident in this plot, the genes that appear on the graph have expression values for patient ID:17 that are outliers when compared to the average expression values of all cancer patients in the dataset. The magnitude and direction of the Z-score indicate how much and in which direction (higher or lower) the gene's expression deviates from the average.

ified the top 20 genes based on variability, we construct the graph using the gene correlations. We will use the correlation values between these genes as weights for the edges in the graph. For simplicity, we will only consider positive correlations.

Figure 5 illustrates the connectivity between the top 20 genes based on their correlation values. Here, nodes represent the genes. The size of the nodes is uniform for all genes. Edges are the connections between nodes (genes) which indicate a positive correlation between their expression values. The thickness of the edge corresponds to the strength of the correlation (i.e. a thicker edge indicates a stronger correlation).

From the graph, you can observe which genes have a strong positive correlation with one another in their expression patterns. This can be interpreted as a potential 'communication pattern' between these genes, suggesting that they might be co-regulated or functionally related in some cellular processes.

5 Discussion

Advances in federated learning have led to significant advances in several fields, including genomics. In our study, we used federated learning in conjunction with counterfactual explanations to analyse the intricate patterns underlying gene expression datasets associated with breast cancer. The fusion of federated learning with the counterfactual paradigm provides a sophisticated approach that improves the confidence and optimisation of data analysis.

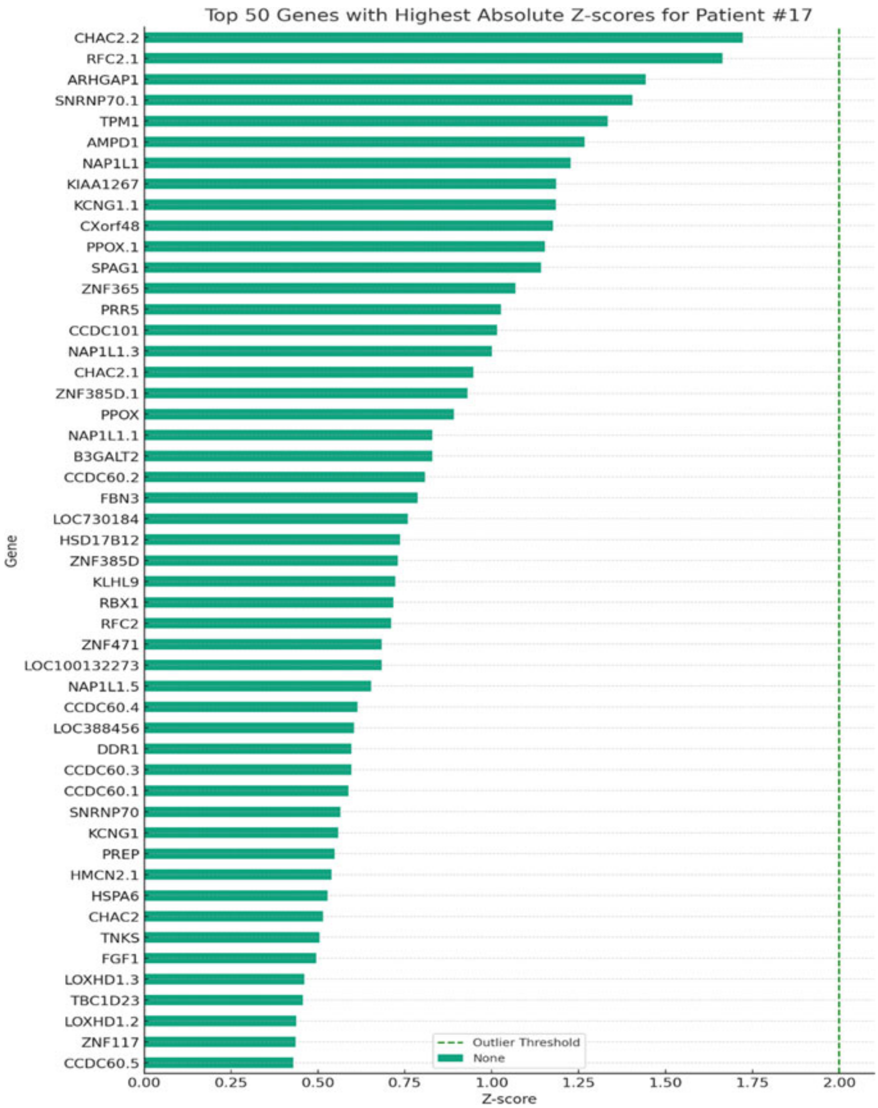

Fig. 4 Top 50 genes with the highest absolute Z-scores for patient ID: 17

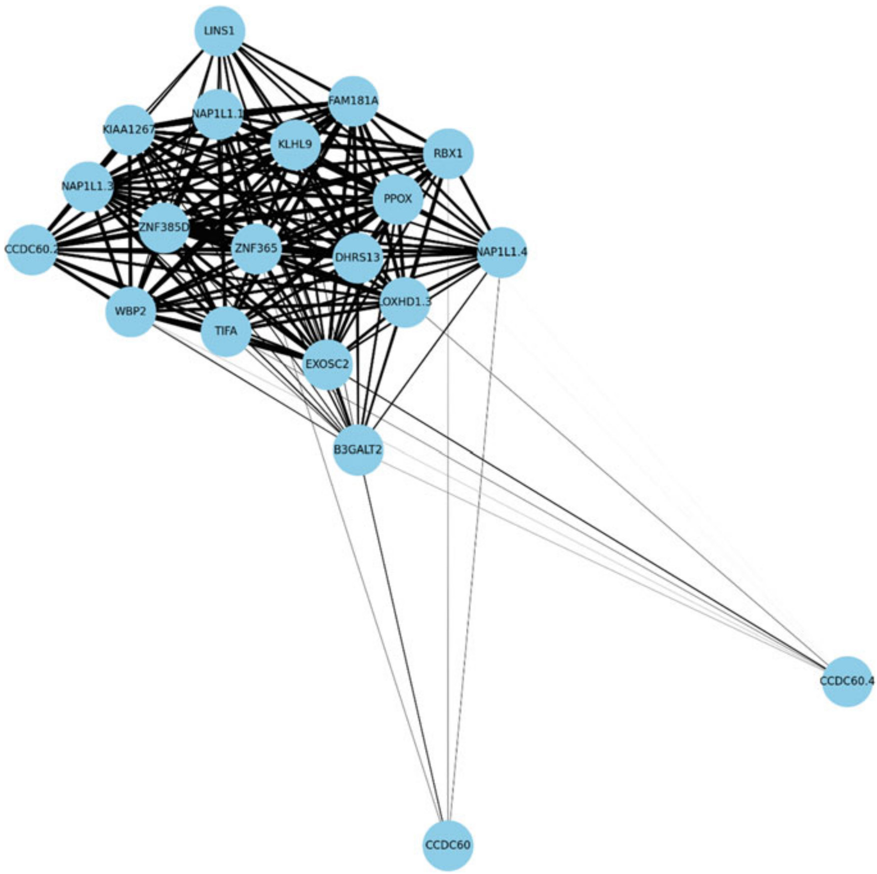

Fig. 5 Top 20 genes for patient ID: 17 based on variability

Histological grading of breast cancer, especially the differentiation of different grades, is a crucial basis for therapeutic decision-making. The importance of distinguishing genes that are differentially expressed across grades cannot be overemphasised. Our results presented in Table 1 highlight genes that show strong differences in expression between cancerous and non-cancerous samples. These genes could be potential biomarkers for breast cancer or even therapeutic targets, although further investigation is needed.

Our correlation analysis (Fig. 2) between different genes highlights the complexity of gene–gene interactions. These correlations could point to possible pathways in which these genes work together to play a role in pathogenesis or disease progression. The following network diagram (Fig. 3) further visualises these interactions and helps researchers prioritise genes that form nodes in these networks for further exploration.

Diving deeper into the individual patient data reveals variability between patients. For example, analysis of patient ID:17 revealed certain genes whose expressions were outliers compared to the average expression in cancer patients, indicating the heterogeneity of breast cancer at the molecular level. Such findings are crucial as they pave the way for personalised medicine, where treatments could be tailored based on an individual's genetic make-up.

6 Conclusion

Our exploration of federated learning combined with counterfactual explanations for gene expression datasets related to breast cancer has led to fascinating findings. Using our methods, we have identified genes that are differentially expressed at different histological breast cancer stages and elucidated possible interactions between them. The granularity of our analysis, which extends to individual patient data, highlights the potential for personalised therapeutic strategies.

While our findings provide a solid foundation, they also pave the way for numerous opportunities for future research. The identified genes and their interactions need to be experimentally validated and their precise role in breast cancer pathogenesis investigated. In addition, as federated learning and counterfactual explanations evolve, so do our methods to ensure we remain at the forefront of using AI for genomics research.

Essentially, our research highlights the immense potential of federated learning in genomics and provides a blueprint for future investigations in this field. Through robust methods and rigorous analysis, we are one step closer to unravelling the complexities of breast cancer, in the hope that our findings will one day translate into improved treatment strategies for patients.

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Chapter 3
Multi-objective Adaptive Guided Differential Evolution for Passively Controlled Structures Equipped with a Tunned Mass Damper

Salar Farahmand-Tabar and Sina Shirgir

1 Introduction

Currently, metaheuristic algorithms (MHs) play a crucial role in tackling highly intricate optimization problems [1–18]. These algorithms are categorized into two primary groups: single-Objective (SO) optimization, and multi-objective (MO) optimization where the number of objective functions is equal to 1, and ≥ 2, respectively [19, 20]. However, handling multi-objective optimization problems (MOOPs) is considerably more challenging than addressing single-objective problems. Among the available strategies to manage the complexity of multi-objective problems, the conversion into single-objective problems has traditionally been an attractive approach. However, in recent years, characterized by the era of digital transformation, there has been a substantial rise in both the quantity and complexity of optimization problems. Moreover, there are heightened expectations for optimization processes to yield more efficient solutions. Indeed, in scenarios where objective functions are conflicting, pinpointing a single global optimal solution is not feasible [21]. In such cases, the process of optimization strives to identify the optimal Pareto set of solution candidates, each excelling in at least one objective function compared

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to its competitors. This task involves exploring nondominated solutions, managing an archive of these solutions, and continuously updating the archive throughout the search process. Consequently, the significance of multi-objective optimization methods, particularly those dealing with conflicting objective functions, has grown, attracting substantial attention from researchers. Specifically, over the last five years, there has been a remarkable growth in research investigations on the advancement of MO optimization algorithms.

This chapter introduces the multi-objective adaptive guided differential evolution (MOAGDE) method to address MOOP. MOAGDE is positioned as a strong and stable contender among multi-objective optimization algorithms. Its core components include the adaptive guided differential evolution (AGDE) algorithm and an archive holding nondominated solutions characterized by crowding-distance metrics. AGDE, a robust variation of the DE, serves as the foundation for MOAGDE. Similar to other successful DE methods such as LSHADE [22], LSHADE-CNEPSIN [23], and LSHADE-SPACMA [24], AGDE is rooted in solid principles and was originally designed for single-objective optimization.

What sets algorithms such as AGDE apart from other metaheuristic methods is the adaptability and flexibility of their mutation and crossover operators to suit optimization problems with diverse characteristics and requirements. This property has allowed pioneering evolutionary algorithms such as GA and DE to excel in optimizing various problem types, from combinatorial problems to global optimization tasks with continuous variables and constraints. In addition to being a contemporary evolutionary search algorithm, AGDE boasts a powerful new crossover operator. Consequently, it was chosen as the basis for developing the MOEA in this study. The dynamic crossover rate enhances AGDE's convergence and diversity capabilities. To extend the effective search capabilities of single-objective AGDE to the multi-objective context, modifications were introduced to search operators and the selection process to align with the principles of Pareto-based archiving. The current research conducted a thorough analysis encompassing both AGDE's search process life cycles and the procedures related to Pareto-based archiving. This analysis involved the careful choice of reference positions to effectively guide the search process. Moreover, the individuals directing the process of search and participating in the operator of crossover were exclusively chosen from the pool of nondominated solutions. The crossover operator design was customized specifically for this scenario, with the crowding distance method serving as the fundamental criterion for making selections among the nondominated solutions during the archive management process. These enhancements result in MOAGDE demonstrating robust convergence and diversity capabilities across various multi-objective problem types.

One significant contribution of this study is the application of the MOAGDE algorithm to address a real-world structural control problem through multi-objective optimization. Specifically, the study focuses on optimizing the mechanical parameters of tuned mass dampers (TMD) as an engineering design problem. To assess the multi-objective optimization approach, the optimal design of TMD was explored in both SO and MO modes. The seismic responses and structural performance of TMD-equipped structures designed with various objective functions were analyzed and compared. These objective functions aimed to minimize drift and acceleration responses. The study employed a 10-story shear frame as a structural model under white noise random seismic excitation. Design variables included the mass, damping, and stiffness of the TMD. Numerical examples were conducted in two sets: single-objective and multi-objective problems. Performance indices (J indices) were employed to assess the seismic performance of TMD-equipped structures.

The subsequent sections build upon this foundation, commencing with an extensive review of existing literature on TMD-based optimal control methods in Sect. 2. Following this, multi-objective adaptive guided differential evolution introduces the framework of the AGDE method, subdivided into its fundamental principles in Sect. 3.1 and its application to multi-objective optimization in Sect. 3.2. Section 4 outlines for optimum design of TMDs and describes the structured approach employed in this study for the design of TMD systems. Moving forward, numerical study in Sect. 5 presents the implementation and methodology, while the analysis results and interpretation of obtained data are discussed in Sect. 6. Finally, conclusions as a last section synthesize key findings and outline implications.

2 Related Works on TMD-Based Optimal Control

Tall buildings, due to their flexibility and low damper ratio, face heightened vulnerability to wind and seismic forces. Under normal circumstances, the structure's response remains within safe limits through proper design. However, when subjected to earthquakes or strong winds, there is no assurance that the structural response will remain in the safe range. To address this issue, effective control devices are essential, and one of the most dependable and conventional options is the TMD [25]. A TMD system comprises a configuration involving mass, and springs that are connected to the primary structure. This system effectively reduces vibration energy by utilizing a combination of the inertial forces generated from mass displacement and the damping properties introduced by the device.

In seismic structural design, the conventional assumption is that structures are rigidly attached or firmly connected to their foundations. However, this assumption holds true only when structures are situated on solid rock or extremely rigid soil. Structures built on deformable soil can exhibit significantly different seismic responses compared to the fixed base assumption. The seismic response of such structures is affected by local site conditions and the dynamic interaction between the soil and structure, referred to as soil-structure interaction (SSI) [26–28]. Numerous investigations have highlighted the impact of elastic dynamic SSI on structural responses [29–32].

Xu and Kwak [33] conducted a study on the wind-induced motion of slender or tall TMD-equipped structures, taking into account SSI. They utilized a transfer matrix formulation to analyze the interaction between the structure, soil, and mass damper in the frequency domain. Their research highlighted the significant impact of soil compliance on structural responses and the effectiveness of TMDs. In a similar vein, Wu et al. [34] investigated how SSI influences the seismic performance of TMDs installed on structures with flexible foundations. They employed a frequency-independent generic model to represent different soil and structural characteristics within a soil-structure system. Additionally, Liu et al. [35] developed a mathematical model to predict wind-induced oscillations of high-rise buildings equipped with TMDs while accounting for the effects of SSI. Their study revealed that this model accurately simulates soil effects, surpassing models assuming a fixed base.

In designing TMDs, three critical parameters—mass, damping, and stiffness—are taken into account. Den Hartog [36] was among the first to consider optimizing these parameters when dealing with a single-degree-of-freedom (SDOF) undamped system under harmonic loading. Subsequent research extended these investigations to damped primary systems [37]. Wang et al. [38] presented a two-stage optimization method for passive TMDs, aiming to decrease the dynamic responses of the structure while adhering to PTMD limitations. Marano et al. [39] introduced the concept of optimizing the mass ratio of the TMD, a parameter that had been preselected in prior investigations. Arfiadi and Hadi [40] employed GA to optimize both the properties and location of the TMD. Bekdas and Nigdeli [41] employed the harmony search (HS) algorithm for optimizing TMD parameters subjected to harmonic loading, resulting in significantly reduced structural responses, however, these findings were further elaborated by other researchers [42, 43]. The HS was applied by Nigdeli and Bekdas [44] to optimize TMD parameters for reducing shear force and preventing brittle fracture. Farshidianfar and Soheili [45, 46] investigated optimal TMD parameters, considering SSI effects.

Rahai et al. [47] recently questioned the accuracy of the previously proposed SSI model [45, 46]. An optimum design approach was introduced by Xiang and Nishitani [48] for a nonconventional TMD directly connected to the ground through a dashpot. This design aims to mitigate structural resonant behavior. Salvi and Rizzi [49] presented the optimum adjustment of free parameters for passive TMDs installed on frame structures under specific seismic excitations. Mohebbi et al. [50] delved into the optimization of the design and assessment of multiple TMDs to mitigate damage in nonlinear steel structures during earthquakes. Kaveh et al. [51] studied the optimal parameters of TMDs subjected to seismic excitation using the charged system search method. They also explored the effectiveness of an optimized fuzzy controller for decreasing responses in a building equipped with a semiactive TMD under seismic load cases [52].

This chapter focuses on exploring the optimum parameters of TMDs under earthquake conditions, taking SSI effects into account. The optimization variables include mass, damping, and stiffness of the TMD, while the objective functions comprise the simultaneous minimization of the maximum structural acceleration and displacement. To accomplish this, the MOAGDE algorithm is utilized. For the SSI system, the Lagrangian method [53] is used to derive the equations of motion. These equations are solved using a time-domain analysis according to the Newmark method. The study investigates the optimum TMD parameters for a 10-story shear frame subjected to the El-Centro earthquake. The results highlight the significant influence of SSI effects on the optimal TMD parameters, emphasizing the importance of their inclusion in TMD system design.

3 Multi-objective Adaptive Guided Differential Evolution

This section details the steps for converting metaheuristic search (MHS) algorithms focused on single objective into multi-objective optimization algorithms according to the Pareto principle. The key difference between population-based multi and single-objective optimization methods lies in population updating. Population management in single-objective optimization relies on the fitness values of solution candidates. In contrast, multi-objective optimization considers the dominance relationships among solution candidates. Consequently, multi-objective optimization algorithms encompass the establishment and handling of an archive where solutions that outperform others within the population are preserved. This archive for multi-objective optimization comprises solutions that are not dominated by others. To transition MHs from single- to multi-objective optimization algorithms, the population management process needs restructuring. This restructuring involves integrating the creation and management of an archive for nondominated solutions into the MH process.

The management of the archiving process is based on the crowding-distance sorting approach and the Pareto-based dominance. By employing the framework depicted in Fig. 1, MHs can be adapted from single- to multi-objective optimization. In this manner, MHs originally designed as single-objective algorithms become capable of tackling MOOPs characterized by conflicting objective functions.

The framework presented in Fig. 1 primarily results from the integration and hybridization of two procedures: the Pareto-based archiving, which is responsible for managing nondominated solutions within the multi-objective search space, and MH search procedures, which simulate the nature-inspired approaches. Considering both processes enables the design of multi-objective MHs. To differentiate between the components of the mentioned procedures, Fig. 1 illustrates the archiving steps based on Pareto principles. The elements of the MH algorithm are represented in blue blocks. The set of solution candidates is defined by the P-vector, while the collection of nondominated solutions is labeled the A-vector. The related values of the objective function are indicated by the O_A-vector, and O_p-vector, respectively.

The process of Pareto-based optimization is initiated by creating the A, O_A, P, O_P vectors. Afterward, the Pareto-based archiving process initiates in parallel with the lifecycle of the search process within the algorithm. This phase involves assessing the effectiveness of newly found positions within the search space. Among these positions, nondominated solution candidates are determined based on the following criteria:

Pareto Dominance: Considering the D-dimensional solution candidates in the population ($\overrightarrow{x} = (x_1, x_2, ..., x_D)$, and $\overrightarrow{y} = (y_1, y_2, ..., y_D)$), and with m objective functions in optimization process $\{o_1(\overrightarrow{x}), o_2(\overrightarrow{x}), \ldots, o_m(\overrightarrow{x})\}$, for $\overrightarrow{x}$ to dominate

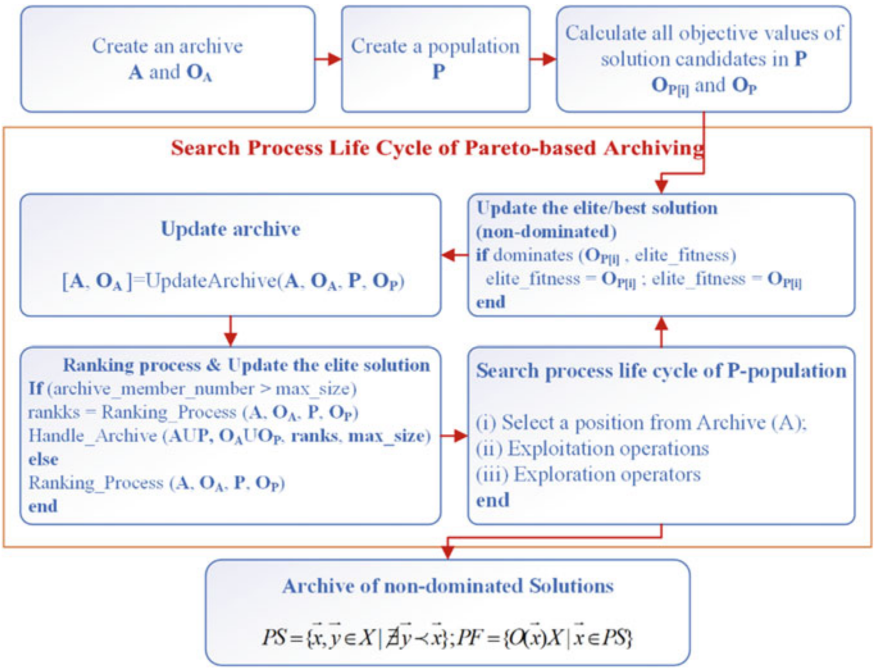

Fig. 1 Multi-objective search process using Pareto-based strategy

$\overrightarrow{y}$ (expressed as $\overrightarrow{x} \prec \overrightarrow{y}$), it must hold true that $\forall i \in \{1, 2, t, m\}$, $o_i\left(\overrightarrow{x}\right) \le o_i\left(\overrightarrow{y}\right)$, while also having at least one $\forall j \in \{1, 2, t, m\}$ where $o_j\left(\overrightarrow{x}\right) \le o_j\left(\overrightarrow{y}\right)$.

Pareto Optimality: If the solution ($\overrightarrow{x}$) is not dominated by any other solution, it is termed Pareto-optimal: $PS = \left\{\nexists \overrightarrow{y} \in X \middle| \overrightarrow{y} \prec \overrightarrow{x}\right\}$.

Consequently, the A-archive undergoes updates during this second step. The process continues until specific termination conditions are met, often relying on a maximum fitness evaluation as a key stopping point. When these termination criteria are satisfied, two sets are obtained: the Pareto set (PS) set, which includes solution candidates that are not dominated by others, and the Pareto front (PF) set, denoting the objective function values for these solution candidates.

$$PS = \left\{\overrightarrow{x}\, and\, \overrightarrow{y} \in X \middle| \nexists \overrightarrow{y} \prec \overrightarrow{x}\right\};\ PF = \left\{O(\overrightarrow{x}) \middle| \overrightarrow{x} \in PS\right\} \tag{1}$$

Originally, the AGDE was proposed to tackle single-objective optimization problems. To create MOAGDE, which is introduced in this chapter and capable of addressing MOOPs, certain adaptations to the AGDE were necessary. The process of transforming AGDE into MOAGDE was influenced by the framework shown in Fig. 5. This section provides a detailed explanation of how MOAGDE was enhanced to handle MOOPs.

3.1 Fundamentals of the AGDE

The AGDE [54] builds upon the foundation of the DE [55]. Its primary objective is to improve the balance between exploration and exploitation in the optimization procedure. Algorithm 1 provides an overview of the general steps of the AGDE algorithm for single-objective problems.

Algorithm 1. General steps in DE and AGDE [54, 55]

1.**Begin**
2. P: randomly create a population
3. **for** $i = 1 : n$ (n is the number of solution candidates)
4. evaluate the fitness values of the candidates in P
5. **end**
5. **while** (search process life cycle)
6. selection process: r_1, r_2, r_3,
7. search operators:
8. Crossover
9. Mutation
10. Updating process:
11. Compare the solution candidates based on their fitness values
12. Next generation until stopping criterion
13. **end**

Algorithm 1 showcases a process in which the steps leading to the search process life cycle share striking similarities with procedures observed in other MHs. Both DE and AGDE procedures [55], which are evolutionary-based methods similar to the GA [56], encompass three primary phases in their search-process life cycle: the selection process, crossover, and mutation steps. Nevertheless, there are two noteworthy distinctions between AGDE and DE.

The first distinction lies in the process of selection, Line 7 in Algorithm 1. In this phase, solution candidates are chosen from the P-population to direct the search process. In the DE, the three reference vectors (r_1, r_2, r_3) governing the search are randomly selected from P, ensuring that they are distinct. In contrast, the AGDE algorithm randomly picks vector r_1 from P, while vectors r_2 and r_3 are determined using a greedy strategy. In this approach, solution candidates within P are ranked according to their fitness values, and the best candidate, denoted as P_{best}, is chosen and assigned as r_2. Likewise, the solution candidate with the lowest fitness value among those in P, referred to as P_{worst}, is chosen and designated as r_3.

The second distinction between the AGDE and DE algorithms lies in the crossover operator (Algorithm 1 - Line 8). The parameter CR regulates the impact of the crossover operator during the search process. Altering the CR value facilitates a shift between exploration and exploitation: higher CR values boost exploration while diminishing exploitation capabilities. Conversely, reducing the CR value allows for a more delicate neighborhood search but hampers diversity. Hence, the precise adjustment of the CR parameter is crucial. The CR parameter in the AGDE is adapted dynamically during the search process [54].

3.2 AGDE for Multi-objective Optimization

In conventional single-objective MHs such as AGDE, selecting candidate solutions that guide the search procedure relies on their fitness values (Algorithm 1 - Line 7). These fitness values are used for comparing and evaluating the performance of candidate solutions within the population. In single-objective MHS algorithms, such as AGDE, or during the selection process [57], fitness values are directly employed for making updates to the population.

However, in situations involving more than one objective function, particularly when these objectives are in conflict, it becomes impractical to compare solution candidates solely according to their fitness values. Evaluating the approximate quality of the obtained Pareto set presents significant challenges [57]. In such cases, it becomes necessary to consider the dominance relationships between solution candidates.

Given these explanations, the design process of MOAGDE [58] is outlined as follows:

- The archive (A) is generated: An archive is established to maintain a sorted collection of nondominated solutions, with its determined size. The Pareto dominance relationship serves as a reference. If the number of nondominated solutions exceeds the archive's maximum limit, a sorting method based on crowding distance is employed [59].
- The process of selection is updated: Solution candidates r_1, r_2, and r_3, which guide the process of the search in AGDE, are now chosen from the archive (A) rather than the population.
- Information about dominance is prioritized over fitness value: To assess the performance of solution candidates' previous and new positions within the MO search space, the Pareto-optimality information is utilized.
- PS and PF sets are established: Upon completing the process of exploration, a collection of nondominated solutions (PS) and their related PF values are generated instead of obtaining just a single best solution.

Based on the explanations provided, Algorithm 2 presents the pseudocode of MOAGDE.

After a thorough analysis of Algorithm 2, it becomes evident that the multi-objective search process using Pareto-based strategy was applied in the development of the MOAGDE. There are two sections within the algorithm which correspond to the AGDE steps, and the steps related to nondominated sorting utilizing the Pareto archiving approach based on the crowding distance. The process of selection, mutation, and crossover are integrated, constituting the three phases of the AGDE search process (lines 16–20). These operators in the MOAGDE have been restructured, and their design variations are detailed below:

In the AGDE, the operator of mutation functions based on Eq. (2).

$$X_i^{G+1} = X_i^G + F.(X_{P_{best}}^G - X_{P_{worst}}^G) \tag{2}$$

In Eq. (2), G represents the iterations, and F is a parameter that adjusts the impact of the difference between X_{p-best} and $X_{p-worst}$. In the AGDE, the variable F randomly selects values within the range of [0.1, 1]. Additionally, within the AGDE, all four solution candidates, namely X_i, X_r, X_{p-best}, and $X_{p-worst}$, are chosen from the solution candidates within the P-population. Specifically, X_i is selected in order, X_r randomly, and X_{p-best} represents the best solution candidate, while $X_{p-worst}$ signifies the worst solution candidate.

The mutation operator in MOAGDE is designed as provided in Eq. (3).

$$X_i^{G+1} = X_i^G + F \cdot (X_{P_{best}}^G - X_{P_{worst}}^G) \tag{3}$$

Equation (3) features X_i, representing a solution candidate chosen from the P-population, in line with the AGDE approach. However, X_{r1} denotes a solution candidate chosen from the A-archive using the greedy method considering the crowding distance, while X_{r2} and X_{r3} signify solution candidates drawn from the A-archive utilizing the method of probabilistic roulette selection, also based on crowding distance.

In Eq. (3), X_i represents a solution candidate chosen from the P-population, consistent with the AGDE. However, X_{r1} signifies a solution candidate chosen from the A-archive using the greedy selection approach relying on crowding distance, while X_{r2} and X_{r3} denote solution candidates drawn from the A-archive utilizing the probabilistic roulette selection approach which is also based on crowding distance.

Algorithm 2. The pseudocode of the MOAGDE [58]

1. **Initialization**
2. A, O_A: generate an archive with defaults
3. P:generate random population

---

4. **while**(search-process life cycle)
5. **for** $i = 1 : n$ (n is the number of solution candidates in P)
6. Compute the values of objectives for solution candidates

$O_{P[i]} =$ F_Objective ($P_{[i]}$)

7. Update the best nondominated solution in P by Pareto dominance

If dominates ($O_{P[i]}, elite\_fitness$) then
$elite\_fitnes = O_{P[i]}$ and $elite\_position = P_{[i]}$
end if

8. **end for**

---

9. ***Updating archive*** $[A, O_A]= f_{ArchiveUpdate}(A, O_A, P, O_P)$
10. $[M, O_M]= Mergethearchive$ (A, O_A) and the population (P, O_P)
11. **for** $i = 1 : n$ (n is total quantity of solution candidates in A and P)
12. **for** $j = 1 : n - 1$
13. Generate PS and PF: Determine nondominated solutions in $[M, O_M]$ by Pareto dominance and

end for
end for

14. **If** (Archive maximum number of members exceeded)
15. Ranking/sorting process: compute the crowd for each solution candidate in $[M, O_M]$ and determine the number of their nearest neighbors

Archive handling: Utilize the crowd info and elitist/roulette selection technique to handle archive in in [A, O_A]
end if

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16. ***Search process of Metaheuristic***
17. Selection: Choosing r_1, r_2, r_3 vectors from $\boldsymbol{A}$ to guide the search process
18. Search operators:
19. Crossover
20. Mutation
21. Next iteration until stopping criterion
22. **end while Return** PS, PF

To summarize, the MOAGDE and AGDE algorithms employ different strategies for selecting solution candidates. AGDE employs a mixed method of random and greedy selection to choose candidates from the P-population. Conversely, the MOAGDE selects solution candidates that guide the search process from the A-archive, relying on their crowding-distance rank for this selection. The archive includes nondominated solutions organized and ranked based on Pareto-based and crowding-distance criteria.

4 Outline for Optimum Design of TMDs

The equation describing the motion of a structural model with N degrees of freedom [60–63], featuring a shear frame system with a TMD placed at the top floor, is expressed as follows:

$$M\ddot{X} + C\dot{X} + KX = P(t) \tag{4}$$

where K, M, and C, are the stiffness, mass, and damping matrices, respectively, with the following formulations:

$$K = \begin{bmatrix} k_1 + k_2 & -k_2 & & & \\ -k_2 & k_2 + k_3 & -k_3 & & \\ & -k_3 & . & . & \\ & & . & . \; k_N + k_d & k_d \\ & & & k_d & k_d \end{bmatrix} \tag{5}$$

$$M = \begin{bmatrix} m_1 & & & & \\ & m_2 & & & \\ & & \ddots & & \\ & & & m_N & \\ & & & & m_d \end{bmatrix} \tag{6}$$

$$C = \begin{bmatrix} c_1 + c_2 & -c_2 & & & \\ -c_2 & c_2 + c_3 & -c_3 & & \\ & -c_3 & . & . & \\ & & . & . \; c_N + c_d & c_d \\ & & & c_d & c_d \end{bmatrix} \tag{7}$$

$$X(t) = \left[x_1 \; x_2 \ldots x_N \; x_d \right] \tag{8}$$

Here, in the equation of motion:

- m_i, c_i, k_i and represent the mass, damping, and stiffness of the i th floor (where $i = 1, 2, \ldots, N$), respectively.
- m_d, c_d and k_d are the mass, damping, and stiffness of the tuned mass damper, respectively.
- x_i represents the lateral displacement of the i th floor.

Through the state-space equation, the solution to the equation of motion is as follows:

$$\begin{cases} \dot{Z}(t) = AZ(t) + BP(t) \\ \\ Y(t) = RZ(t) + QP(t) \end{cases} \tag{9}$$

In this context, $Z(t)$ represents the state vector with the system states of A and B. Additionally, the system matrices are represented as R and Q, and their definitions are contingent upon the nature of the anticipated output, Y.

$$Z(t) = \begin{bmatrix} X(t) \\ \dot{X}(t) \end{bmatrix} \tag{10}$$

$$A = \begin{bmatrix} 0 & I \\ -M^{-1} \cdot K & -M^{-1} \cdot C \end{bmatrix} \tag{11}$$

$$B = \begin{bmatrix} 0 \\ M^{-1} \end{bmatrix} \tag{12}$$

It is important to highlight that if the external loading corresponds to the base acceleration, often associated with earthquake loading, the vector $P(t)$ can be expressed as:

$$P(t) = M\{1\}_{N+1}\ddot{x}(t)_g \tag{13}$$

The behavior of a TMD-equipped structure is significantly affected by the mechanical properties of TMD components, including mass, stiffness, and damping (as depicted in Fig. 2). Consequently, the design of a TMD can be conceptualized within the framework of an optimization problem. In this context, metaheuristic methods are commonly employed for the optimal design of TMD systems. These approaches are population-based methods that efficiently handle optimization problems by systematically navigating and exploring the search space through a sampling process.

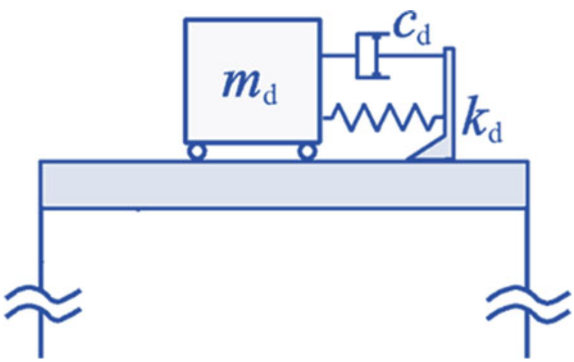

Fig. 2 Configuration of TMD

In these methods, the parameters that are not known a priori are chosen as design variables, and the primary purpose is to minimize an objective function. Additionally, constraints are imposed on the design variables to ensure that they fall within an acceptable range. The formulation of the TMD design optimization problem can be defined as follows:

Find: design Variables $= \{m_d.k_d and c_d\}$

to minimize:

$$\text{Objective Functions}\begin{cases} OF1 = \sum_{i=1}^{n}\left(\frac{\text{Max. Controlled Drift}_i}{\text{Max. UncontrolledDrift}_i}\right) \\ OF2 = \sum_{i=1}^{n}\left(\frac{\text{Max. Controlled Acceleration}_i}{\text{Max. Uncontrolled Acceleration}_i}\right) \end{cases}$$

$$\text{Subject to}: \begin{cases} m_{dmin} \le m_d \le m_{dmax} \\ k_{dmin} \le k_d \le k_{dmax} \\ c_{dmin} \le c_d \le c_{dmax} \end{cases}$$

The objective function in this context is established by considering the maximum ratio of drift in the controlled structure to that of uncontrolled structure. The MOAGDE algorithm, as presented, is used to address this optimization problem. Additionally, once the TMD has been optimally designed using this method, a comparative analysis is performed to investigate the seismic performance of TMD-equipped structures designed using various methodologies.

5 Numerical Study

From comparative investigations on the optimal design of TMDs, this study selected a benchmark example. It involves the shear frame of a ten-story building, which has been chosen to investigate the efficiency of the proposed optimum design strategies utilizing multi-objective metaheuristic methods. This structural model is characterized by various mechanical properties for each story, including stiffness, mass, and damping. The structural behavior of this model is assumed to be linear, with an elastic stiffness of $K = 650{,}000$ kN/m. Each story has a mass of 360 tons, resulting in an initial natural frequency of $f_{initial} = 1.0108$ Hz, based on the elastic stiffness and mass. Additionally, it is assumed that the first vibration mode of the structural model has a damping ratio of 5%, with a linear viscous damping coefficient of $c = 6200$ kN·s/m. Table 1 presents the mechanical properties of each story in the structural model.

The optimal TMD design problem is conducted in two cases. In the first case, the design of the TMD is considered as a single-objective optimization with the defined objective functions separately. In the second case, the problem is considered as multi-objective optimization with both objective functions. In both cases, the lower and upper bounds for the mass of TMD are selected equal to 50 and 150 tons. For stiffness, the lower and upper bounds are selected as 0 and 10,000 kN/m, respectively.

Table 1 Mechanical properties of the ten story building structures

Story	Mass ($\times 10^3$kg)	Stiffness ($\times 10^3$kN/m)	Damping ($\times 10^3$kN.s/m)
1	360	650	6.2
2	360	650	6.2
3	360	650	6.2
4	360	650	6.2
5	360	650	6.2
6	360	650	6.2
7	360	650	6.2
8	360	650	6.2
9	360	650	6.2
10	360	650	6.2

Additionally, for damping, the lower and upper bounds values are defined as 0 and 3000 kN.s/m, respectively.

The structural model was subjected to analysis in both uncontrolled and controlled scenarios utilizing a nonlinear dynamic method. The ground acceleration input applied to the model was represented by white noise with a peak ground acceleration (PGA) of 0.4 g, denoted as $W(t)$ and illustrated in Fig. 3. This particular record was chosen as an illustrative example to showcase the methodology. To design a control system, the design record for the particular site can be used in the analysis and optimization. This may involve adjusting the acceleration scale factor or using multiple ground acceleration records relevant to the site under consideration.

The assessment of control performance utilizes various performance indices, comparing the controlled and uncontrolled structural responses. Various criteria are considered, with the first half representing the maximum values and the remaining

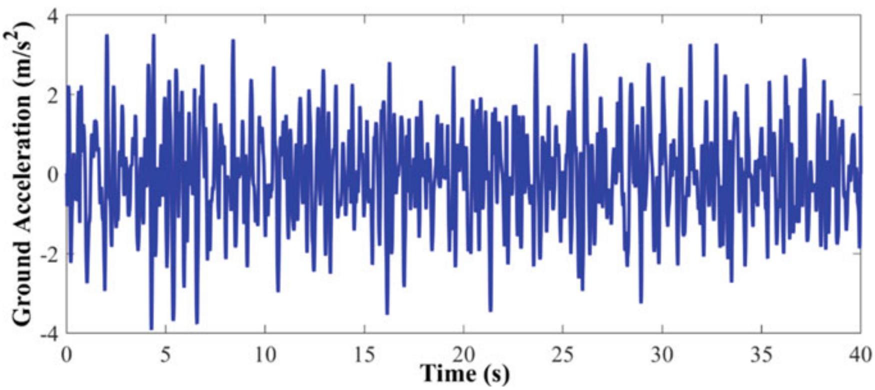

Fig. 3 Time-history ground motion acceleration (white noise with PGA = 0.4 g)

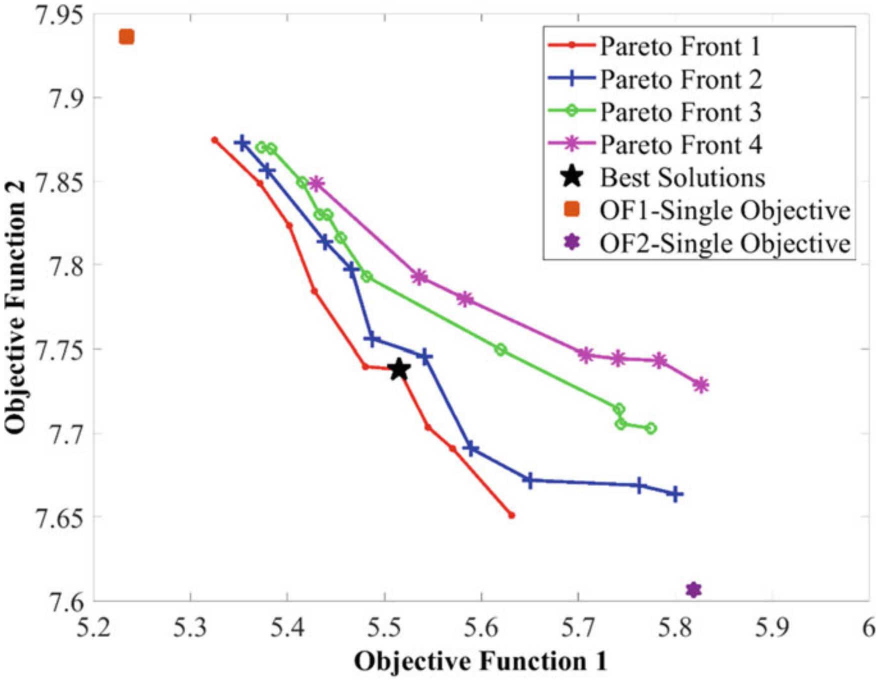

Fig. 4 Pareto solutions of the structure with optimum design of TMD

reflecting the structural norm response. These criteria are succinctly presented in Table 2 and encompass the following:

- δ^{max}: Max. interstory drift (uncontrolled).
- $\ddot{x}_u^{max}$: Absolute acceleration value of the roof.
- F_b^{max}: Max. base-shear force.
- $d_i(t)$: Interstory drift of the ith level (Controlled).
- $\ddot{x}_{ai}(t)$: Absolute acceleration of the ith level.
- m_i: Seismic mass of the ith level.

In addition, the application of the norm operator $\|.\|$ is used to determine the norm values of the responses.

6 Results and Discussion

Table 3 presents the results achieved from the optimum design of the mass damper in both single- and multi-objective modes. The design optimization is based on objective functions related to acceleration and relative displacement responses. The results reveal that, in the single-objective optimization mode, the desired objective function

Table 2 Performance criteria for the controlled building

• Peak interstory drift: $J_1 = max\left\{\frac{\underset{t.i}{max}(d_i(t))}{\delta^{max}}\right\}$	• Peak level acceleration: $J_2 = max\left\{\frac{\underset{t.i}{max}\{\ddot{x}_{ai}(t)\}}{\ddot{x}_u^{max}}\right\}$
• Peak base shear force: $J_3 = max\left\{\frac{\underset{t}{max}\left\Vert\sum_i m_i\ddot{x}_{ai}(t)\right\Vert}{F_b^{max}}\right\}$	• Normed interstory drift: $J_4 = max\left\{\frac{\underset{t.i}{max}(\Vert d_i(t)\Vert)}{\Vert\delta^{max}\Vert}\right\}$
• Normed level acceleration: $J_5 = max\left\{\frac{\underset{t.i}{max}\Vert\ddot{x}_{ai}(t)\Vert}{\Vert\ddot{x}_u^{max}\Vert}\right\}$	• Normed base shear force: $J_6 = max\left\{\frac{\underset{t}{max}\left\Vert\sum_i m_i\ddot{x}_{ai}(t)\right\Vert}{\Vert F_b^{max}\Vert}\right\}$

experiences a reduction. Conversely, in the multi-objective optimization mode, there is a reduction in both objective functions, falling in between. Furthermore, in the optimal design mode within the multi-objective framework, the designed values for the mass damper parameters prove to be more suitable. Analyzing the performance indicators for controlling the seismic response of the TMD-equipped structure under white noise excitation, it becomes evident that designing the damper in the multi-objective mode exhibits a remarkable ability to reduce seismic responses, especially the base shear response.

Through multi-objective optimization, a Pareto front that encapsulates a range of optimal solutions is successfully identified, each offering a unique trade-off between our considered design objectives (Fig. 1). Among this diverse set of solutions, we have pinpointed the one that stands out as the most favorable, marked here on the Pareto front. This particular solution represents an exceptional balance between the reduction of seismic responses and the optimization of TMD parameters. Furthermore, to provide a wider perspective on the design space, we also obtained two additional solutions from single-objective optimization, one on each side of the Pareto front. These solutions, while not considered Pareto-optimal, serve as valuable reference points, showcasing the extremes of performance with respect to each individual objective. Such a comprehensive analysis empowers engineers and decision-makers with a holistic view of the available design options, facilitating informed choices that align with project-specific priorities and constraints.

Table 3 Results of optimization

Parameter	Dim	Single objective		Multi-objective
		OF1	OF2	OF1 & OF2
OF1	–	5.2338	5.8190*	5.5149
OF2	–	7.9361*	7.6065	7.7379
m_d	10^3 kg	149.99	146.17	143.379
k_d	10^3kN/m$	5.5051	8.0478	6.2807
c_d	10^3kN · s/m	0.5005	0.9847	0.6247
J_1	–	**0.47913**	0.56097	0.51937
J_2	–	0.54351	0.53048	**0.52542**
J_3	–	0.6718	0.67732	**0.64896**
J_4	–	**0.6149**	0.69346	0.6417
J_5	–	**0.67219**	0.72432	0.68961
J_6	–	0.67172	0.72281	**0.66767**

*The result that is achieved using the solution of the other objective

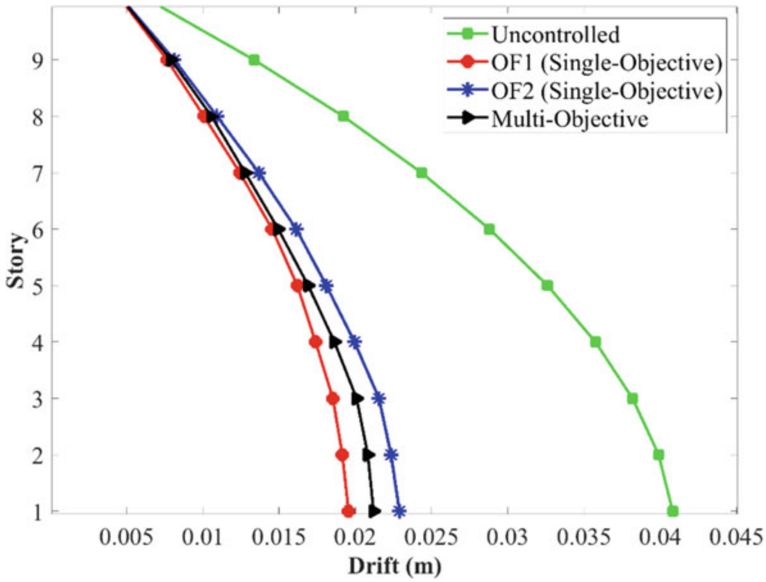

Fig. 5 Comparison of the maximum drift response of the structure under white noise excitation in different modes of mass damper design

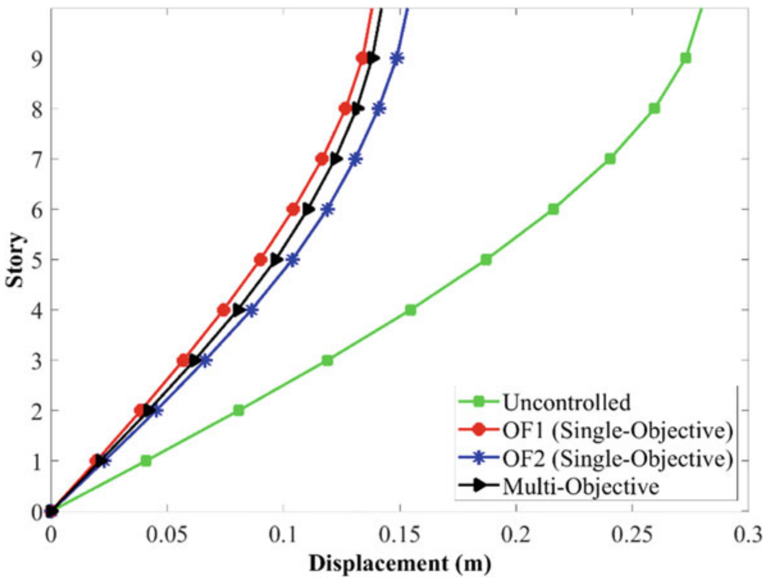

Fig. 6 Comparison of the maximum lateral displacement response of the structure under white noise excitation in different modes of mass damper design

Figures 5, 6 and 7 illustrate a comparison between the maximum seismic responses of an uncontrolled structure and a TMD-equipped structure. These structures are designed using single-objective modes based on displacement and acceleration objective functions, as well as a multi-objective mode. In Figs. 1 and 2, the maximum response of the lateral displacement and drift of the structural model is presented in both uncontrolled and controlled states, with mass dampers designed using different approaches.

As shown, the performance of the damper designed using the multi-objective mode falls between those designed with a single-objective approach. To further highlight this performance, Fig. 3 provides a detailed comparison of the maximum acceleration response of the structure. This comparison confirms the efficiency of the multi-objective approach in simultaneously reducing both acceleration and drift responses.

7 Conclusions

In this research, the application of the algorithm named Adaptive Guided Differential Evolution (MOAGDE) was evaluated for solving an optimal engineering design problem in a multi-objective mode. Specifically, the optimum design of TMD mechanical parameters was chosen as an engineering design problem. To investigate the performance of the multi-objective optimization method, the optimal design of

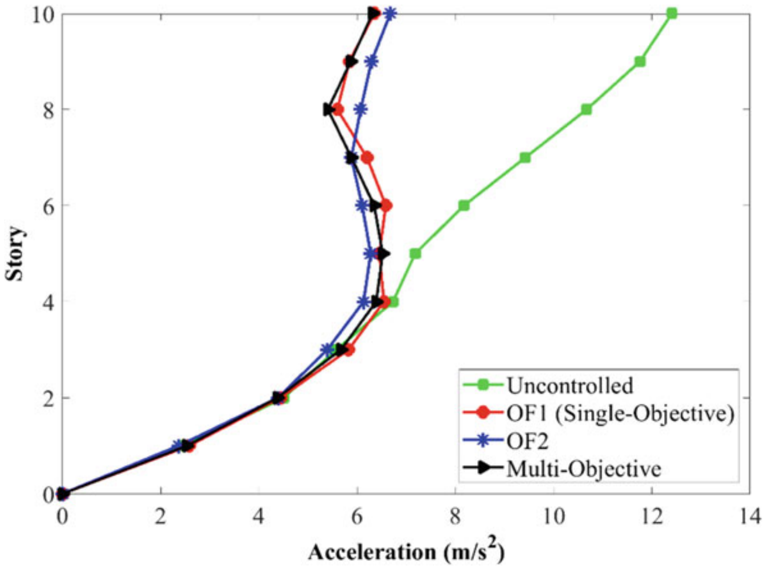

Fig. 7 Comparison of the maximum acceleration response of the structure under white noise excitation in different modes of mass damper design

the TMD was conducted in both single- and multi-objective modes. The seismic responses of the TMD-equipped structure—designed with different objective functions—were compared. The objective functions were defined as the sum of the ratios of the controlled response to the uncontrolled response across all stories. The optimal damper design problem aimed to reduce drift and acceleration responses.

For numerical study, a structural model of a 10-story shear frame under white noise random seismic excitation was utilized. The mechanical parameters of TMD, including stiffness, damping, and mass, were selected as design variables. Numerical examples were solved in two groups: single- and multi-objective problems. Performance indices (J indices) were utilized to investigate the seismic performance of TMD-equipped structures.

The results indicate that in the single-objective optimization mode, each objective function experiences greater reduction compared to the multi-objective optimization mode. However, in the multi-objective optimization mode, both objective functions exhibit intermediate reductions, resulting in a favorable reduction in seismic responses. Additionally, in the multi-objective optimization mode, the designed values for the parameters of the mass damper were found to be more suitable.

Overall, the results demonstrate that TMDs designed using a multi-objective approach and AGDE algorithm exhibit appropriate performance in reducing seismic responses, including drift, lateral displacement, acceleration, and base shear.

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Chapter 4
Evolutionary Approaches for Multi-objective Optimization and Pareto-Optimal Solution Selection in Data Analytics

Vijay Harkare, Ramchandra Mangrulkar, Onkar Thorat, and Sachin R. Jain

1 Introduction

Currently, within a dynamic and data-driven global context, the profound influence of data analytics stands as an indisputable fact. Data analytics, defined as the systematic process encompassing data examination, purification, transformation, and subsequent interpretation, is designed to yield significant insights and support informed decision-making. As a multifaceted discipline, it has generated transformative effects across various industries, enabling organizations to effectively harness the extensive reservoirs of data at their disposal. The fundamental processes within data analytics include data collection, data preparation, data modelling, and data visualization. Illustrated in Fig. 1 is the collaborative interplay of these processes, aimed at achieving knowledge extraction through means of data analytics. Each of these stages employs a diverse array of algorithms, methodologies, and models while contending with

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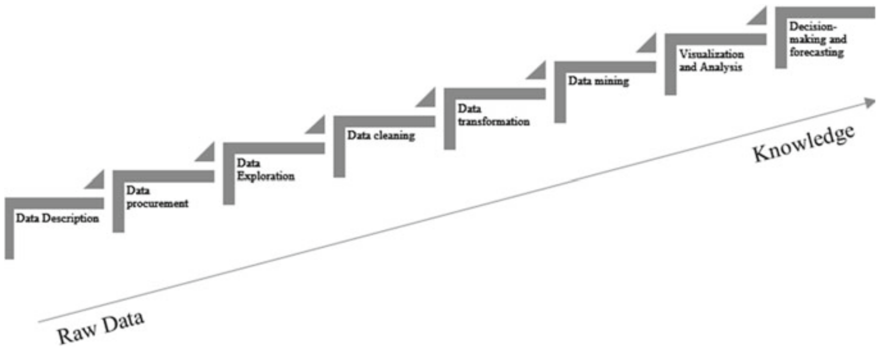

Fig. 1 Knowledge extraction using data analytics

limitations in terms of both resources and time. Consequently, optimization emerges as a pivotal element. Optimization, in this context, entails the precise fine-tuning of algorithms and models to optimize both efficiency and accuracy, ensuring that data-driven solutions possess not only informativeness but also actionable quality. It serves as the cornerstone that empowers businesses to derive the utmost value from their data, ultimately fostering innovation and enhancing competitiveness within the contemporary landscape.

Despite the widely recognized importance of optimization within data analytics, its practical implementation frequently proves to be a challenging endeavour. This complexity primarily arises from the fact that optimization in these processes is far from straightforward, often necessitating consideration of multiple objectives rather than a singular one. For example, each of the processes mentioned earlier needs to be optimized with respect to both the resources and time taken to complete it, thus requiring two-objective optimization. If more than one resource needs to be optimized, then the number of objectives continues to increase, thus giving rise to multiple objectives for optimization. These problems are technically termed multi-objective optimization problems or MOOPs.

Multi-objective optimization problems (MOOPs) in data analytics are prevalent across various domains, each presenting a unique set of challenges. In customer relationship management, a company may aim to simultaneously increase customer satisfaction, reduce churn rates, optimize marketing expenditure, and enhance product quality. In supply chain management, businesses strive to balance objectives such as minimizing transportation costs, maximizing delivery speed, reducing carbon emissions, and ensuring product freshness. In healthcare analytics, optimizing patient treatment plans requires considering objectives such as minimizing healthcare costs, reducing patient waiting times, improving treatment effectiveness, and ensuring patient safety. Additionally, in financial portfolio management, investors seek to maximize returns, minimize risks, reduce tax liabilities, and maintain liquidity. These MOOPs demand sophisticated approaches to navigate the intricate trade-offs between multiple objectives, ultimately enabling data-driven decisions that align with specific goals and constraints in diverse fields.

Before the advent of nature-inspired and evolutionary algorithms, solving multi-objective optimization problems (MOOPs) relied on more traditional methods. The older methods often involved mathematical programming techniques such as weighted sum or epsilon-constraint strategies. These techniques aimed to transform multi-objective optimization problems (MOOPs) into single-objective optimization problems. However, they had clear shortcomings. They frequently demanded that decision-makers allocate subjective weights to objectives, a task that could be challenging and might not accurately mirror real-world priorities. Furthermore, these approaches grappled with intricate, nonlinear relationships among objectives and struggled to efficiently explore the complete Pareto frontier. Consequently, they proved less effective when dealing with MOOPs featuring multiple conflicting objectives.

Expanding upon traditional methods, the incorporation of nature-inspired algorithms (NIAs) represented a notable step forward in handling multi-objective optimization problems (MOOPs). NIAs derive inspiration from natural phenomena and processes, encompassing genetic algorithms emulating biological evolution, particle swarm optimization inspired by bird flock behaviour, and simulated annealing grounded in metallurgical annealing processes. Among these NIAs, evolutionary algorithms, in particular, have surfaced as one of the most efficient approaches for MOOPs.

Evolutionary algorithms leverage principles from natural selection, reproduction, and mutation to iteratively explore and enhance solutions. They excel when confronted with intricate, multi-objective situations and offer several advantages. First, they adeptly search for varied solutions throughout the Pareto frontier, ensuring a thorough exploration of trade-offs between objectives. Second, their need for minimal problem-specific knowledge renders them versatile for diverse applications. Third, these algorithms demonstrate adaptability, allowing them to adapt to changing problem landscapes during optimization. Their capacity to strike a balance between exploration and exploitation renders them exceptionally robust in identifying optimal or near-optimal MOOP solutions, even when dealing with nonlinear relationships or a high number of conflicting objectives. Consequently, they have evolved into indispensable instruments in the field of data analytics, facilitating informed, data-driven decision-making across varied domains while mitigating the shortcomings associated with conventional approaches.

The rest of this chapter is organized as follows: Sect. 2 gives an overview of multi-objective optimization, Sect. 3 highlights various MOOPs in various domains of data analytics, Sect. 4 provides a brief introduction of various nature-inspired algorithms, Sect. 5 describes the use of evolutionary algorithms for solving MOOPs, and finally, Sect. 6 concludes the discussion on the use of evolutionary algorithms for solving various MOOPs in data analytics.

2 Overview of Multi-objective Optimization (MOO)

Multi-objective optimization, typically referred to as MOOP, stands as a foundational concept in the field of optimization and decision-making. In contrast to the singular focus of single-objective optimization, which aims to pinpoint a solitary optimal solution, MOOP confronts the intricate task of concurrently optimizing multiple objectives, which frequently exhibit conflicting priorities. These objectives span a broad spectrum, encompassing endeavours such as profit maximization and cost minimization within the business sphere, as well as the optimization of numerous performance metrics in the realms of engineering and scientific exploration. MOOP assumes a pivotal role across diverse domains precisely because it mirrors the multifaceted complexity encountered in real-world scenarios, where decision-makers grapple with the delicate equilibrium of diverse objectives and associated trade-offs. The ideal process of multi-objective optimization is summarized in Fig. 2.

In this section, the details of multi-objective optimization and the terminology associated with it are explained. Furthermore, a formal definition of multi-objective optimization and the process involved in the formulation of MOOPs is highlighted in this section. Finally, various traditional methods for solving MOOPs and for the process of Pareto-optimal solution selection are described.

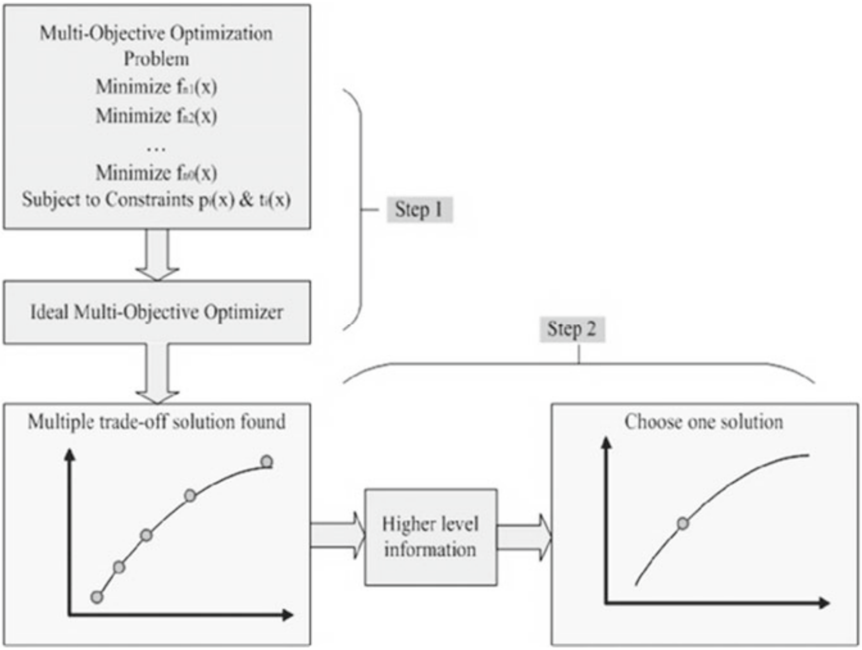

Fig. 2 Multi-objective optimization (Ideal) procedure [9]

Table 1 Multi-objective optimization terminologies

Term	Description
Multi-objective optimization	Optimization challenges encompassing mathematical problems where the aim is to concurrently optimize multiple objective functions [1, 2]
Pareto optimization	Another term for multi-objective optimization [1, 2]
Nondominated solution	A solution where none of the objective functions can be improved in value without degrading some of the other objective values [1, 2]
Pareto-optimal solution	Another term for a nondominated solution [1, 2]

2.1 Terminology in MOO

The terminologies mainly associated with multi-objective optimization are summarized in Table 1.

2.2 Definition and Formulation

Multi-objective optimization, often referred to as Pareto optimization, multi-objective programming, vector optimization, multicriteria optimization, or multi-attribute optimization, exists as a discipline within the realm of multiple-criteria decision-making. It is primarily concerned with the mathematical facet of optimization, wherein the primary focus resides in scenarios entailing more than a single objective function necessitating concurrent optimization. The practical applications of this field span various scientific disciplines, encompassing domains such as engineering, economics, and logistics, where the attainment of optimal decisions invariably involves navigating trade-offs among two or more conflicting objectives [1, 2].

In instances characterized by nontrivial multi-objective optimization quandaries, it is imperative to acknowledge the inherent absence of a singular solution capable of simultaneously optimizing all the involved objectives. In such cases, the objective functions are inherently conflicting. A solution attains the distinctions of being nondominated, Pareto optimal, Pareto efficient, or noninferior when it is impossible to enhance the value of any objective function without concomitantly diminishing the values of other objectives. In the absence of supplementary subjective preference information, there may exist a multitude of Pareto optimal solutions, potentially constituting an infinite set, each of which is accorded equal merit and standing [1, 2].

Mathematically, a multi-objective optimization problem can be formulated as shown in Eq. 1.

$$Maximize\ or\ Minimize\, f_m(x),\ m = 1, 2, \ldots, M \tag{1}$$

Subject to:

$$g_i(x) \geq 0, i = 1, 2, \ldots, I$$

$$h_k(x) = 0, k = 1, 2, \ldots, K$$

$$x_j^{(L)} \leq x_j \leq x_j^{(U)}, i = 1, 2, \ldots, n$$

where M is the number of objectives to be optimized, and x is the feasible set of vectors. The feasible set is usually defined by some constraints in functional forms such as $g_i(x)$ and $h_k(x)$.

Now that the definition and concept of MOO are clear, it is time to move on to Pareto-optimal solution selection.

The selection of a Pareto optimal solution hinges on the principle of dominance. A solution earns the designations of being nondominated or Pareto optimal when it is impossible to enhance the value of any objective function without simultaneously diminishing the values of other objectives. In a formal sense, a state attains the status of being Pareto-optimal if no alternative state exists where improvements can be made to the well-being of at least one participant without simultaneously diminishing the well-being of any other participant [3].

2.3 Traditional Methods for MOO

Multi-objective optimization, involving the simultaneous optimization of multiple objective functions, poses formidable challenges for conventional optimization techniques and algorithms due to inherent conflicts among diverse optimization objectives and functions [4]. Prior to the emergence of nature-inspired and evolutionary algorithms, a number of traditional approaches were employed to address these complex problems. Here, we outline some of these conventional methods:

1. Weighted Sum Method: This approach linearly combines a set of objectives into a single objective by assigning each objective a predetermined weight, reflecting its relative importance.
   - Advantages: It offers simplicity in implementation.
   - Limitations: Setting appropriate weight vectors to attain Pareto-optimal solutions within specific regions of the objective space can be challenging. Furthermore, in nonconvex objective spaces, it may fail to identify certain Pareto-optimal solutions.
2. ε-Constraint Method: This method selects one primary objective while constraining the remaining objectives within user-defined limits.

- Advantages: Effectively suited for situations where one specific objective takes precedence.
- Limitations: The solution obtained is highly susceptible to variations in constraint values and relies heavily on user knowledge [5].

These conventional methodologies, while historically significant, have ceded ground to more advanced algorithms that have come to the forefront with the evolution of computing capabilities. These contemporary algorithms demonstrate enhanced capabilities in addressing problems with increased complexity, encompassing greater numbers of variables, objectives, and nonlinear characteristics. Their application has yielded notable improvements wherever they have been employed [2].

Despite their commendable success in solving many of the then existing problems, they suffered from a few limitations, as noted below:

1. Addressing Nonconvex and Discontinuous Pareto Fronts: Traditional approaches encounter difficulties when dealing with nonconvex or discontinuous Pareto fronts [1, 6].
2. Handling High-Dimensional Objective Spaces: In scenarios involving numerous objectives, these methods exhibit inefficiency as their computational complexity increases exponentially. This makes them less suitable for problems with a high number of objectives [1, 6].
3. Dependence on Prior Knowledge: Many of these techniques necessitate prior knowledge about the problem, such as objective importance rankings or specific constraint values. This reliance becomes a significant drawback when such information is unavailable [1, 6].
4. Limited Solution Generation: Traditional methods typically yield only one solution at a time, whereas modern evolutionary algorithms can produce a set of Pareto-optimal solutions within a single iteration [1, 6].
5. Parameter Sensitivity: The performance of these methods is notably sensitive to their parameter configurations, diminishing their robustness and practical utility [1, 6].

2.4 Traditional Methods for Pareto-Optimal Solution Selection

Before the rise of nature-inspired and evolutionary algorithms (NIAs), various traditional techniques were employed for the selection of Pareto optimal solutions in multi-objective optimization problems. Here are some of these conventional methods:

1. Technique for Order Preference by Similarity to Ideal Solution (TOPSIS): TOPSIS is a widely used approach that exhibits a clear inclination towards specific objectives, aiding in the ranking and selection of solutions from the Pareto optimal set [7].

2. Elimination and Choice Translating Reality (ELECTRE): ELECTRE stands as another prevalent technique employed for ranking and choosing solutions within the Pareto optimal set [7].
3. Analytic Hierarchy Process (AHP): AHP finds application in the ranking and selection of solutions within the Pareto optimal set [7].

These traditional methodologies, although historically significant, have yielded ground to novel algorithms that have emerged with advancements in computing capabilities. These robust algorithms have proven more adept at handling greater numbers of variables, objectives, and nonlinearities, showcasing considerable enhancements in their application domains [8].

Similar to the traditional algorithms for multi-objective optimization, the algorithms for Pareto-optimal solution selection also suffer from a few limitations similar to those of the traditional algorithms for MOOs.

Now that the traditional techniques for MOO and Pareto-Optimal Solution Selection are clear, it is time to discuss the need for these techniques (and many others as well) in Data Analytics, and MOOPs present themselves in various domains of data analytics.

3 MOO Problems (MOOPs) in Data Analytics

After going over the fundamentals of MOOPs, it is time to look at MOOPs in the context of Data Analytics. This segment presents a concise synopsis of Data Analytics, elucidating its constituent processes. Additionally, it details how MOOPs in data analytics can be visualized and solved utilizing multi-objective evolutionary methods.

3.1 Overview of Data Analytics

Data analytics is the process of examining, cleaning, transforming, and interpreting data to uncover valuable insights and support informed decision-making. It serves as a cornerstone in today's data-driven world, offering organizations the means to leverage their vast data resources for competitive advantage. This multifaceted discipline involves various stages, from data collection and preparation to modelling and visualization, enabling businesses to extract actionable knowledge from complex datasets. Data analytics has a profound impact across diverse sectors, facilitating innovation, enhancing operational efficiency, and guiding strategic choices. Its versatility empowers decision-makers to navigate the complexities of our data-rich environment and harness the power of information to drive success. The fundamental processes in data analytics have already been highlighted in Fig. 1 in the Introduction section.

3.2 MOO View of Data Analytics Problems

Multi-objective optimization plays a crucial role in addressing a wide range of challenges in data analytics across various domains. Here are some problems identified in data analytics along with examples:

1. Feature Selection in Machine Learning: In feature selection, one often aims to simultaneously optimize multiple objectives, such as maximizing predictive accuracy, minimizing model complexity, and reducing computation time. Research by Deb et al. [10] explores multi-objective feature selection for classification.
2. Resource Allocation in Cloud Computing: Optimizing resource allocation in cloud environments involves balancing objectives such as minimizing costs, maximizing resource utilization, and ensuring Quality of Service (QoS).
3. Portfolio Optimization in Finance: In financial portfolio optimization, investors seek to maximize returns while minimizing risks. Multi-objective optimization helps find diversified portfolios that balance these conflicting objectives.
4. Supply Chain Management: Optimizing supply chain operations involves multiple conflicting objectives, such as minimizing transportation costs, reducing delivery times, and ensuring product quality.
5. Image Processing: In medical image segmentation, objectives include minimizing both false positives and false negatives while maximizing the accuracy of segmentation.
6. Social Network Analysis: Community detection in social networks often involves optimizing objectives such as maximizing modularity and minimizing the number of cut edges.

These examples illustrate how multi-objective optimization techniques are applied to real-world problems in data analytics, providing decision-makers with a range of solutions that consider multiple conflicting objectives and trade-offs.

In the next section, the advent of nature-inspired algorithms is described along with their application to MOO.

4 Overview of Nature-Inspired Algorithms (NIAs)

Nature-inspired algorithms (NIAs) mimic various natural occurrences and processes. Due to their inherent capability to model complex environments, NIAs have been widely used for solving MOOPs. This section explains the basic principles of various NIAs, such as evolutionary algorithms, swarm optimization algorithms, ant colony optimization algorithms, and neural network algorithms, along with their applications in various domains, such as engineering, data mining, image processing, and machine learning.

4.1 Ushering into a New Era of NIAs

Transitioning into a new era of nature-inspired algorithms (NIAs) signifies a significant shift in problem-solving methodologies, particularly in the domain of optimization. NIAs draw inspiration from the intricate and efficient processes observed in the natural world, applying these principles to computational challenges. Unlike traditional approaches, which often rely on simplifications and heuristics, NIAs leverage the elegance of evolution, the collaborative behaviour of swarms, and the adaptability of physical processes to tackle complex, real-world problems. This paradigm shift has unleashed a new wave of algorithms that excel in handling high-dimensional spaces, nonlinearity, and the simultaneous optimization of multiple objectives—areas where conventional methods often falter. With their versatility and ability to explore vast solution spaces, NIAs have found applications across diverse domains, from engineering and logistics to finance and healthcare, ushering in a promising era of innovation and enhanced problem-solving capabilities.

The following subsections give an overview of various NIAs, such as evolutionary algorithms, swarm optimization algorithms, ant colony optimization algorithms, and neural networks. There are many other NIAs available that also perform optimization tasks, but for the reader to get clarity on how NIAs solve optimization problems, these should suffice.

4.2 Evolutionary Algorithms

Evolutionary algorithms (EAs) belong to a class of optimization techniques that draw inspiration from the intricate workings of natural evolution, including concepts such as reproduction, mutation, recombination, and selection. These algorithms are a subset of a broader category known as evolutionary computation, representing a versatile population-based metaheuristic approach to optimization challenges [11, 11]. EAs prove particularly valuable in tackling intricate problems that surpass the boundaries of current knowledge and conventional problem-solving capabilities, employing a form of synthetic digital evolution [12]. What makes them especially intriguing is their capacity to yield novel solutions, capitalizing on attributes or effects that were previously undisclosed within the given problem context [12].

The fundamental operational sequence of an evolutionary algorithm adheres to the following stages [11, 11]:

1. Initialization: Commence by generating an initial population of individuals in a randomized manner, constituting the inaugural generation.
2. Evaluation: Assess the fitness of each individual within the population.
3. Selection: Identify the fittest individuals to serve as parents for the upcoming reproduction phase.
4. Reproduction: Employ crossover and mutation operations to breed new individuals, thereby giving rise to offspring.

5. Replacement: Replace the least fit individuals in the population with the newly generated individuals.
6. Termination Check: Continuously monitor for the fulfilment of termination criteria, such as reaching a maximum number of generations. If the criterion is not met, return to the evaluation step.

This iterative process persists until the predetermined termination criteria are satisfied [11, 11]. The process is depicted in detail in Fig. 3.

In terms of practical applications, EAs have found utility in a multitude of scientific fields, including engineering, economics, and logistics, where the imperative lies in making optimal decisions while navigating trade-offs among two or more competing objectives [11]. Their adaptable framework facilitates cross-domain applicability [12].

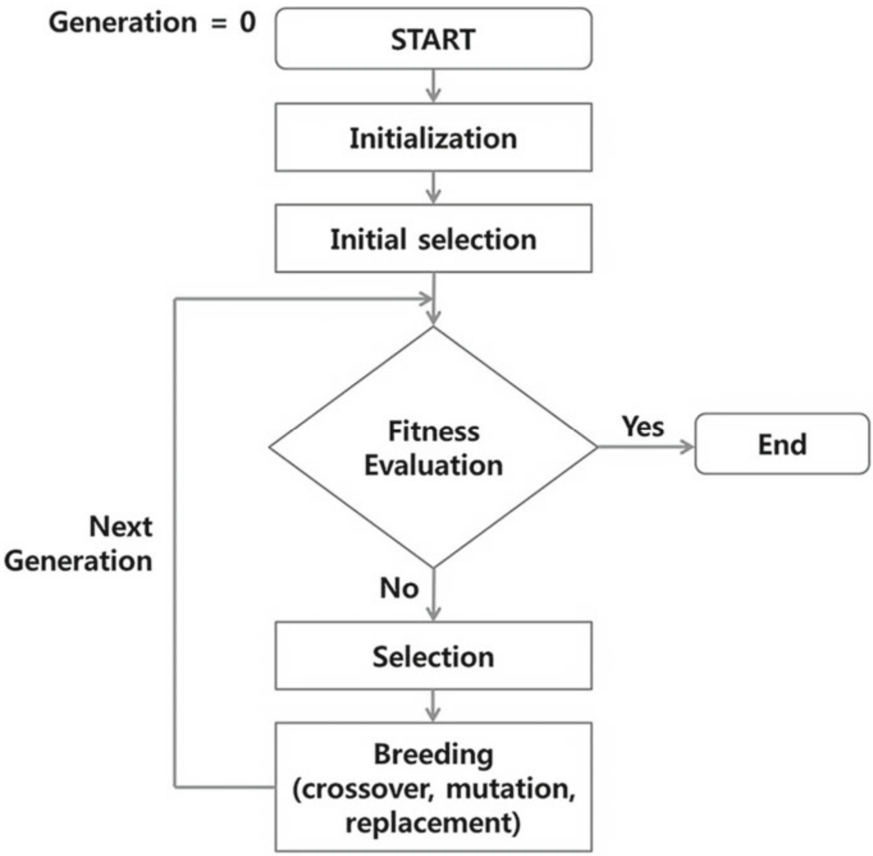

Fig. 3 Flowchart of evolutionary algorithm process [13]

4.3 Swarm Optimization Algorithms

Swarm optimization algorithms are a class of nature-inspired optimization techniques that leverage the collective behaviour of a group of agents, often referred to as a "swarm," to solve complex optimization problems. One of the most well-known swarm optimization algorithms is the particle swarm optimization (PSO) algorithm.

In PSO, each agent, known as a "particle," represents a potential solution to the optimization problem. These particles move through the search space, adjusting their positions based on their own experiences and the experiences of their peers. The algorithm is guided by two key principles:

1. Individual Experience (Cognitive Component): Each particle remembers its own best position in the search space, which corresponds to the best solution it has found thus far. It uses this information to guide its movement towards that optimal position.
2. Social Experience (Social Component): Additionally, particles are influenced by the best positions discovered by their neighbours within the swarm. This enables the swarm to collectively converge towards promising regions in the search space.

The process of particle swarm optimization (a type of swarm optimization) is detailed in Fig. 4.

The movement of particles is controlled by adjusting their velocities and positions iteratively. PSO aims to strike a balance between exploration (searching for new, unexplored regions) and exploitation (refining solutions in promising areas) to find an optimal or near-optimal solution.

Kennedy et al. [14] laid the foundation for PSO and described its basic principles. Since that time, PSO has found extensive use in a range of domains, encompassing engineering, data mining, image processing, and machine learning. Its efficiency in navigating solution spaces and uncovering optimal or nearly optimal solutions has made it particularly valuable.

It is worth highlighting that while PSO stands as a prominent swarm optimization algorithm, the realm of swarm intelligence boasts other variations and algorithms, including Ant Colony Optimization (ACO) and Bee Colony Optimization (BCO), each distinguished by its own attributes and areas of application.

4.4 Ant Colony Optimization

Ant colony optimization (ACO) is an optimization algorithm inspired by the foraging behaviour of ants. Dorigo et al. [16] initially introduced ACO, and it has since gained widespread recognition and influence as an optimization technique.

In the ACO framework, the optimization problem is represented using a graph, commonly known as a "graph-based representation." Here is an outline of how ACO operates:

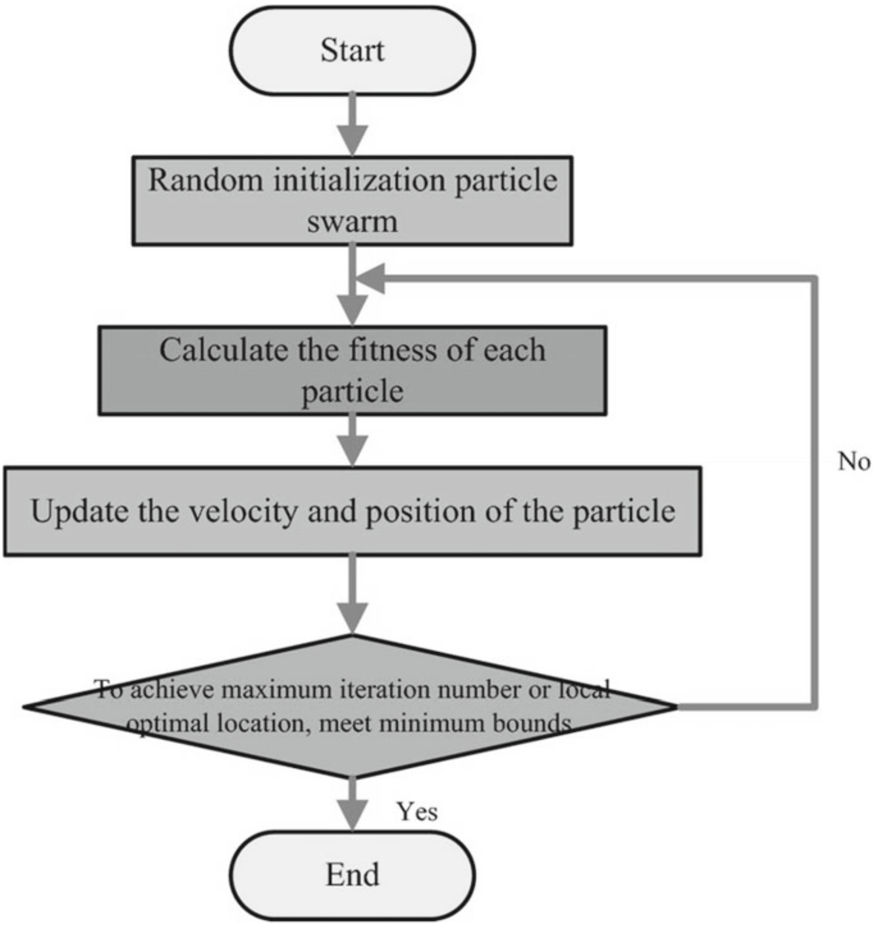

Fig. 4 Flowchart for the particle swarm optimization algorithm [15]

1. Problem Representation: The optimization problem is translated into a graph, where nodes symbolize potential solutions or decision points, and edges signify the connections between these solutions.
2. Ant Agents: ACO employs a group of artificial agents known as "ants." Each ant represents a potential solution to the problem and begins its journey from an initial node within the graph.
3. Solution Construction: Ants traverse the graph by selecting edges using a probabilistic decision-making process. Their decisions are guided by a blend of "pheromone" levels on edges and a heuristic function that evaluates the solution's quality at each node.
4. Pheromone Update: As ants traverse edges, they leave behind "pheromone" on those edges. Pheromone serves as an indicator of the desirability of a specific

path. Edges with higher pheromone levels become more appealing to subsequent ants.
5. Evaporation: Over time, the pheromone levels on all edges slowly diminish, mimicking the gradual fading of scent trails in the natural world. This ensures that the algorithm retains its exploratory nature.
6. Solution Evaluation: When an ant completes its journey and reaches the end of the graph, the quality of the solution it has constructed is evaluated.
7. Pheromone Update (Global): The pheromone levels on edges are updated globally, with the best solutions receiving a higher boost in pheromone. This encourages the algorithm to converge toward better solutions.
8. Termination: The process continues for a predetermined number of iterations or until a termination criterion is met.

The process followed by the ant colony optimization algorithm is depicted in Fig. 5.

The ACO algorithm is guided by the principle of "collective intelligence," where individual ants contribute to the overall optimization process through their exploration and exploitation of the solution space. As ants repeatedly explore the

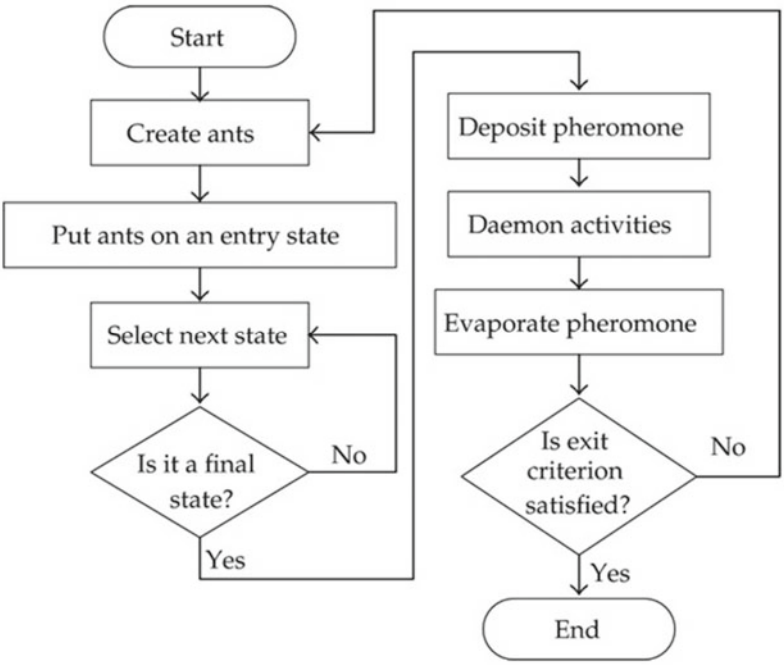

Fig. 5 Flowchart of the ant colony optimization algorithm [17]

graph, they tend to converge toward better solutions by following paths with higher pheromone levels and better heuristic information.

Since its inception, ACO has been successfully applied to a wide range of optimization problems, including the travelling salesman problem, vehicle routing, and network routing.

4.5 Neural Networks

Neural networks belong to a category of machine learning models inspired by the structure and operations of the human brain. They have become a foundational tool in diverse domains, encompassing image recognition, natural language processing, and the realm of optimization. In this overview, an explanation of neural networks, their operational steps, and their role in enhancing optimization will be presented.

Overview of Neural Networks

A neural network comprises interconnected artificial neurons or nodes systematically arranged into layers. These layers are categorized into three primary types:

1. Input Layer: This initial layer receives input data and subsequently transmits it to the following layer.
2. Hidden Layers: Positioned between the input and output layers, these intermediary layers process the input data by utilizing weighted connections and employing activation functions to generate output.
3. Output Layer: As the ultimate layer in the network, it generates the network's output, which can serve various purposes, such as classification, regression, or optimization.

Figure 6 depicts a basic neural network with one hidden layer.

Working Steps of Neural Networks

1. Initialization: The weights and biases of the connections between neurons are initialized randomly.
2. Forward Propagation: Input data are fed into the input layer and processed through the hidden layers via weighted connections and activation functions. The output is computed layer by layer until the final output layer is reached.
3. Loss Calculation: The output of the neural network is compared to the target or ground truth, and a loss (error) is calculated to measure how far off the predictions are from the actual values.
4. Backpropagation: The network uses an optimization algorithm, such as gradient descent, to adjust the weights and biases in the opposite direction of the gradient of the loss function. This step iteratively minimizes the loss and updates the model parameters.
5. Training: Steps 2–4 are repeated for multiple iterations or epochs to optimize the network's performance on the training data.

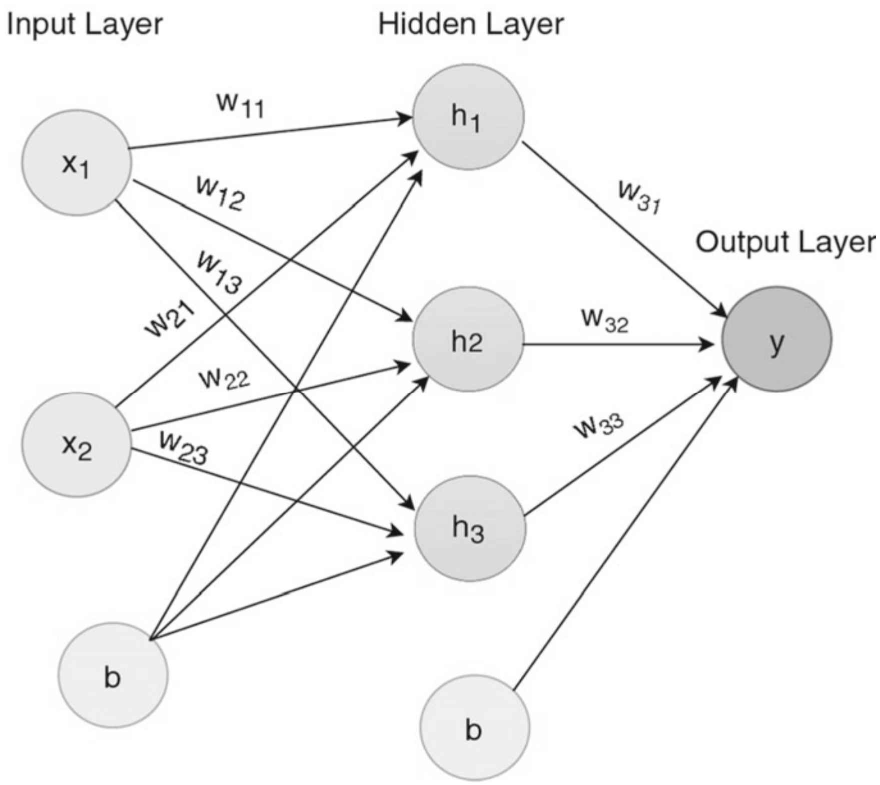

Fig. 6 Neural network with 1 hidden layer [22]

6. Prediction: Once trained, the neural network can be used for making predictions on new, unseen data.

Figure 7 summarizes the training and backpropagation phases of a neural network.

Neural Networks and Optimization

Neural networks assume a pivotal role in optimization challenges, often serving as function approximators or surrogate models. Their applications in optimization encompass the following avenues:

1. Hyperparameter Tuning: Neural networks find application in optimizing the hyperparameters of other machine learning models or algorithms. Research conducted by Snoeyink et al. [18] delves into the utilization of neural networks for the optimization of hyperparameters.
2. Optimization in Reinforcement Learning: In the realm of reinforcement learning, neural networks are harnessed to approximate value functions or policy functions, aiming to optimize the behaviour of agents. The work by Mnih et al. [19] introduces the application of deep Q-networks for reinforcement learning.

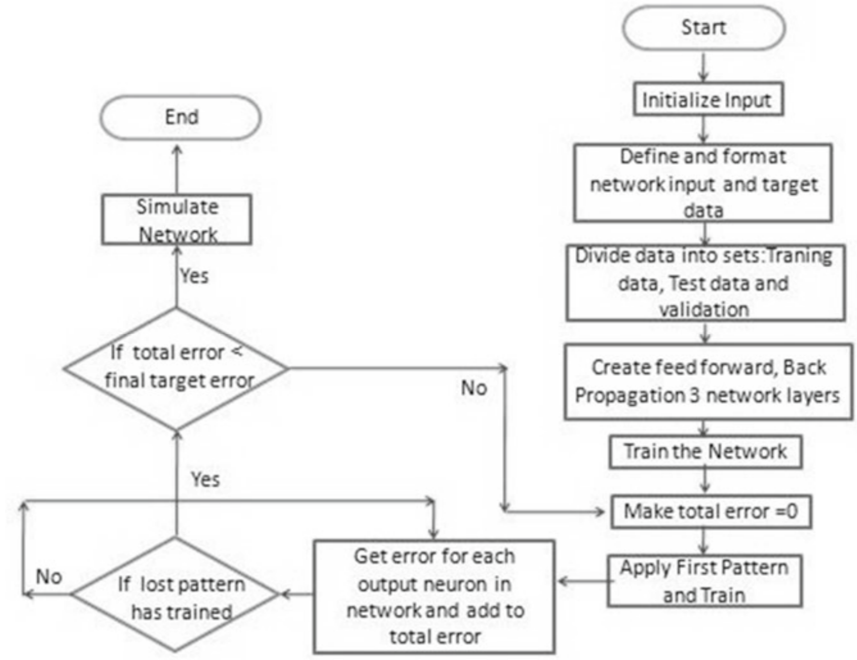

Fig. 7 Flowchart for training an artificial neural network (ANN) [23]

3. Optimization in Deep Learning: Neural networks themselves are subject to optimization, employing various techniques such as stochastic gradient descent and its derivatives. The research by Kingma and Ba [20] introduces the Adam optimization algorithm, tailored for training deep neural networks.
4. Optimization in Image Registration: Neural networks can optimize the intricate process of image registration, facilitating the alignment of diverse images or scans, particularly in medical or computer vision applications. The research by Balakrishnan et al. [21] demonstrates the integration of neural networks into the optimization of image registration.

These examples vividly illustrate the integral role that neural networks play in optimization across a spectrum of domains, contributing to the quest for optimal solutions in complex problem spaces.

4.6 Overview of MOO Using NIAs

Multi-objective optimization using nature-inspired algorithms presents an alluring approach to tackling intricate real-world conundrums where the simultaneous consideration of multiple conflicting objectives is imperative. These algorithms derive inspiration from natural phenomena and processes, mirroring biological evolution, swarm dynamics, and physical principles to navigate their pursuit of optimal or near-optimal solutions. The incorporation of nature-inspired algorithms in multi-objective optimization has garnered considerable attention, owing to their adeptness in navigating high-dimensional solution spaces, accommodating nonlinear relationships, and managing the inherent trade-offs within multi-objective quandaries.

Within the domain of nature-inspired algorithms for multi-objective optimization, a prominent category is evolutionary multi-objective optimization (EMO). EMO algorithms, exemplified by NSGA-II (Nondominated Sorting Genetic Algorithm II) as introduced by Deb et al. [10], draw inspiration from the principles underpinning biological evolution. These algorithms embark on a quest through a populace of candidate solutions, iteratively evolving this population across generations to unearth a diverse spectrum of Pareto-optimal solutions. The bedrock of EMO is founded on the concept of Pareto dominance, which deems a solution superior if it cannot be enhanced in one objective without compromising another. Algorithms such as NSGA-II shine in furnishing a plethora of solutions that harmonize multiple objectives, equipping decision-makers with valuable choices that navigate trade-offs effectively.

Another avenue within the realm of nature-inspired approaches is the adaptation of Particle Swarm Optimization (PSO) to address multi-objective optimization. Research by Mostaghim and Teich [24] extends the purview of the PSO algorithm to embrace multi-objective problem solving. In the domain of PSO-based multi-objective optimization, the particles within the swarm collaborate harmoniously, fathoming their way toward a constellation of Pareto-optimal solutions. The algorithm's prowess in calibrating the equilibrium between exploration and exploitation renders it an efficacious tool for endeavours involving multi-objective optimization.

Furthermore, the vista of nature-inspired multi-objective optimization (NMOO) encompasses a diverse repertoire of algorithms that draw inspiration from an array of natural processes. These encompass simulated annealing, ant colony optimization, and firefly algorithms, among others, aimed at surmounting multi-objective challenges. These algorithms aspire to encapsulate the quintessence of diverse natural phenomena in their quest to unearth solutions that satisfy multiple objectives. Consequently, they furnish a versatile arsenal for confronting multifarious optimization tribulations.

This completes the discussion on nature-inspired algorithms and their optimization capabilities. This section also provides an overview of how NIAs can be effectively applied to solve some complex and perplexing MOOPs that would have been unsolvable with traditional methods alone.

As stated in the chapter thus far, among all NIAs, evolutionary algorithms (EAs) have proven to be most effective for multi-objective optimization. The next section dives deeper into this discussion and focuses on the concept of evolutionary multi-objective optimization, which utilizes EAs to find solutions to MOOPs. Additionally, the next section also discusses the application of EAs for the Pareto-optimal solution selection problem.

5 Evolutionary MOO (EMOO)

Evolutionary multi-objective optimization (EMOO) represents a critical paradigm in the field of optimization, specifically tailored to tackle problems with multiple conflicting objectives. In an increasingly complex and interconnected world, decision-makers often face the challenge of balancing diverse objectives, ranging from minimizing costs and maximizing performance to optimizing resource allocation. EMOO is a powerful approach to navigate these multifaceted optimization landscapes, as it seeks to uncover a set of solutions known as the Pareto front, where no single solution dominates the others in all objectives.

Among the diverse array of nature-inspired algorithms (NIAs), EMOO stands out as one of the most effective methods for solving multi-objective problems. While NIAs such as Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO) draw inspiration from the natural world, EMOO leverages principles akin to biological evolution, reproduction, and natural selection. This unique approach enables EMOO to efficiently explore the Pareto front, providing decision-makers with a comprehensive view of trade-offs between objectives and facilitating the selection of optimal or near-optimal solutions.

In the following subsections, we will delve deeper into the principles underlying EMOO, explore its application in various domains, and examine the key challenges and advancements in this dynamic field. By the end of this section, readers will gain a profound understanding of how EMOO serves as a cornerstone in addressing complex multi-objective optimization problems across diverse domains.

5.1 Formulation of EMOO Problems

Evolutionary multi-objective optimization (EMOO) entails the utilization of evolutionary algorithms for the resolution of challenges characterized by multiple optimization objectives. Such multifaceted problems are prevalent in real-world scenarios, primarily because trade-offs invariably exist, and achieving improvements in one aspect may entail compromises in another. For instance, within the realm of embedded system design, enhancing performance often necessitates increased costs. In structural engineering, the quest for greater stability is intricately linked with the construction's weight, while in machine learning, elevating model accuracy requires careful consideration of model complexity [25].

The work outlined by Deb in [26] serves as a valuable resource for understanding the mathematical formulation and principles underlying EMOO, providing a solid foundation for researchers and practitioners in this field.

Formulating evolutionary multi-objective optimization (EMOO) involves defining the mathematical framework and the key components of the algorithm. Below, the formulation of EMOO is explained step by step, incorporating mathematical notation where necessary:

1. Step 1: Define the Objective Functions (F_1, $F_2\ldots$, F_n):
   In EMOO, the optimization problem begins with the specification of multiple objective functions, denoted as F_i, where i ranges from 1 to n. These objective functions quantify the quality of solutions in various aspects relevant to the problem at hand.
2. Step 2: Define the decision variables (X_1, $X_2\ldots$, X_m):
   Next, decision variables, represented as X_i, where i ranges from 1 to m, are identified. These variables correspond to parameters or choices that influence the solutions.
3. Step 3: Formulate the Multi-objective Optimization Problem:
   The mathematical formulation of the multi-objective optimization problem involves minimizing or maximizing the vector of objective functions $F = [F_1(X), F_2(X),..., F_n(X)]$ while adhering to any constraints on the decision variables.
4. Step 4: Incorporate evolutionary algorithms:
   Evolutionary algorithms such as genetic algorithms (GAs) or differential evolution (DE) are integrated into EMOO to explore the solution space effectively.
5. Step 5: Define Population and Operators:
   A population of individuals, each representing a set of decision variables (X), is defined. Operators such as crossover and mutation are applied to generate new individuals.
6. Step 6: Implement Selection Mechanisms:
   Selection mechanisms, including nondominated sorting, crowding distance, and Pareto dominance, are used to determine which individuals proceed to the next generation based on their nondominance status.
7. Step 7: Iterate for Convergence:
   The EMOO algorithm iteratively evolves the population across multiple generations to explore and converge toward a diverse set of nondominated solutions along the Pareto front.

Figure 8 depicts the process of EMOO for feature selection and multilabel classification diagrammatically.

5.2 Overview of Approaches and Techniques Involved in EMOO

Many different approaches and techniques have been experimented with for the task of EMOO. Some of them are explained below:

1. Weighted Approach: In the weighted approach, the initial multi-objective problem is transformed into a single-objective problem through the application of appropriate weights to each objective. These weights are combined to form a unified objective function, offering the flexibility to fine-tune the weights to discover various solutions along the Pareto front.

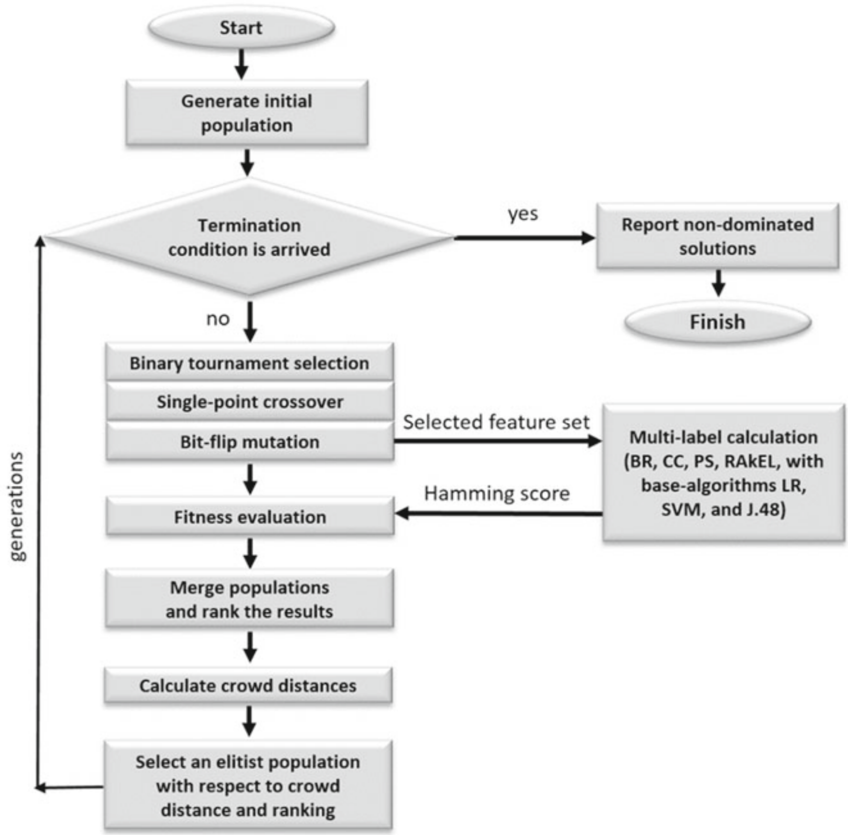

Fig. 8 Flowchart for the EMOO algorithm [27]

2. Lexicographical Approach: In the lexicographical approach, objectives are prioritized, and optimization is carried out based on their order of importance. The solution initially focuses on optimizing the most critical objective. If multiple optimal solutions exist, subsequent optimization considers the next most crucial objective, and so forth.
3. Pareto Approach: The widely recognized Pareto approach yields a comprehensive set of nondominated solutions. A solution is deemed Pareto optimal when there is no alternative solution that can improve one criterion without simultaneously worsening at least one other criterion.

These diverse approaches and techniques within the realm of evolutionary multi-objective optimization provide versatile tools for addressing complex real-world problems with multiple conflicting objectives. While the weighted approach offers customization through objective weighting, the lexicographical approach prioritizes objectives. The Pareto approach, on the other hand, yields a collection of nondominated solutions, allowing decision-makers to explore a range of trade-offs. The choice of approach depends on the specific problem and the decision-maker's preferences, underscoring the importance of understanding and applying these techniques effectively to navigate the complexities of multi-objective optimization.

The next subsection focuses on the application of EMOO in the vast field of data analytics.

5.3 EMOO in Data Analytics

In the dynamic field of data analytics, evolutionary multi-objective optimization (EMOO) has emerged as a key player. It provides a robust framework to tackle the multifaceted challenges that often arise in data-driven decision-making. Within the world of data analytics, balancing various objectives is a common puzzle to solve. For example, optimizing machine learning models can involve a delicate dance between accuracy, computational efficiency, and model interpretability. EMOO is a valuable tool, giving data analysts the power to explore a diverse range of solutions that cater to these varied goals. This flexibility fosters well-informed decision-making and fuels innovation in the ever-evolving landscape of data-driven insights.

Take, for instance, a scenario in financial analytics where a portfolio manager seeks to fine-tune investment decisions. They aim to maximize returns while minimizing risks simultaneously. EMOO can assist in generating a spectrum of investment portfolios, each offering a different balance between potential returns and risk levels. This empowers the manager to make choices aligned with their risk tolerance and investment objectives.

In healthcare analytics, EMOO can be a game changer for optimizing treatment plans for patients with chronic illnesses. Here, the objective is to find a sweet spot between minimizing treatment side effects and maximizing therapeutic effectiveness. EMOO provides the means to explore a variety of treatment options, enabling healthcare professionals to make tailored decisions that benefit their patients' well-being. EMOO can generate a set of treatment plans, each offering a different compromise between these objectives, enabling healthcare professionals to tailor treatments to individual patient needs.

These are just a couple of examples showcasing how EMOO empowers data analysts to address complex multi-objective challenges in the world of data analytics, enabling them to make informed decisions while navigating the intricacies of data-driven insights.

Now that MOO, specifically EMOO, is clear, it is now time to move on to the discussion of Pareto-optimal solution selection using EAs, which is discussed in the next subsection in detail.

5.4 Evolutionary Pareto-Optimal Solution Selection

In the realm of evolutionary multi-objective optimization (EMOO), the process of selecting Pareto-optimal solutions takes centre stage. This process resembles a meticulous sorting task with the aim of finding the cream of the crop among a group of potential solutions. The solutions, referred to as individuals, each have a unique set of characteristics that influence various objectives. The twist here is that multiple objectives are at play, and these objectives often pull in different directions.

During the selection process, each individual's performance is scrutinized closely. The key question is how well each performs across all these conflicting objectives. If one individual emerges as the best, and no other solution in the group outperforms it in every aspect, it is granted the esteemed title of being Pareto-optimal. In essence, this means it is a solution that just cannot be tweaked to make it better in one aspect without making something else worse—quite a remarkable achievement!

However, the aim is not to settle for just one shining star. In EMOO, the goal is to identify a whole constellation of them. Therefore, this selection process unfolds repeatedly, allowing for the gathering of a diverse collection of Pareto-optimal solutions. Each of these solutions represents a unique trade-off between the conflicting objectives.

This process is not about settling for a one-size-fits-all solution. Instead, it is about offering decision-makers a rich array of options, each finely tuned to specific needs and priorities. It is akin to browsing through a menu at a restaurant, where one can pick the dish that perfectly suits their taste. Only here are solutions crafted for complex real-world problems spanning various fields, from engineering to finance and beyond.

Given below are some advantages and limitations of the evolutionary approaches for Pareto-optimal solution selection.

Advantages

1. Their primary strength lies in their knack for finding solutions that are near the global optimum [28].
2. Since 1985, several evolutionary methods for multi-objective optimization have emerged, demonstrating the ability to simultaneously seek multiple Pareto-optimal solutions in a single simulation run [29].

Limitations

1. In numerous real-world optimization problems that are computationally intensive, even when using these approaches, there can be a significant number of calls to the objective function to find a satisfactory solution [28].
2. Evolutionary multi-objective optimization (EMO) algorithms inherently possess stochastic elements. Despite extensive efforts to discover a well-balanced set of trade-off solutions, they frequently fall short of achieving the desired distribution [30].

3. Beyond the inherent stochasticity in the algorithms, the complexity posed by problems near the Pareto-optimal (PO) front presents another challenge in obtaining a uniformly spread set of trade-off solutions [30].
4. Despite the variety of approaches, there is a noticeable absence of comprehensive comparative studies in the literature [29].

This completes the discussion on the application of EAs to Pareto-optimal solution selection.

To provide the reader with some idea about the real-world applicability of all of these approaches and techniques, particularly the application of EMOO in real-world problems, the next subsection provides a description of the latest research on EMOO and highlights some key SOTA (State-Of-The-Art) EMOO models.

5.5 Latest Research in EMOO

Below is a comprehensive overview of recent developments in the field of evolutionary multi-objective optimization (EMOO), noteworthy state-of-the-art (SOTA) models, challenges and future directions:

Recent Progress in EMOO

1. Addressing Numerous Objectives: Recent research has prioritized the effective handling of problems with a multitude of objectives, which can be daunting for traditional methods such as the nondominated sorting genetic algorithm II (NSGA-II) and strength Pareto evolutionary algorithm 2 (SPEA2). The computational complexity and maintenance of diversity in the objective space have posed challenges. To counter this, newer algorithms such as NSGA-III and MOEA/D have emerged as efficient solutions for tackling many-objective problems.
2. Embracing Hybrid Approaches: A notable trend involves the integration of EMOO with various optimization techniques, leading to hybrid EMOO algorithms. By combining EMOO with local search methodologies or other metaheuristics such as Particle Swarm Optimization (PSO), researchers have succeeded in enhancing the exploration and exploitation capabilities of these algorithms [31].
3. Adaptation to Dynamic and Uncertain Environments: In recognition of real-world complexity, EMOO has expanded its horizons to accommodate dynamic multi-objective optimization problems (DMOPs) and uncertain multi-objective optimization problems (UMOPs). These scenarios mimic situations where objectives or constraints may undergo alterations over time or incorporate elements of uncertainty.

State-of-the-Art Models

1. NSGA-III: An extension of the popular NSGA-II, NSGA-III specializes in the resolution of many-objective problems. It introduces a novel reference-point-based strategy to maintain diversity within the objective space.

2. MOEA/D: MOEA/D adopts a unique decomposition approach by breaking down a multi-objective problem into a set of scalar optimization subproblems, which are simultaneously optimized. This approach leverages information from neighboring subproblems, resulting in robust performance.
3. SPEA2-SDE: A variant of SPEA2, the SPEA2-SDE model incorporates shuffled differential evolution (SDE), a global optimization algorithm. This amalgamation has demonstrated impressive efficacy when applied to benchmark problems.

These SOTA models each possess distinctive strengths, making them suitable for varying multi-objective optimization contexts. The selection of a model hinges on the specific characteristics of the problem at hand, including the number of objectives, the nature of decision variables, and the presence of dynamic or uncertain elements. Researchers and practitioners now have a diversified toolkit at their disposal to address a wide spectrum of complex optimization challenges.

Challenges and Limitations

Evolutionary algorithms, while potent, come with their unique set of obstacles. The challenge of scalability is of utmost importance. As the complexity of the problem intensifies, the computational demands of these algorithms can become daunting, making them less suitable for large-scale problems. Moreover, the preservation of diversity within the population is critical to prevent an early convergence to suboptimal solutions, but achieving this in a real-world setting can be a formidable task. The issue of parameter tuning also presents a significant challenge, as the efficiency of evolutionary algorithms can be heavily influenced by their parameter configurations, and pinpointing the optimal configuration can be a complex undertaking. Furthermore, in the context of multi-objective optimization and Pareto-optimal solution selection, the size and diversity of the Pareto-optimal set can pose additional challenges. Selecting the most preferred solutions from a large and diverse Pareto-optimal set can be a non-trivial task, especially when the decision maker's preferences are not well-defined or change over time.

Future Directions

There are several intriguing avenues for future exploration in the field of evolutionary multi-objective optimization. One such area is the realm of dynamic and uncertain environments. Many of the current evolutionary algorithms operate under the assumption of a static and deterministic problem. However, in numerous real-world situations, the objective functions and constraints can fluctuate over time or involve elements of uncertainty. The development of evolutionary algorithms that can adapt to these changes and deliver robust solutions is a complex yet crucial task. Another area of interest is problems with many objectives, where the number of objectives exceeds three. These problems introduce new challenges in terms of solution representation, diversity preservation, and preference articulation, necessitating innovative approaches and techniques [32]. Lastly, the incorporation of the decision maker's preferences into the search process through interactive and preference-based methods offers a promising direction. This allows the search process to be guided towards the most preferred solutions. These represent just a few of the open-ended questions in this dynamic and rapidly evolving field, and we eagerly anticipate the creative solutions that future research will yield.

6 Conclusion

In this chapter, the exploration ventured through the intricate landscape of multi-objective optimization problems (MOOPs) within the domain of data analytics. Commencing a comprehensive overview of multi-objective optimization in Sect. 2, fundamental concepts and challenges related to the simultaneous optimization of multiple conflicting objectives were examined. Subsequently, the journey delved into the realm of MOOPs across diverse domains of data analytics, as outlined in Sect. 3. Real-world examples spanning business, engineering, and scientific contexts illuminated the omnipresence and significance of MOOPs in contemporary decision-making processes.

The narrative took a transformative trajectory in Sect. 4 with the introduction of nature-inspired algorithms (NIAs). Drawing inspiration from the intricate workings of the natural world, these algorithms emerged as formidable tools for addressing MOOPs. In Sect. 5, evolutionary algorithms emerged as pivotal entities within the realm of NIAs. These algorithms, reminiscent of the forces governing natural selection, adaptation, and mutation, have fundamentally reshaped the approach to MOOPs.

As the chapter draws to its culmination in Sect. 6, it marks the threshold of a new era in data analytics. The fusion of evolutionary algorithms and MOOPs has not only augmented the capacity to unearth optimal or nearly optimal solutions but has also facilitated the selection of Pareto-optimal solutions tailored to multifarious real-world requisites and preferences. Through comprehensive explorations, the intricate interplay between these evolutionary methodologies and the intricate, multifaceted challenges posed by MOOPs was unveiled.

In conclusion, the ever-evolving landscape of data analytics and multi-objective optimization has become a novel horizon. As the complexity of decision-making scenarios continues to intensify, the synergy between evolutionary algorithms and MOOPs stands as a guiding beacon. It empowers decision-makers to navigate the labyrinthine landscape of modern trade-offs and uncertainties, endowing them with a versatile toolkit for charting a path forward. While this chapter concludes its journey, the road ahead is illuminated by the promise of discovery, innovation, and transformative solutions for the challenges that loom large in data analytics.

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Chapter 5
Multi-objective Lichtenberg Algorithm for the Optimum Design of Truss Structures

Salar Farahmand-Tabar

1 Introduction

Engineering design challenges are typically intricate and multifaceted, making their solutions highly demanding [1, 2]. In such domains, various objectives, collectively known as multi-objective (MO) goals, are often interconnected, and they frequently conflict with one another, necessitating a trade-off [3, 4]. Consequently, rather than yielding a single optimal solution, these situations result in a set of two or more potential solutions, which is a departure from problems with a single objective function. Optimization is an essential component of system design, aiming to maximize or minimize system outcomes while conserving resources. In the context of structural optimization, these conflicting goals generally involve minimizing costs or weight while simultaneously maximizing strength or performance, presenting a formidable design challenge [5–11]. In recent decades, various deterministic optimization approaches have been proposed to address such design issues. However, these methods face difficulties in finding optimum solutions due to issues such as complex programming approaches, premature convergence, inefficiency in handling nonlinear problems, and challenges related to nonconvex functions [12]. Inaccurate derivative calculations can often mislead optimizers, leading to incorrect solutions and ill-conditioned linear and quadratic programming subproblems.

These shortcomings have prompted many researchers to turn to more gradient-free and robust algorithms such as metaheuristics (MHs) for addressing challenging design problems. MHs are computational techniques that take inspiration from various natural phenomena and offer advantages such as simplicity, minimal parameter requirements, and adaptability. Consequently, these algorithms have found applications across a wide spectrum of industrial and engineering sectors, effectively

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addressing real-world optimization problems [13]. Furthermore, in the last two decades, MHs have gained prominence in solving structural optimization problems with multiple conflicting objectives. In contrast, to single-objective design problems that yield a single solution [14–19], multi-objective design issues typically result in a set of optimum solutions referred to as the Pareto optimum set [20, 21]. Managing a design problem with multiple objective functions generally involves two steps. First, it entails exploring all potential Pareto optimum solutions by a powerful optimizer. Second, it requires selecting one or more solutions from the achieved Pareto optimum set, often involving a decision-making process. Hence, the ideal solution should emerge from the Pareto optimum set while also being in accordance with the designer's preferences [22].

While metaheuristics were initially designed for single-objective optimization and were originally termed evolutionary algorithms, their adaptation for multi-objective optimization can be achieved by leveraging the inherent characteristics of MH searches, which rely on populations or sets of design solutions and incorporate randomization [23]. The nondominated sorting operator, also known as nondominated sorting, plays a pivotal role in the transition of existing single-objective MHs into multi-objective variants. Initially known as multi-objective evolutionary algorithms (MOEAs) and later broadened to encompass diverse theoretical approaches to MH-based search, the first-generation multi-objective metaheuristics (MOMHs) included MO genetic algorithms (MOGA) [24], nondominated sorting GA (NSGAII) [25], and strength Pareto evolutionary algorithms (SPEA2) [26]. Subsequently, numerous MOMHs were developed alongside their single-objective counterparts. Methods for enhancing single-objective metaheuristics to encompass multi-objective capabilities involve the incorporation of nondominance strategies [27], decomposition-based methods [28], elitism strategies [29–32], grid-based approaches [33], preference-based approaches [34], population-based guided methods [35], and opposition-based learning strategies [36–38]. The standout characteristic of multi-objective metaheuristics (MOMHs) is their capacity to navigate through a Pareto front in a single optimization process. However, achieving this entails a potent blend of exploitation and exploration, which becomes increasingly complex when dealing with numerous objective functions, commonly known as many-objective optimization. Consequently, the quest for improved MOMHs remains an ongoing challenge.

Recently, a new metaheuristic, drawing inspiration from natural occurrences such as Lichtenberg Figures (LF) and lightning storms, has been introduced in its single-objective iteration. Pereira and colleagues [39] provide comprehensive insights into the generation of the Lichtenberg algorithm (LA), its validation, and its application for tackling complex single-objective constrained problems. This algorithm has demonstrated success in various applications, including crack identification [40], damage assessment in composites [41], and the optimum design of carbon fiber isogrid lower limb prostheses [42]. The success of LA in its single-objective version can be attributed, in part, to its unique combination of stochastic population-based trajectory search incorporating LF into the optimal solution from the previous iteration. During each iteration, these fractal-based patterns, known as Lichtenberg figures, are triggered with varying sizes and rotations, allowing them to cover the entire search space, from small regions to the entire domain. Points are randomly chosen from this structure and computed in the objective function. This distinctive search strategy holds promise for a multi-objective method.

This chapter proposes the Multi-Objective Lichtenberg algorithm (MOLA) for structural optimization, focusing on two conflicting objectives: minimizing the mass and displacement of the structure. For the performance investigation of the proposed MOLA, two challenging constrained truss structures, 60- and 200-bar truss structures, were investigated in this study. The subsequent sections progressively focus on this domain, initiating with an extensive survey of prior research on multi-objective truss optimization in Sect. 2. This is followed by an elucidation of the multi-objective Lichtenberg algorithm in Sect. 3 meticulously outlining its methodology and application in the optimization framework. Subsequently, multi-objective truss optimization problem in Sect. 4 describes the specific problem statement and objectives addressed within this study. Moving ahead, the analysis of the numerical examples is discussed scrutinizing the results of two distinct truss structures, the 60-bar, and the 200-bar configurations, in Sects. 5.1 and 5.2 respectively. Finally, Sect. 6 remarks the key findings, highlights significant insights from the study.

2 Related Works on Multi-objective Truss Optimization

Table 1 updated from [43] for the context of this chapter, provides a summary of key information related to structural optimization problems. This table encompasses details such as the dimensionality of the problem (planar or space structure), the structural type (truss, tower, bridge, etc.), the specific objective functions targeted, constraints involved, and the algorithms referenced in the reviewed literature. The distinctive innovation highlighted in this paper, compared to the proposals, discussions, and approaches outlined in Table 1, lies in its holistic consideration of objectives, constraints, and automatic member grouping as a unified objective.

In Table 1, W represents the overall weight or mass, f_1 denotes the initial natural vibration frequency, u signifies the maximum displacement, λ^m pertains to the constraint for member m within the structure regarding buckling, σ stands for the permissible stress, and $NCST$ indicates the count of distinct cross-sections. FRF refers to the frequency response function, while $\overline{FRF}$ represents the average value of the crest parameters of FRF concerning the natural vibration frequencies f_1, f_2, and f_3. FT signifies the crest parameter for force transmissibility concerning f_i, and $\overline{FT}$ denotes the mean value of FT crest parameters at f_1, f_2, and f_3. CMA is the average mass under constraints, $SDCV$ represents the variability in constraint violations, RI denotes the index of reliability incorporating RC reliability constraints, TPE relates to the total potential energy, LCC represents the life-cycle costs, GC represents the geometric constraints, and λ_1 is the initial load factor concerning global stability and the elastic critical load.

Table 1 Literature review of multi-objective truss optimization

Structure type	Method	Objective function	Constraints	References
Planar truss	SLDM-NSGA-II	W, u	σ	[44]
	MOPSO	W, u	σ	[45]
	CoSAR	W, RI $W, f_1 + f_2 + f_3$ $W, 1/\sum_{i=1}^{3} FRF(f_i)$ $W, 1/\sum_{i=1}^{3} FT(f_i)$	RC λ^m Ad hoc	[46]
	AMISS-MOP	W, u	σ	[47]
Planar truss bridge	PAES, NSGA-II, SPEA2, PBIL, PSO	$W, 1/f_1$	σ	[48]
Truss tower	SPEA2, PBIL, AMOSA	$M, 1/\sum_{i=1}^{3}(F_i u_i)$	σ	[49]
Space truss	KSR	W, u	σ	[50]
	gM-PAES	W, u	σ	[51]
	MO-CLFBA, MO-FA	W, u	σ	[52]
	CSS	W, u	σ	[53]
	MOAS, MOACS	W, u	σ	[54]
	AICSAMO	W, u	σ	[55]
	MOASOS	W, u	σ	[27]
	GDE3, GDE3 + APM	W, u	σ	[56]
	MOCBO	W, u	σ	[57]
	MOVPS, MOCBO	W, u	σ	[58]
	MOMASOS	W, u	σ	[59]
	MSOGP	W, u	σ	[60]
	MOMHTS	W, u	σ	[61]
	MOHTS	W, u	σ	[62]
	MOWCA, MOEA/D NSGA-II, MODA MOSWCA	Mass, Compliance	σ, u	[63]
	User preference oriented R-GDE3, R-DE3 + APM R-NSGA-II	W, u	σ	[64]
	GDE3	W, f_1 W, λ_1 W, u	σ, λ_1, u σ, f_1, u σ, λ_1, f_1	[43]
	GDE3, SHAMODE, SHAMODE-WO, MM-IPDE	W, f_1, u W, λ_1, u W, f_1, λ_1 W, f_1, u, λ_1	σ, f_1 σ, u σ σ	[65]
	FC-MOPSO	W, u, f_1, TPE W, u, f_1 W, u	σ σ, u σ	[66]

3 Multi-objective Lichtenberg Algorithm

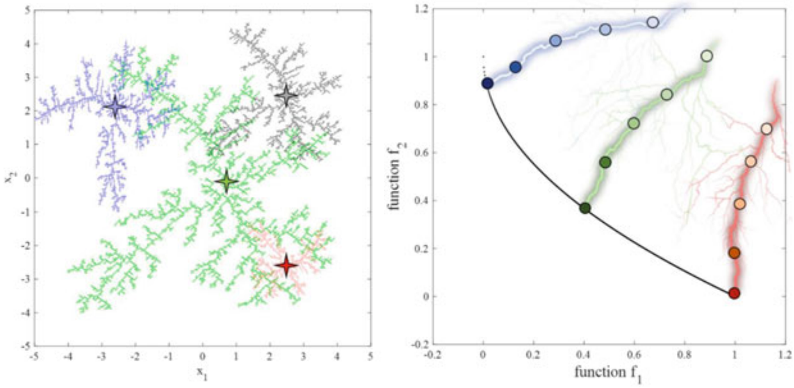
Pereira et al. [39] have recently introduced a novel metaheuristic algorithm called the Lichtenberg algorithm (LA). This algorithm draws inspiration from natural phenomena such as lightning storms and Lichtenberg Figures (LFs). LA has demonstrated its effectiveness in various applications, including damage identification, the detection of cracks in composite materials, and optimum design of carbon fiber-reinforced polymers [40–42]. The LA operation revolves around the creation of LF within the search space, with the points within this figure serving as candidates for evaluating objective function(s).

Lichtenberg initially studied the phenomenon of electric discharges spreading through dielectric (resistant) materials, leading to intricate and branching figures. Because of the inherent variations in materials, it is unfeasible to ascertain resistances at individual points, leading to unpredictable figure growth in each scenario, even when the material and electrostatic conditions are similar.

This random growth process is characterized by numerous physical and mathematical uncertainties, making precise calculations practically unattainable. The direction of lightning propagation depends on a multitude of variables, including air temperature, humidity, pressure, density, soil conditions beneath the cloud, dielectric medium properties, cloud density, particle velocities, particle types, and the presence of oxides, among others. Despite the intricacies involved, the result is a random event that adheres to the path of least resistance. The LF can be created by a stochastic growth process involving numerous particles, creating a cluster. Due to the stochastic nature of the algorithm, each run can yield distinct figures, ultimately resulting in a complete numerical construction of the LF.

The algorithm adopts the growth model based on diffusion limited aggregation (DLA) theory. It creates a binary matrix (comprising 0 s and 1 s), functioning as a map, with an initial particle (denoted by the number 1) fixed at the center. The cluster is built by assigning values of one to the matrix elements within the cluster, while empty spaces are marked with zeros. Each element in the matrix with a value of one signifies the presence of a particle within the cluster, and the initial program input specifies the total number of particles (N_p) in the cluster. This matrix can be visualized as a monochromatic bitmap image. The construction area of the figure is determined through the creation radius (R_c), and the dimensions of the matrix are configured to be twice the R_c for both rows and columns.

Particles are distributed randomly throughout the matrix, and when they come into contact with the cluster (initially consisting of a single particle at the center), they have a probability of adhering, known as the stickiness coefficient (S). This coefficient regulates the cluster's density. Particles move in a random manner, both in a radial direction and as if they were on a Cartesian plane from the center, ultimately settling in positions that correspond to matrix elements with rows and columns. A particle can only be introduced if it lands adjacent to another particle, a condition verified by a lateral check. If a particle reaches a radius just slightly larger than R_c, it is removed, and a new particle starts its random trajectory. This sequence continues



(a) (b)

Fig. 1 Fundamental exploration approach of MOLA within the design and objective domains [39]

until all the predefined particles (N_p) are incorporated into the cluster or until it reaches its construction limit.

Each clustered particle can be mapped to coordinates on a Cartesian plane, enabling the LF to be plotted in varying sizes, orientations, and starting positions. During each iteration, the LF can be depicted with various rotations and sizes that are randomly chosen. This approach enhances the algorithm's capabilities for both exploration and exploitation while averting biased assessments of the search space. In the optimizer, there is another parameter called refinement (ref), a customizable input that falls within the range of 0–1. This parameter generates a secondary Lichtenberg Figure (red) in every iteration, with its size scaling from 0 to 100% of the size of the primary Lichtenberg Figure (blue), as depicted in Fig. 1.

The "ref" parameter, which corresponds to a smaller-scale figure, is introduced to enhance the local search. When ref is set to 0, only one Lichtenberg Figure (LF) influences the optimizer in each iteration, depicted in blue, representing the global LF. It is important to note that the objective function(s) in MOLA are not computed using every LF (Lichtenberg Figure) point. Instead, the population parameter, which is established at the initiation of the algorithm, commonly equals ten times the number of design variables in the problem. These LF points, forming the population, are carefully selected from within the evolving LF structure, which undergoes changes in each iteration. These population points are visually represented as black dots and are rigorously validated to ensure that they always remain within the search space. This unique approach gives rise to a hybrid algorithm in LA, combining characteristics of two algorithmic types commonly found in the literature: trajectory-based and population-based. This hybrid aspect, absent in any metaheuristic algorithms introduced in the single-objective version, significantly enhances both exploration and exploitation capabilities, revealing substantial potential for multi-objective methods.

The switching factor is another parameter denoted as M, governing the LF in the input data of the optimizer. M can assume values of 0, 1, or 2. When M is set to

0, a figure is created at the program's outset and remains consistent throughout all iterations. When M equals 1, a new figure is created in each iteration. Finally, M set to 2 employs a previously stored figure in the optimizer, obviating the need for new figure generation.

This algorithm supports the creation of both 2D and 3D Lichtenberg figures, rendering it suitable for problems involving two or three decision variables. For cases involving more variables, the algorithm employs a projection or mirroring technique to extend the figures to accommodate n variables. The number of iterations (N_{iter}) is a configurable parameter in the LA, typically initialized to 500 iterations.

Every point assessed within the search space yields solutions within the objective space that are subsequently compared through principles of the Pareto dominance. Solutions that remain undominated by others remain in the solution set, while those that are dominated are removed. The set of nondominated solutions in each iteration constitutes the present Pareto front of the problem, progressively converging toward the true Pareto front over consecutive iterations.

In the MOLA, all points present on the current Pareto front are considered as potential points for LF generation. One of these points is chosen randomly to produce an LF in each iteration. This approach facilitates targeted exploration in the variable space, focusing on regions that exhibit superior objective function values, as illustrated in Fig. 1. The recommended parameter ranges for MOLA are provided in Table 2. The pseudocode of MOLA and LF creation are presented according to Algorithms 1 and 2.

Table 2 Recommended parameters for LA [39]

Parameters	Any dimensions	Three-dimensional
R_c	50–200	50–150
N_p	$> 10^3$ and $< 10^6$	$> 10^5$ and $< 10^6$
S	0 to 1	0 to 1
pop	(10 a 40)$\times$ d	(10 a 40)$\times$ d
ref	0–1	0–1
M	0, 1 or 2	0 or 1
N_{iter}	$> 10^2$ and $< 10^3$	$> 10^2$ and $< 10^3$

Algorithm 1 Pseudo-Code of MOLA

Set search space and objective functions– J_i, lower & upper bounds
Set number of populations, iterations, and parameters– N_{iter}, $Pop, LF–Ref$, M
Set LF Parameters – R_c, N_p, S
if $M = 0$, load LF, end if
if $M = 1$, Generate the LF, end if
while ($iter < N_{iter}$) do
if $M = 2$, Generate the LF, end if
$X_{trigger}$ = center of the search space
if $ref = 0$
Implement random scaling and rotation
Generate random population using LF, X_i ($i = 1, 2, \ldots, Pop$)
else
copy LF to generate a second LF of size $ref * LF$ ($Local$)
Implement the same random scaling and rotation
Generate local and global random population using LF,
$Xglobal_i$ ($i = 1, 2, \ldots, 0.4 * Pop$), $Xlocal_j$ ($j = 1, 2, \ldots, 0.6 * Pop$)
$X_i = Xglobal_i + Xlocal_j$
end if
Compute J_i for each X_i of the problem
Find nondominated and dominated solutions using J_i
Generate the current Pareto front considering the nondominated solutions
Store nondominated solutions during iterations and eliminate the others
Randomly choose a nondominated solution & fitness vector (X_{ND})
$X_{trigger} = X_{ND}$
$Iter = iter + 1$
end while
return Pareto front

Algorithm 2 Lichtenberg Figure generation

Generate a zeros-sized matrix of R_c and position a unitary particle at its center
While $(i < N_p)$ do
  Place a unitary particle randomly within the matrix
  if the plotted particle t is adjacent to another unitary particle
    if $rand < S$
      A new unitary particle is positioned in the matrix; $i = i + 1$
    else; The initially placed (plotted) unitary particle is removed
    end if
  end if
  if the cluster of unitary particles reaches R_c
    The simulation is terminated
  end if
end while
$X =$ coordinates of all unitary particles in Cartesian space.

4 Multi-objective Truss Optimization Problem

In essence, practical engineering design problems [67] can be effectively represented as multi-objective problems characterized by inherent conflicts among objectives. Take, for instance, the trade-off between enhancing structural strength, which impacts both structural mass and its cost. Moreover, these design challenges typically exhibit multimodality, nonlinearity, and often entail vast search spaces, rendering them inherently complex. The design complexity of such problems arises from the multitude of potential solutions, in contrast to the single optimum solution typically found in single-objective problems. Designers are faced with the daunting task of selecting a single solution from the optimum solutions set, known as the Pareto optimum set, based on predefined criteria—a task that is itself highly challenging.

Furthermore, in many instances, these Pareto optimal solutions remain largely unexplored for a given design problem. Hence, there is a crucial need to seek out multiple optimal solutions rather than confining the design process to a single solution. The mathematical representation for multi-objective problem of truss optimization is typically formulated as follows:

$$Find\ Z = \{Z_1, Z_2, ..., Z_m\}$$

$$to\ optimize\ the\ structural\ mass\ and\ maximum\ nodal\ deflection$$

$$minimize\ \ f_1(Z) = \sum_{i=0}^{m} Z_i \rho_i L_i\ and$$

$$
\begin{aligned}
&maximize\ f_1(Z) = max\left(\left|\delta_j\right|\right) \\
&Subjected\,to: \\
&Behavior\ constraints(\text{Stress}): g(Z): |\sigma_i| - \sigma_i^{max} \leq 0 \\
&Construction\ constraints(\text{Cross} - \text{sectional area}): \\
&Z_i^{min} \leq Z_i \leq Z_i^{max}
\end{aligned} \tag{1}
$$

In this context, Z_i represents a design variable vector, ρ_i stands for the density of the elements' mass, and L_i denotes their length. Additionally, Ei and σi represent the module of elasticity and stress of the i th element, respectively. The permissible lower and upper limits are denoted by the superscripts max and min, respectively.

When dealing with constrained problems, a highly efficient strategy involves converting these design complexities into dynamic unconstrained optimization problems utilizing a penalty method.

$$
f(X) * (1 + \varepsilon_1 * \complement)^{\varepsilon_2}, \complement = \sum_{i=1}^{m} C_i, C_i = |1 - \frac{|\sigma_i^{max} - \sigma_i|}{\sigma_i^{max}}| \tag{2}
$$

Users have the flexibility to define the values of the parameters ε_1 and ε_2, with both of them being set to 3 in this case.

5 Results and Discussion on Numerical Examples

The performance assessment of the algorithms under consideration involves their application to two complex structural examples [59–62]. Additionally, the evaluation involves the utilization of standard multi-objective convex test functions to gauge the search efficiency of the algorithms. For this study, 50,000 fitness evaluations and a population size of 100 are employed, in line with previous research practices. To obtain results, each algorithm undergoes 10 independent runs.

Four performance indicators, hyper volume (HV), generational distance (GD), inverted generational distance (IGD), and space-to-extend (STE), are evaluated to measure the algorithm's performance in the search process. Lower values across all indicators denote better performance, except for HV, where higher values signify better results.

5.1 The 60-Bar Truss Structure

To evaluate the algorithm's performance, the first example focuses on the 60-bar truss structure, as illustrated in Fig. 2. Detailed design specifications can be found in Table 3. Consistent with previous studies, the structural elements in this example are grouped into 25 members.

The figures in this study play an important role in illustrating the effectiveness of the multi-objective truss optimization algorithm (MOLA) compared to the NSGAII in solving complex multi-objective truss optimization problems. Figure 3 showcases the best Pareto fronts, which have been evaluated based on two key performance indicators: the HV and IGD.

First, let us focus on the Pareto fronts depicted in Fig. 3. These fronts provide a visual representation of the trade-offs between conflicting objectives for both MOLA and NSGAII. What immediately stands out is that the Pareto front resulting from the joint consideration of HV and IGD for MOLA appears superior when compared

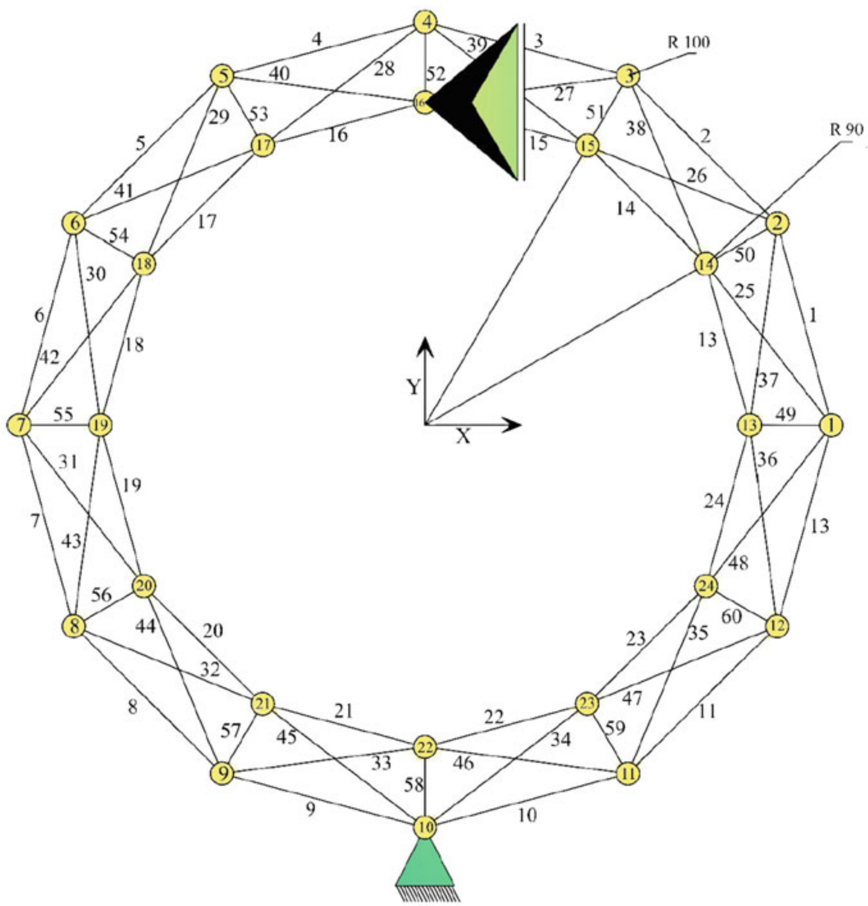


Fig. 2 The 60-bar ring truss

Table 3 Considerations for designing the 60-bar truss

Design variables	$A_{i,} i = 1, 2, \ldots, 25$
Constraints	$\sigma_{max} = 40\, ksi$
Density	$\rho = 0.1\, lb/in^3$
Young modules	$E = 104\, ksi$
Size variables	$A_{i,} \in S, i = 1, 2, \ldots, 25$ $S = [0.5, 0.6, 0.7, \ldots, 4.9] in^2$
Loading conditions	***Case 1***: $P_{x1} = -10$ *Klb*, $P_{x7} = 9$ *Klb* ***Case 2***: P_{x15}=$P_{x18} = -8$ *Klb*, P_{y15}=$P_{y18} = +3$ *Klb* ***Case 3***: $P_{x22} = -20$ *Klb*, $P_{y22} = 10$ *Klb*

to the Pareto front obtained with NSGAII. This suggests that MOLA is more effective in optimizing the 60-bar truss structure when considering multiple, potentially conflicting, objectives.

Taking a closer look at the performance indicators presented in Fig. 3, several noteworthy observations can be made. The HV values, which are a measure of the volume covered by the Pareto front, are significantly higher for MOLA than for NSGAII. This implies that MOLA achieves a broader coverage of the Pareto front, indicating a higher diversity of solutions, which is often a desirable trait in multi-objective optimization.

In addition to HV, the values of the generational distance (GD) and space-to-extend (STE) are lower for MOLA when compared to NSGAII. A lower GD signifies that the solutions found by MOLA are closer to the true Pareto front, highlighting the ability of algorithm to generate solutions that are not only diverse but also closer to the optimal trade-off between objectives. A lower STD indicates that the solutions produced by MOLA are more consistent, with less variation, which is crucial for reliable optimization.

Collectively, these results underscore the exceptional performance and reliability of the MOLA in handling the multi-objective optimization problem of the 60-bar truss structure. The ability of MOLA to produce a superior Pareto front, combined with higher HV, lower GD, and STD values, demonstrates its effectiveness in providing well-rounded solutions that balance multiple objectives effectively.

5.2 The 200-Bar Truss Structure

The 200-bar truss (Fig. 4) consists of bars organized into 29 groups. The members have elasticity modulus and density values of 30,000 ksi and 0.283 lb/in3, respectively. The optimization problems are subject to constraints related to stresses, which are restricted to $\pm$ 10 ksi.

The following load cases are considered:

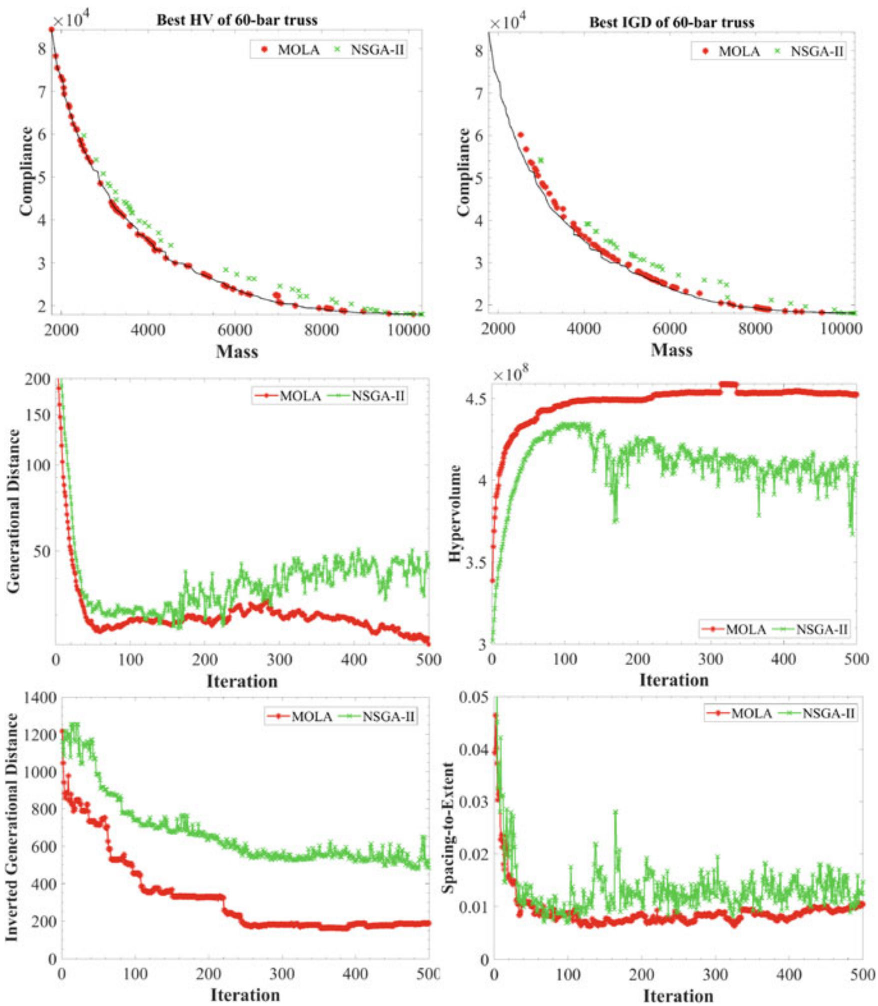


Fig. 3 Optimum result of the 60-bar truss: Pareto fronts and performance indicators

- A 1.0 kips load was applied in the positive x-direction at nodes 1, 6, 15, 20, 29, 34, 43, 48, 57, 62, and 71.
- A 10 kips load was applied in the negative y-direction at nodes 1, 2, 3, 4, 5, 6, 8, 10, 12, 14, 15, 16, 17, 18, 19, 20, 22, 24, …, 71, 72, 73, 74, and 75.
- Load cases (1) and (2) acting simultaneously.

The lower and upper bounds for the continuous case of the cross-sectional areas are determined within the range of 0.1–20 in^2 which can be discretized into the set {0.1, 0.15, 0.2, 0.25, 0.3, …, 20} in^2, resulting in 399 options. In the dynamic case, a non-structural mass of 220.458 lb is applied to nodes 1 to 5.

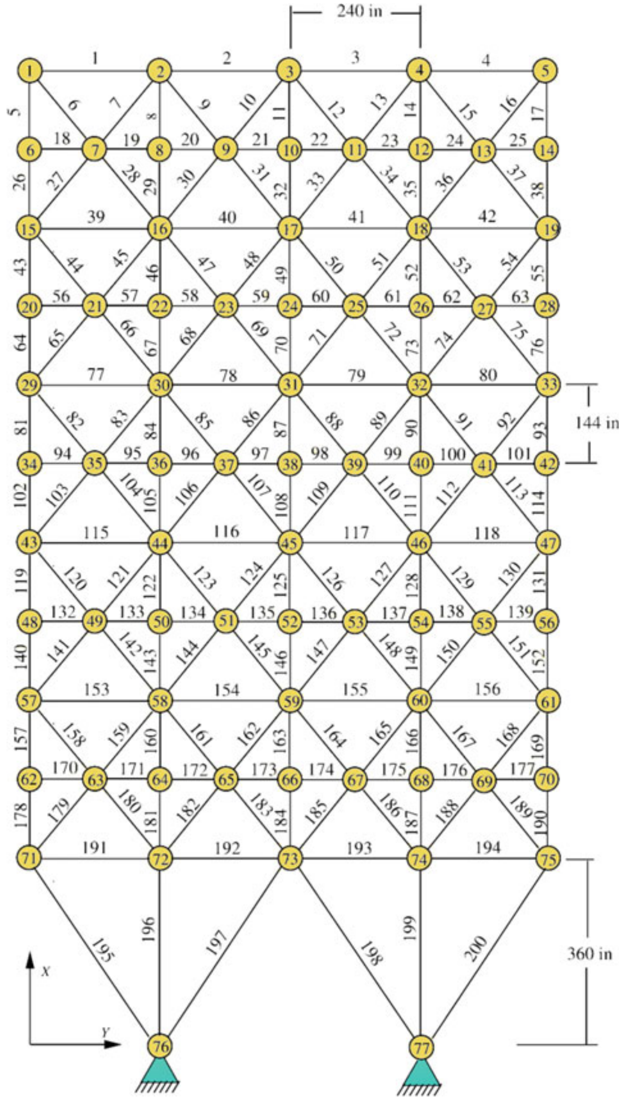


Fig. 4 The 200-bar truss

Figure 5 displays the optimal Pareto fronts derived from the multi-objective truss optimization problems concerning the 200-bar truss, based on evaluations using the HV and IGD indicators. Additionally, Fig. 5 presents a comprehensive analysis of the performance indicators. Notably, the results highlight that when both HV and IGD are considered, the Pareto front achieved by the MOLA surpasses that attained by the NSGAII.

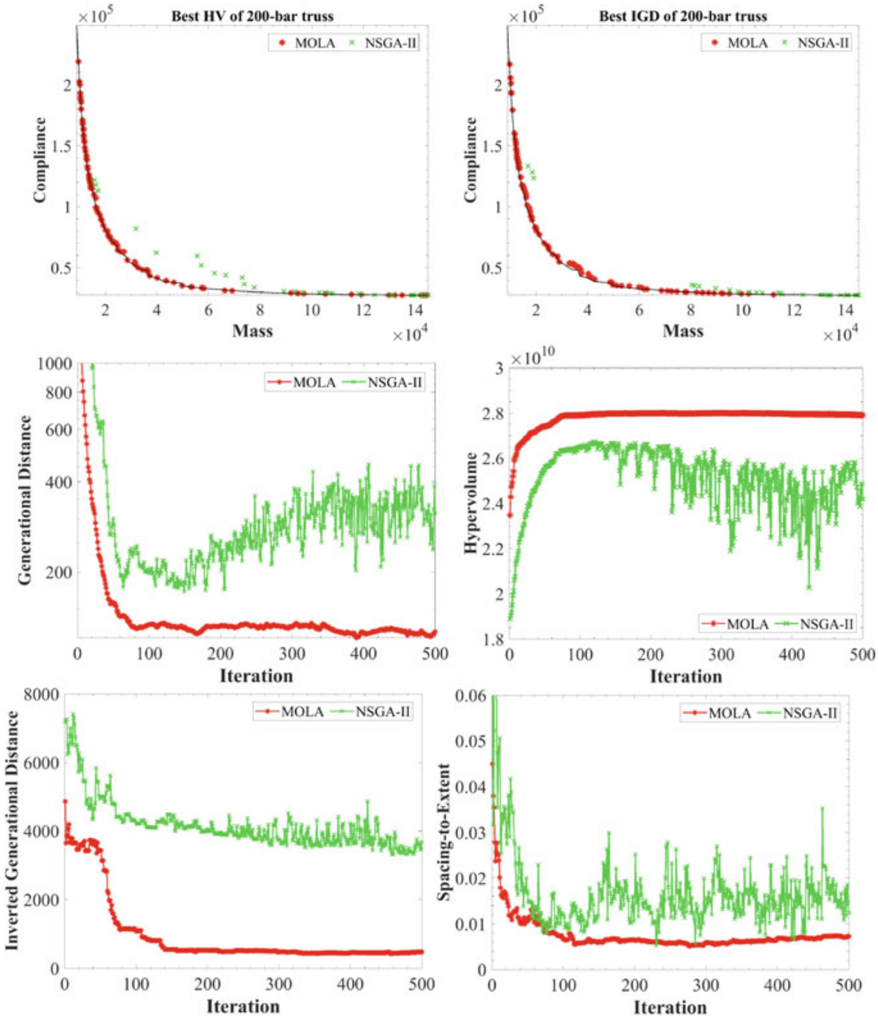


Fig. 5 Optimum result of the 200-bar truss: Pareto fronts and performance indicators

A thorough examination of the performance indicators reveals compelling insights. Specifically, MOLA demonstrates significantly higher HV values compared to NSGAII. This implies that MOLA excels in providing more extensive coverage of the Pareto front, which is indicative of a diverse range of solutions—a key characteristic in multi-objective optimization endeavors.

Furthermore, focusing on the GD and STE values, it is evident that MOLA outperforms NSGAII in both respects. The lower GD values for MOLA indicate that its solutions are closer to the true Pareto front, emphasizing its capacity to generate solutions that offer a superior trade-off between objectives. Additionally, the reduced STE values signify that solutions of MOLA are more consistent and exhibit less variability, underscoring its reliability.

These results signify the remarkable performance and reliability of the MOLA when applied to the intricate multi-objective optimization challenges associated with the 200-bar truss structure. MOLA not only excels in producing an enhanced Pareto front but also achieves higher HV values while maintaining lower GD and STD values. These findings establish MOLA as a robust and effective tool for addressing the complexities of multi-objective optimization in the realm of structural engineering, particularly for larger-scale truss systems.

6 Conclusions

The study introduced and explored an optimization algorithm known as the Lichtenberg algorithm for the optimization of truss structures considering multiple objectives. MOLA draws inspiration from natural phenomena of lightning storms and the creation of Lichtenberg Figures within the search space, with the points within this figure serving as candidates for evaluating objective function(s). To assess the algorithm's performance in the multi-objective cases, it was applied to two challenging structural optimization problems, featuring design variables that align with practical engineering constraints. The results achieved from the MOLA were compared with those of well-established optimization algorithms. Performance indicators such as the hypervolume, generational distance, and spacing-to-extent of the Pareto front, were used to assess the quality and diversity of the nondominated solution set. The results revealed the superiority of MOLA. Overall, MOLA demonstrated notable improvements in terms of hypervolume and spacing-to-extent of the Pareto front. These results signify the effectiveness of MOLA in efficiently addressing large, real-world, and intricate optimization problems within the field of structural engineering.

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Chapter 6
Performance Analysis of a Multi-objective Function-Based PID Controller for System Frequency Regulation

D. Boopathi, K. Jagtheesan, Sourav Samanta, and Kanendra Naidu

1 Introduction

In recent days, people have moved towards modern life, which refers to a well-developed and equipped environment. The equipped environment needs more electric power. The population is also a factor in increasing electricity demand. That demand should be balanced by enhancing plant capacity and the number of plants throughout the country. When the power system is under a balancing process, it creates more quality issues. The power quality includes frequency deviations and voltage fluctuations. There are many methods to address the voltage oscillation. However, there are very few methods for frequency damping control, and the most popular is load frequency control (LFC). With the support of the secondary controller, the LFC can be implemented in the power system. The performance of the secondary controller has been improved by adopting various optimization techniques. Several researchers are handling the LFC with the support of a tuned secondary controller. A detailed literature review is conducted to analyse the exact need for the work.

Weighted sum of ITAE and ISTE objective function-based ABC tuned PID controller designed for the LFC of an interconnected thermal power network [1]. An equal three-area interconnected power network has been investigated for LFC with the atom search algorithm-based FOPID controller. The preeminence of the ASO–FOPID controller was proven with the GA, GWO, and SCA methods. ITAE,

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ISE, ITAE, and IAE cost functions were involved in the optimization [2]. An interconnected thermal and PV power plant is examined by various classical controllers that are tuned with the Jaya optimization algorithm [3]. The ITAE cost function-based slime mould algorithm (SMA)-optimized PI controller implemented in an interconnected thermal and PV power network is investigated for LFC. The superiority of the SMA–PI controller is validated by comparing the result with the WOA, GA, and FA-PI controller [4]. PV composed of two areas of thermal power plants was examined by ITAE objective function-based modified whale optimization. As a result, the MWOA response was better than that of the classical PID controller [5].

An isolated multiple-source power system comprised of thermal, wind, solar, and energy sources has been investigated for LFC. PSO-PID controller employed as a subordinate controller for this system. The efficacy of the PSO-PID controller was shown by a comparative analysis of its outcomes against those of conventional I, PI, and PID controllers [6]. A two-area multisource power grid was examined by a PSO-tuned fuzzy logic controller for LFC. IAE, ITAE, and ITSE objective functions are involved in the process of tuning the gain parameters. The supremacy of the FLC is shown by the comparison of PID and FLC controller results [7]. An isolated multisource power grid was investigated for LFC by adopting a PSO-PID controller. The exact enhancement and the PSO-PID controller supremacy have been proved by validating the results of GA, conventional, and DE algorithms [8]. An ALO-based PI controller was utilized for LFC for an interconnected power system [9]. MA-tuned PID controller designed for a multimicrogrid power system LFC. The results of the GA, DE, and PSO techniques were compared with the proposed method to prove supremacy [10].

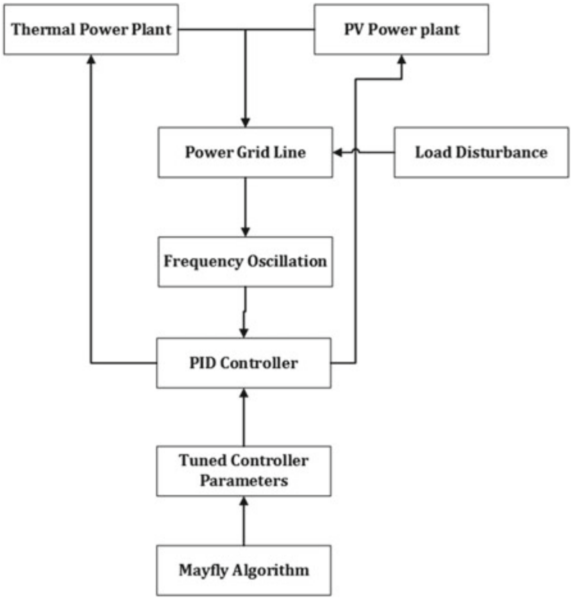
A single-area zero carbon power plant combined with energy stowage units is tested for LFC by using the ACO-PID controller. The exact supremacy of the ACO-PID controller was proven with the results of the PSO technique [12, 13]. ACO-tuned PID controllers have been implemented for single-area nuclear [14] and integrated nuclear power systems [15] to perform LFC. The ITAE objective function-based multi-sever optimizer (MVO) is implemented to optimize the MPC for a multisource power system as a secondary controller. The efficiency of the MPC has been proven by comparing the results of the genetic algorithm and intelligent water drop methods [16]. A grey wolf optimization tuned PI−(I + DD) controller was developed to perform LFC in three area thermal power plants with PV. The proposed controller performed 40–45% better than other controllers [17]. An interconnected thermal and PV power plant system was investigated for frequency regulation with the support of a secondary controller (PID), which is tuned by the PSO-optimized Ziegler Nichols method. The result was compared with PI, fuzzy, and fuzzy PID controllers [18]. Chaos game optimization-based PID controller developed for two area thermal + PV plant LFC [19]. The gravitational search algorithm tuned active disturbance-rejection control was designed for two area thermal power plants with PV and energy storage units to execute LFC [20]. A detailed report of the survey is shown in Table 1.

Table 1 Summary of survey

References	System	Controller	Objective function	Technique
[1]	Two area—thermal	PID	ISTE + ITAE	ABC
[2]	Three area—thermal + hydro + gas	FOPID/PID	IAE, ISE, ITAE, and ITSE	Atom search algorithm
[3]	Two area—thermal + PV	I, PI, PID, IDD	ITAE and IAE	Jaya optimization
[4]	Two area—thermal + PV	PI	ITAE	Slime mould algorithm
[5]	Two area—thermal + PV	PID	ITAE	Modified whale optimization algorithm
[6]	Single area—thermal + solar + wind	PID	ITAE	PSO
[7]	Two area—thermal + gas + wind + PV	FLC	IAE, ITAE, and ITSE	PSO
[8]	Single area—thermal + gas + hydro	PID	IAE, ISE, ITAE, and ITSE	PSO
[9]	Multi area—thermal + hydro + PV	PI	ITAE	Antlion optimizer
[10]	Microgrid—thermal + PV + wind	PID	ITAE	Mayfly algorithm
[12]	Single area—wind + ESU	PID	ITAE	Ant colony optimization
[16]	Single area multi source	MPC	ITAE	Multi verse optimizer
[17]	Three area—thermal + PV	PI-(I + DD)	ITAE	Grey wolf optimizer
[18]	Two area—thermal + PV	PID	ITAE	PSO + ZN
[19]	Two area—thermal + PV	PID	ITAE	Chaos game optimization
[20]	Two area—thermal + PV + Energy storage units	Active disturbance rejection control	ITAE	Gravitational Search Algorithm

The review identifies the research gap in the field of study. From this review, there is a need for renewable energy sources (RES) to replace the conventional method of electric power generation. The penetration of RES has been high in recent days for generating electric power generation. In particular, PV is the most common because of the easily available and mass production of electric power. In this chapter, the

Fig. 1 Graphical representation of the chapter



void is supplied by harmonizing a thermal power plant's power demand with that of a PV plant. The proposed PV power plant is incorporated with a conventional thermal power plant. The proposed chapter expresses the study of the weighted sum of the PID controller against frequency fluctuation at sudden load variation in the power grid. The graphical representation of the chapter is given in Fig. 1.

1.1 Organization of the Chapter

The organization of the proposed chapter has 6 sections, including the introduction and conclusion. The details of the sections are as follows:

Section 1 deals about introduction and literature review. System of investigation is discussed in Sect. 2. The proposed optimization algorithm and tuned controller gain parameters are in Sect. 3. Section 4 illustrates the controller structure and objective function utilized in the chapter. The performance discussion of the proposed controller is elaborated in Sect. 5.

Section 6, about the conclusion of the chapter with results.

1.1.1 Significance of the Chapter

A balanced non-polluting (solar)-incorporated integrated power network was constructed for investigation. A popular secondary controller (PID) is employed as a secondary controller in the proposed system for LFC. The Multi-objective function-based computational method is utilized for tuning the PID controller and the result of the MA-tuned weighted sum of PID is validated with MA-tuned PI and PID controllers for supremacy.

2 System of Investigation

A coal-based power plant integrated with a PV unit is taken for investigation for frequency regulation analysis of the proposed secondary controller [11, 18, 19]. The Laplace transfer functions of the proposed system are given in Eqs. 1–6 [11]. The conventional method of electricity production of thermal power plants is one of the noticeable causes of environmental pollution. To reduce pollution and overcome the scarcity of coal, renewable sources are introduced in the proposed system (Fig. 2).

$$\text{Thermal steam governor} = \frac{1}{1 + sT_g} \tag{1}$$

$$\text{Reheater} = \frac{1 + sK_r}{1 + sT_r} \tag{2}$$

$$\text{Steam turbine} = \frac{1}{1 + sT_t} \tag{3}$$

$$\text{Power generator} = \frac{K_p}{1 + sT_p} \tag{4}$$

$$\text{Tie line} = \frac{2\pi T_{12}}{S} \tag{5}$$

$$\text{PV power plant with MPPT} = \frac{a + sb}{s^2 + sc + d} \tag{6}$$

3 Controller Approach with Objective Function

The PID controller demonstrates effective control over the system frequency and tie-line power flow inside the examined multiarea power system. Under nominal loading circumstances, each location within the system receives and sustains its designated load capacity while adhering to the specified constraints of the power line. During sudden loading conditions, the system stability is affected. To address this issue, the PID controller has been implemented in the power system as a secondary controller. The PID controller comprises three distinct controllers, namely, the proportional

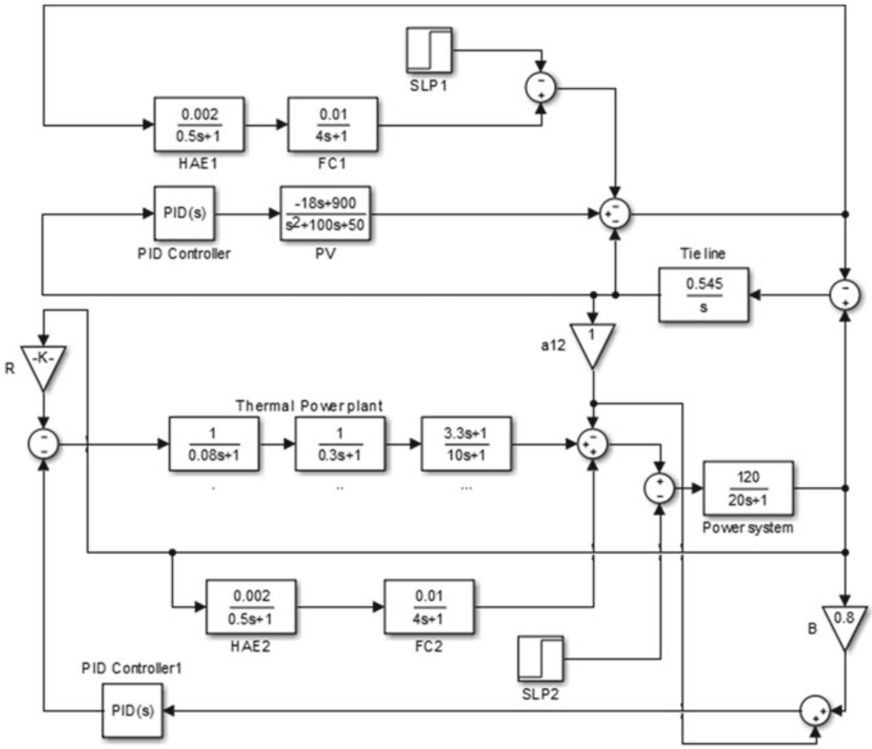


Fig. 2 Laplace function model of the proposed system

controller, integral controller, and derivative controller. Concerning the error values, the controller produces the control signal to the system. In this chapter, each controller produces different control signals as follows:

$$U_1 = K_{p1} \cdot ACE_1 + K_{i1} \cdot \int ACE_1 dt + K_{d1} \cdot \frac{dACE_1}{dt}$$

$$U_2 = K_{p2} \cdot ACE_2 + K_{i2} \cdot \int ACE_2 dt + K_{d2} \cdot \frac{dACE_2}{dt}$$

The gain levels of the proportional (P), integral (I), and derivative (D) controllers are denoted as K_p, K_i, and K_d, respectively. The conventional control structure of the proportional-integral-derivative (PID) controller is shown in Fig. 3. The ITAE goal function is used in this chapter to optimize the controller. The objective function is represented by Eq. 7. The symbol 'J' is used to represent the performance index, whereas the variable 't' specifies the simulation duration. [21, 22]

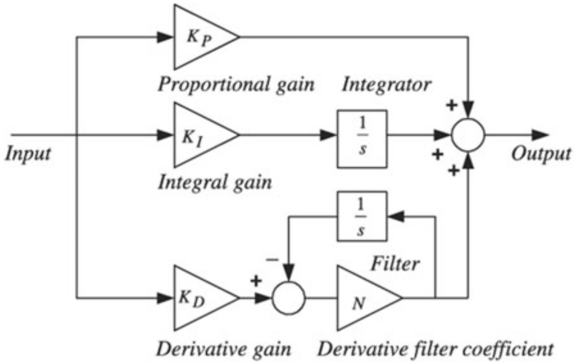


Fig. 3 Classical structure of the PID controller

$$J = \int_0^t t.|ACE|dt \tag{7}$$

4 Mayfly Algorithm

Zervoudakis and Tsafarakis introduced the Mayfly algorithm for several optimization problems. Mayflies are classified as insects belonging to the taxonomic order Ephemeroptera, which is a subdivision of the ancient taxonomic group Palaeoptera. There are approximately 3000 species of mayflies worldwide. Their appellation derives from the fact that they are most prevalent in May in the United Kingdom. After budding from the egg, mayfly nymphs are visible to the unaided eye and spend several years developing as aquatic nymphs before they are ready to emerge as adults. The adult may fly start their mating process with nuptial dance. Male flies follow the female and dance with its tail, and the atterrated female flies' mate with the male fly. After completing the mating process, the female flies drop eggs in water. The operation flow of the MA is given in Fig. 4.

The may fly algorithm is summarized as follows:

Step 1: Initialization:

Position initialization of male $x_i = [x_1, x_2 \ldots x_n]$, female position $y_i = [y_1, y_2, \ldots, y_n]$ and its corresponding velocity values $v_i = [v_1, v_2, \ldots v_n]$.

Step 2: Male movement:

The position of each may fly is adjusted to reach its best position P_{best} and neighbors' best position $g_{best.}$

Step 3: Female movement:

The adjusted location is determined based on the Cartesian distance between males and females.

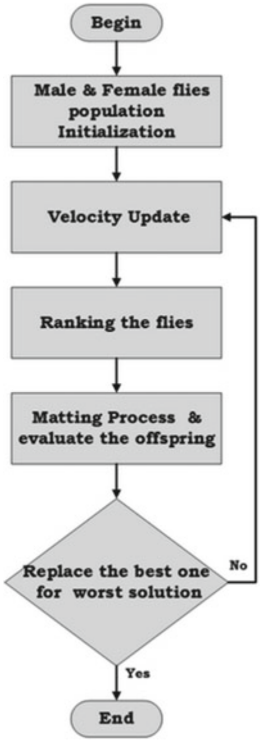


Fig. 4 Operational flow of MA

Step 4: Mayfly mating:

After completion of the mating process between male and female flies, it bordures two types of off springs as follows

Off spring 1: L*male + (1-L) * female.

Off spring 2: L*female + (1-L) * male.

where L denotes the random number of males and females.

Step 5: Update solutions:

After completion of the above process, the solution is generated in which sources are replaced by the new best source. The respective values of g_{best} and p_{best} are updated. The process was repeated until it reached its stop conditions.

The MA algorithm is effectively used for fine-tuning the gain parameters of the controller. The resulting optimized values are documented in Tables 2 and 3.

Table 2 Optimized classical controller parameters

Parameters/Controller	K_{P1}	K_{I1}	K_{D1}	K_{P2}	K_{I2}	K_{D2}
PI	0.2457	0.0177	–	0.99	0.1387	–
PID	0.3816	0.0924	0.8094	0.99	0.99	0.100

Table 3 Optimized weighted sum of PID controller parameters

Weight 1	Weight 2	K_{P1}	K_{I1}	K_{D1}	K_{P2}	K_{I2}	K_{D2}
1	0	0.381287	0.092712	0.806761	1	1	0.103501
0.9	0.1	0.381293	0.092712	0.806776	1	1	0.103492
0.8	0.2	0.387133	0.094451	0.82087	1	1	0.102591
0.7	0.3	0.381272	0.092708	0.806727	1	1	0.103523
0.6	0.4	0.381264	0.092706	0.806702	1	1	0.10354
0.5	0.5	0.38146	0.09281	0.806574	1	1	0.104436
0.4	0.6	0.381213	0.092698	0.806575	1	1	0.103616
0.3	0.7	0.376718	0.091239	0.795404	1	1	0.106468
0.2	0.8	0.370593	0.089251	0.781498	1	1	0.10906
0.1	0.9	0.362459	0.086763	0.764611	1	1	0.112456
0	1	0.198427	0.045971	0.402008	1	1	1

5 Outcome and Discussion

The optimized parameters are used in the controllers, and the performance is analysed in this section. This section has two different analysis cases 1, about the response analysis of the weighted sum of PID in various weights. Case 2 is the response analysis of MA-tuned classical controllers (PI and PID) with a weighted sum PID controller.

5.1 Case 1: Performance Analysis of Weighted Sum of PID Controller

The proposed weighted sum of PID controller parameters is tuned by the MA with various weights. Each weight is provided a different response against the frequency oscillation. By comparing a few of the weighted parameters, the best and finest weighted sum of the parameters is identified in this section. Figures 5, 6 and 7 refers to the performance comparison of the weighted sum of the PID controller results. The tabulated data in Table 4 present the nominal values derived from the graph. Del F1 are del F2 represented frequency variation in areas 1 and 2. The controlled parameters are Ts, US, and UO, denoted as settling time, peak under, and overshoot, respectively.

The performance of the weighted sum of the PID controller has been tested with three different controller gain parameters with a weighted sum. The results of the comparison are shown in Figs. 5, 6 and 7 and Table 4. In the different weighted sums of the controller, the W1 = 1 and W = 0 sum of the controller gain value is given a better response than another weighted sum. The bar chart (Figs. 8, 9 and 10) is provided for the conformation of the supremacy of the W1 = 0 & W = 1 weighted.

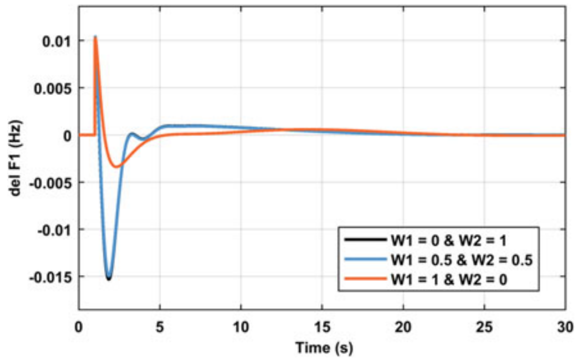


Fig. 5 delF analysis in area 1 for different weights

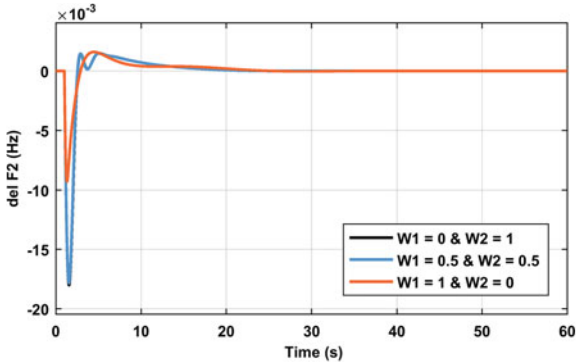


Fig. 6 del F analysis in area 2 for different weights

Table 4 Controlled parameters in the frequency oscillation from the weighted sum of the PID controller

Weight 1	Weight 2	Del F1			Del F2			Tie line		
		T_s(S)	US (Hz) 10^{-3}	UO (HZ) 10^{-3}	T_s(S)	US (Hz) 10^{-3}	UO (HZ) 10^{-3}	T_s(S)	US (Hz) 10^{-3}	UO (HZ) 10^{-3}
0	1	24	15.5	10.4	27	18	1.5	25	4	0.2
0.5	0.5	24	15	10.4	27	17.8	1.5	24	4.1	0.2
1	0	22	3.5	10.2	25	9	1.6	23	4.8	0.5

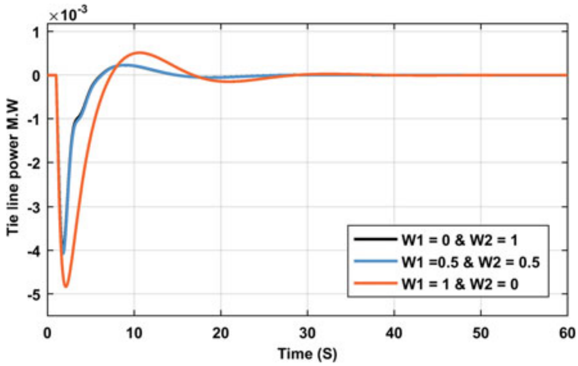


Fig. 7 del P analysis in tie line for different weights

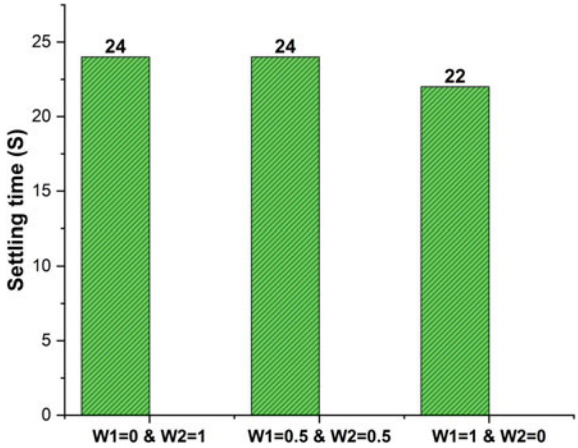


Fig. 8 Column chart for del F1 from weighted sum of PID controller

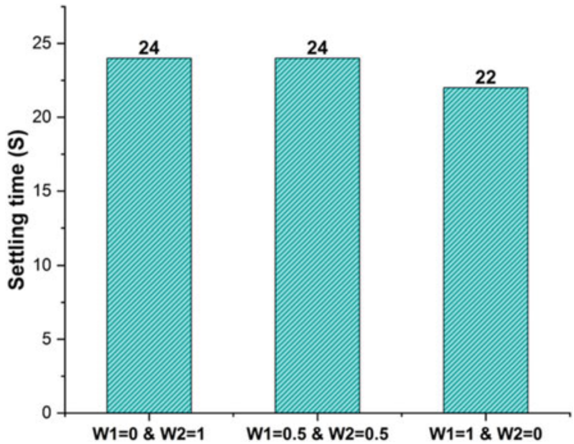


Fig. 9 Column chart for del F2 from the weighted sum of the PID controller

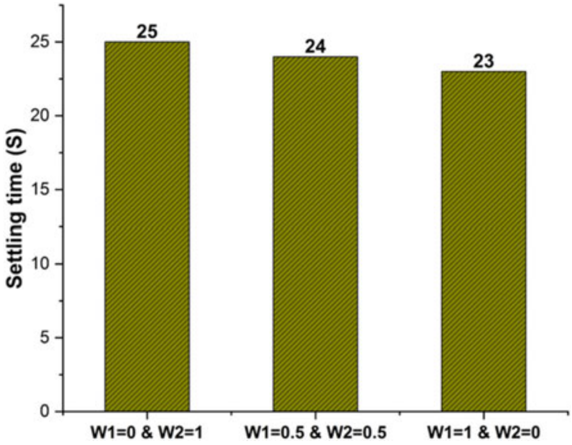


Fig. 10 Column chart for the del tie line power from the weighted sum of the PID controller

5.2 *Case 2: Performance Analysis of Weighted Sum of PID Controller with Classical Controller*

In Sect. 5.1, the analysis demonstrates that the PID controller has superior performance in terms of rapid settling and minimum peak values when compared to other controllers. In this section, the selected weighted sum of PID controller gain values is validated with the classical PI and PID controllers, which are tuned by the MA. The comparison helps to conclude the work. The comparison of results is shown in Figs. 11, 12 and 13, and the corresponding nominal values are published in Table 5.

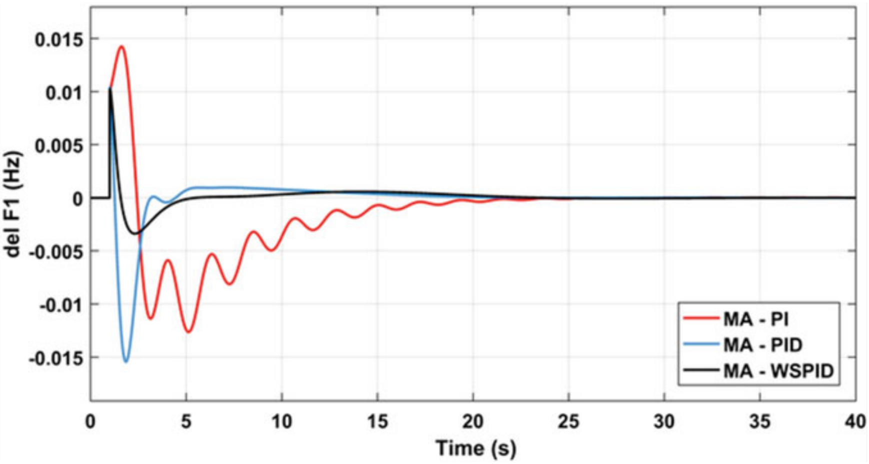


Fig. 11 del F analysis in area 1

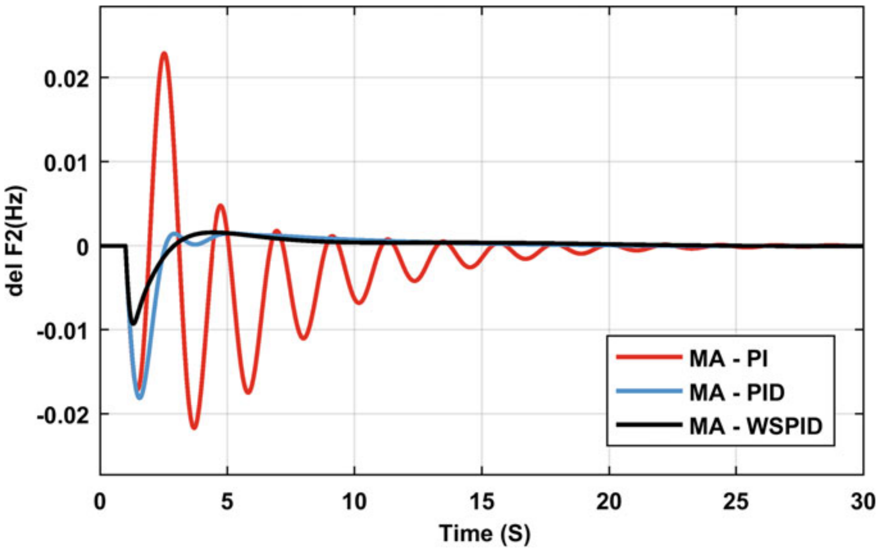


Fig. 12 del F analysis in area 2

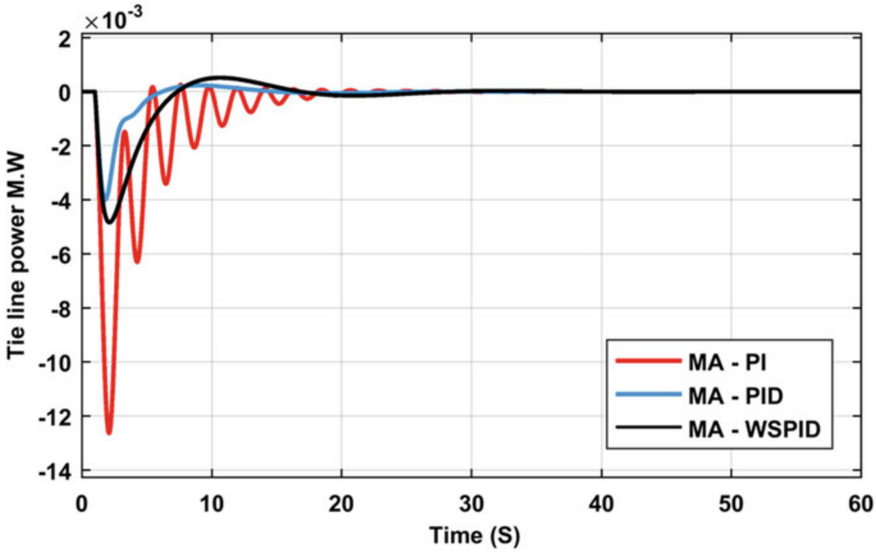


Fig. 13 del P analysis in tie line

Table 5 Controlled parameters in the frequency oscillation from PI, PID, and weighted sum of PID

Parameters/ Controller	Del F1			Del F2			Tie line		
	T_s(S)	US (Hz) 10^{-3}	UO (HZ) 10^{-3}	T_s(S)	US (Hz) 10^{-3}	UO (HZ) 10^{-3}	T_s(S)	US (Hz) 10^{-3}	UO (HZ) 10^{-3}
MA—PI	28	12.2	14	38	22	23	35	12.5	0.1
MA—PID	24	10	15	25	17.5	1.5	24	4	0.2
MA—Weighted sum of PID	22	3.5	10.2	25	9	1.6	23	4.8	0.5

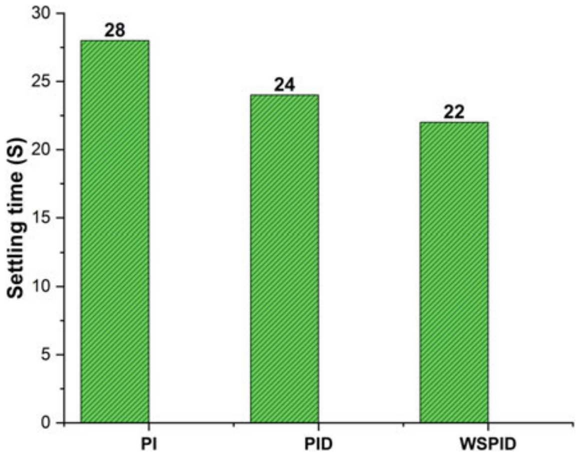


Fig. 14 Column chart for del F1 from the PI, PID, and weighted sum of the PID controller

To conform the supremacy of the weighted sum of the PID controller, the bar chart for the settling time of del F1, F2, and the tie line is shown in Figs. 14, 15 and 16.

The proposed weighted sum of the PID (WSPID) controller provided an improved response over the other two (PI and PID) controllers. The percentage of improvement (%) of T_s is reported in Table 6, and the graphical report is in Fig. 17.

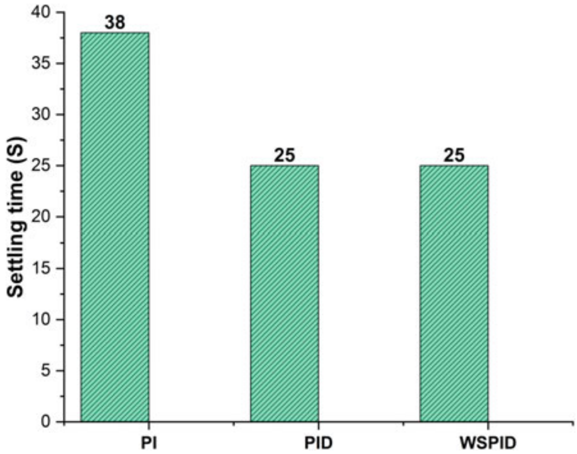


Fig. 15 Column chart for del F2 from the PI, PID, and weighted sum of the PID controller

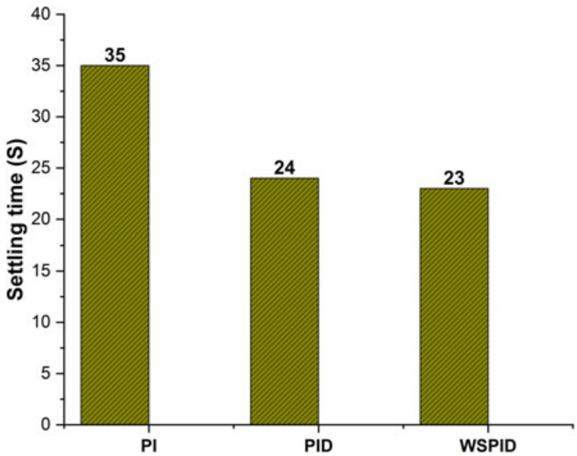


Fig. 16 Column chart for the del tie line from the PI, PID, and weighted sum of the PID controller

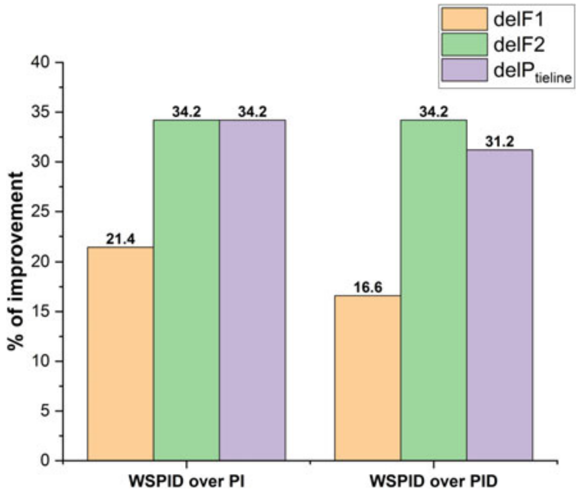


Fig. 17 % of improvement of the WSPID controller over the PI and PID controllers

Table 6 Percentage of improvement

Parameters/Controller	delF1	delF2	del $P_{\text{tie line}}$
WSPID over PI	21.4	34.2	34.2
WSPID over PID	16.6	34.2	31.2

6 Conclusion

The study in the chapter is a performance analysis of the weighted sum of the PID controller against the frequency oscillation in a PV-incorporated thermal power network. The study was carried out in two different cases. In case 1, the best-weighted sum of PID controller gain parameters is selected by comparing the various weighted sums of PID controller gain parameters. In case 2, the best-weighted sum of PID controller gain values is validated with classical PI and PID controllers. During the whole performance analysis, MA was used for optimization. The summarized results of the study are as follows:

- The weight of the PID controller was varied from 0 to 1. Among 11 different weighted sums of PID controller gain values, the W1 = 1 & W2 = 0 weighted sum of the PID controller provided an improved result compared to the others.
- The weighted sum of the PID controller (W1 = 1 & W2 = 0) result compared with PI and PID. The weighted sum of the PID controller gives better results than the PI and PID controllers.

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Chapter 7
Multi-modal Routing in Urban Transportation Network Using Multi-objective Quantum Particle Swarm Optimization

Salar Farahmand-Tabar and Parastoo Afrasyabi

1 Introduction

Every engineer holds the common objective of attaining their goals while minimizing computational resources and costs. These aims encompass a diverse range of outcomes, such as offering people various housing choices, electrical devices, and mechanical tools. Currently, there are two main classes of optimization approaches to fulfill these objectives: local optimizers, which leverage mathematical programming techniques and utilize gradient data to investigate the vicinity of an initial point [1], and global optimizers referred to as meta-heuristics, designed for broad searches without the need for continuous cost functions and variables [2–4]. These global optimizers have broad applications in all fields, especially in engineering [5–10]. They have the potential to improve performance and address various types of problems [11–18]. One of the most challenging problems among these is the routing problem, which, with its intricate logistics and diverse constraints, greatly benefits from the capabilities of these optimizers.

The multi-modal routing problem is complex and features an extensive solution space [19]. Its intricacy escalates with network expansion. This multi-objective problem involves conflicting goals [20], akin to optimization problems unsolvable through precise methods [21]. Meta-heuristic algorithms, like the Genetic algorithm (GA) [22] and its advanced version NSGA-II [23], are adept at tackling such intricate

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challenges [24]. These algorithms have demonstrated efficiency in solving routing problems [25], spanning both discrete and continuous domains. As a result, the GA and NSGA-II prove well-suited for addressing the discrete problem of multi-modal routing [26].

To solve real-world problems for reasons such as constraints in resources, time, etc., it is not possible to use deterministic methods and get a completely correct answer. Optimization methods, find optimal solutions, according to objective function. Meta-heuristic algorithms find the optimal solutions in a reasonable time by randomly searching the problem space [27]. This section introduces the routing problem in the multi-modal network, and problem modeling.

A multi-modal network is a transport network that has more than one mode to move. In the multi-modal, the path found consists of different modes to move from the source point to the destination point. In other words, a set of transportation and displacement methods that passengers can use along the way to reach their destination [28–33].

In this chapter, the multi-modal routing problem encompasses walking, buses, subways, and taxis, all within a multi-objective framework. Objectives encompass factors like path length, congestion, and traffic. Additionally, traveler dwell time in subways, taxis, and buses is constrained. The multi-objective quantum particle swarm optimization (MOQPSO) algorithm is applied to address this problem within the transportation network in the capital of Iran, Tehran. The results are obtained and compared using other algorithms.

2 Related Works on Multi-modal Routing

The origins of the routing problem can be traced back to the inception of the traveling salesman problem (TSP) [34]. As cities grew, transportation networks became increasingly intricate, leading to the emergence of multi-modal transportation networks where paths involve a combination of different modes of transport [35]. This prompted extensive research into finding integrated multi-modal routes within urban transportation networks. Initially, the focus was on analyzing the multi-modal routing problem as a single-objective challenge, typically centered around path length. However, individuals have diverse and non-coextensive preferences when choosing a route, prompting a shift toward viewing the problem of multi-modal routing as a multi-objective optimization issue [36].

Stochastic approaches are well-suited for addressing this problem due to its intrinsic nature as an optimization challenge. Gharib et al. successfully employed the GA to solve the TSP problem [37]. Deng et al. [38] tackled the multi-modal routing problem using the GA, considering a variety of transport modes. Pahlavani et al. addressed the multi-modal routing problem with a focus on objectives such as travel time, cost, and modal transitions, utilizing the NSGA-II algorithm. Their study involved a multi-modal network integrated into Tehran's transportation system, encompassing walking, buses, subways, and taxis paths. Pahlavani et al. prioritized the resulting set of solutions through the TOPSIS method, ultimately demonstrating the suitability of the NSGA-II for this problem [39].

3 Multi-objective Algorithm of Quantum Particle Swarm Optimization

When compared to other evolutionary algorithms (EAs), the PSO algorithm offers significant advantages due to its ease of implementation, a reduced number of adjustable parameters, and the absence of gradient information. As a result, it finds extensive applications in function optimization, neural network training, and various other domains. Numerous experiments have demonstrated that PSO is capable of solving a diverse range of optimization problems, similar to genetic algorithms. In comparison to the standard PSO, the QPSO algorithm exhibits enhanced global search capabilities and requires fewer control parameters. Furthermore, MOQPSO represents an improvement over MOPSO specifically designed for tackling multi-objective optimization problems.

3.1 Basic Concept of Particle Swarm Optimization (PSO)

The initial proposal of PSO was made by Kennedy and Eberhart [40]. They developed the algorithm by emulating the foraging behavior of birds and successfully applied it to solve continuous function optimization problems. In this approach, each bird within the group is represented as a particle, devoid of mass or volume, and serves as a potential solution to the optimization problem. The fitness value of each particle is computed using a fitness function that determines its quality. The speed and trajectory of each particle control its direction and distance of movement. By continuously adjusting their trajectories based on the best-performing particles, the particles navigate toward an optimal solution through successive iterations of the search process.

In a search space with D dimensions and M particles, the position of a particle at the t-th iteration is denoted as $x_i^t = (x_{i1}, x_{i2}, ..., x_{iD})$, and its velocity is represented by $V_i^t = (v_{i1}, v_{i2}, ..., v_{iD})$, where i ranges from 1 to M. In PSO, each particle keeps track of its own best position achieved so far in the search space, denoted as $pbest_i = (P_{i1}, P_{i2}, ..., P_{iD})$. Furthermore, the algorithm maintains a global best position, $gbest = (P_{g1}, P_{g2}, ..., P_{gD})$, which represents the best position attained by any particle thus far. At each iteration, the PSO algorithm updates the positions and velocities of the particles using Eqs. (1) and (2):

$$v_{id}^{t+1} = \omega \cdot v_{id}^t + c_1 \cdot rand\left(P_{id} - x_{id}^t\right) + c_2 \cdot rand\left(P_{gd} - x_{id}^t\right) \quad (1)$$

$$x_{id}^{t+1} = x_{id}^{t} + v_{id}^{t+1} \tag{2}$$

Here, d takes values from 1 to D, where D represents the dimension of the search space. The variable w represents a predefined constant called the inertia weight, while c_1 and c_2 denote the acceleration constants. The term rand denotes a random number that is uniformly generated within the range [0, 1].

3.2 Quantum-Behaved Particle Swarm Optimization (QPSO)

Sun et al. [41] have highlighted distinctive characteristics of Quantum Particle Swarm Optimization (QPSO) when compared to other heuristic algorithms:

(a) QPSO exhibits strong exploratory tendencies, enabling it to navigate the design space effectively without becoming trapped in local minima. This quality proves particularly advantageous for solving complex and chaotic problems.
(b) The concept and equations underlying QPSO are relatively straightforward, making it easier to implement and adapt to various problem domains.
(c) While QPSO demonstrates promising results in the single-objective domain, its exploration in the multi-objective domain is still relatively limited in the literature.
(d) Based on Yang et al. [42], QPSO demonstrates superior convergence behavior compared to the traditional Particle Swarm Optimization (PSO). This makes QPSO an attractive choice for tackling complex multi-modal routing optimization problems.

The QPSO algorithm, developed by Sun et al. [41], builds upon the foundation of the conventional Particle Swarm Optimization (PSO) method. In PSO, particles are updated based on their position and velocity.

In contrast, QPSO takes a different approach by updating particles based on the square of the module of a wave function, represented as $|\psi(x,t)|^2$, which describes the probability density function of particle positions. Equations (3)–(6) specify how the particle states, $x_{ij}(t+1)$, evolve using the Monte Carlo method.

$$x_{ij}(t+1) = P_{ij} - \beta \cdot \left(mbest_j(t) - x_{ij}(t)\right) \cdot ln\left(\frac{1}{u_{ij}(t)}\right),\ for\ \rho \geq 0.5 \tag{3}$$

$$x_{ij}(t+1) = P_{ij} + \beta \cdot \left(mbest_j(t) - x_{ij}(t)\right) \cdot ln\left(\frac{1}{u_{ij}(t)}\right),\ for \rho < 0.5$$

$$P_{ij}(\mathrm{t}) = \varphi_j(\mathrm{t}) \cdot pbest_{ij}(t) + \left(1 - \varphi_j(\mathrm{t})\right) \cdot gbest_{ij}(t) \tag{4}$$

$$mbest_j(t) = \frac{1}{n}\sum_{i=1}^{n} pbest_{ij}(t) \tag{5}$$

$$\beta(t) = [(\beta_1 - \beta_0)(T - t)/T] + \beta_0 \tag{6}$$

In this context, the terminology used includes $pbest$ to denote each particle's local best position, $gbest$ for the global best position encompassing the entire swarm, and $mbest$ representing the mean best position calculated from all individual best positions ($pbest$). n denotes the population size within the swarm, t signifies the current iteration time-step, T corresponds to the total number of iterations, and β is the coefficient of contraction–expansion, bounded by β_1 (upper limit) and β_0 (lower limit). Additionally, ρ, u, and ϕ are random numbers distributed uniformly in the range of 0 and 1.

3.3 Multi-Objective Quantum-Behaved Particle Swarm Optimization (MOQPSO)

In the realm of multi-criteria decision-making, multi-objective optimization serves as a field that delves into the intricacies of balancing conflicting objectives and exploring the various design possibilities. In multi-objective optimization problems, a set of design variables denoted as $X_i = \{X_{i1}, X_{i2}, \ldots, X_{in}\}^T$, is associated with multiple response functions, typically represented as $f(X_i) = \{f_1(X_i), f_2(X_i), \ldots, f_k(X_i)\}^T$.

The Pareto front concept is extensively utilized to tackle multi-objective problems. It is determined such that a solution vector, denoted as $f(X_A)$, dominates another solution vector, $f(X_B)$, if the following conditions are met (in the case of minimization):

- For each objective functionf_j, where j ranges from 1 to k (k being the number of objective functions), $f_j(X_A)$ is less than or equal to $f(X_B)$: $f_j(X_A) \leq f_j(X_B)$ for $j = 1, ..., k$.
- Additionally, there must be at least one index j for which $f_j(X_A)$ is strictly less than $f_j(X_B)$: there exists at least one j such that $f_j(X_A) < f(X_B)$.

When this rule is upheld, both sets of design variable vectors are termed non-dominated by each other, and they collectively constitute the setP_s, defined as $P_s = \{X_i \in X | \nexists X_j \succ X_i\}^T$. This signifies that within the solution space$X$, the set of solution vectors X_i does not possess a dominating vectorX_j. The collection of all non-dominated objective function vectors is referred to as the Pareto front $P_f = \{f(X_i) | X_i \in Ps|\}^T$.

The considered algorithm in this research draws inspiration from the Pareto front archive strategy. The fundamental concept underlying this strategy is to maintain a database storing the best positions. A maintenance routine is executed at the conclusion of each iteration. The algorithm employs three distinct archives, namely:

- The f_P archive is utilized to store the individual local best positions of each particle.
- The f_{G1} archive is employed to store the global best positions of the entire swarm.
- The f_{G2} archive, on the other hand, is tasked with storing global best positions filtered based on a distance criterion, which effectively removes clusters of particles.

The set of design variables associated with these archives is denoted as X_G, X_{G1}, and X_{G2}, respectively. Modifications to the core equations of the quantum particle optimization model (Eqs. 3–6) involve the replacement of $mbest_j(t)$ and $gbest_{ij}(t)$ with guide $_j(t)$ and $pbest_{ij}(t)$, respectively. In essence, the classical $meanbest$ variable used in QPSO is substituted with guide particles in MOQPSO. These guide particles are individually selected for each new solution generated by every particle in each iteration. The guide particle's selection method has proven to be more effective when combining two specific approaches: the $extremeparticle$ method and the $nearestparticle$ method. The probability of using the 'nearest particle' method instead of the $extremeparticle$ method is predefined by the user and denoted as $guid\_prox$.

When selecting extreme particles, the number of calculated extreme particles is considered to be equal to the number of multi-objective functions. To prevent the algorithm from choosing particles that are located very close to each other, thereby promoting diversity along the Pareto front, a geometric mean-based criterion is utilized. The positions of extreme particles are determined from the f_{G1} archive during each iteration, following these rules:

Initially, the cumulative distance in the objective function space between each particle is computed using $d_i(f_{G1i}(X), f_{G1j}(X)) = \sum_{j=1}^{ng_1} \|f_{G1i} - f_{G1j}\|_2$, for$i = 1, ..., ng_1$, where ng_1 represents the number of particles in the f_{G1} archive. The selection of the first extreme particle is according to these calculations.

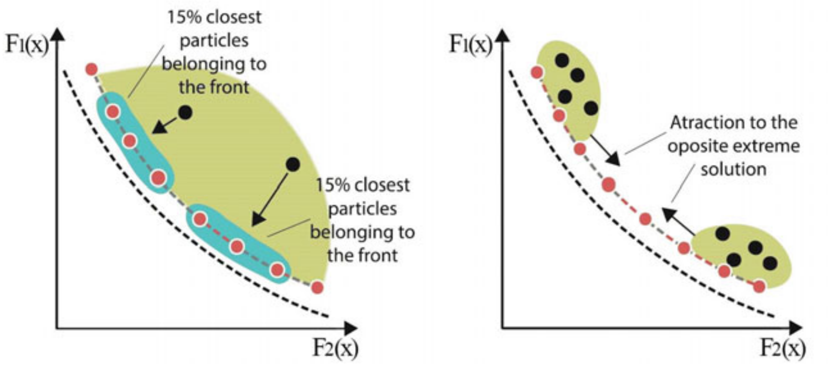
$$
\begin{aligned}
&X_{E1} = \arg \max_{X \in_{G1}} d_i(f_{G1i}(X), f_{G1j}(X)) \\
&f_{E1}(X_{E1}) = \{f_1(X_{E1}),\ f_2(X_{E1}), \ldots f_k(X_{E1})\}^T
\end{aligned} \tag{7}
$$

The selection of the second extreme particle is according to the characteristics of the previous extreme particle, and it is determined as follows:

$$
\begin{aligned}
&X_{E2} = \arg \max_{X \in_{G1}} f_{E1}(X_{E1}) - f_{G1i_2}(X),\ for\ i = 1, \ldots, ng_1 - 1 \\
&f_{E2}(X_{E2}) = \{f_1(X_{E2}),\ f_2(X_{E2}), \ldots f_k(X_{E2})\}^T
\end{aligned} \tag{8}
$$

A geometric mean is utilized and evaluated for the third extreme, as $d_{gi}\big(f_{E1}(X_{E1}), f_{E2}(X_{E2}), f_{G1j}(X)\big) = f_{E1}(X_{E1}) - f_{G1i_2} f_{E2}(X_{E2}) - f_{G1i_2}(X)$, for $i = 1, \ldots, ng_1 - 1$ and the definition of the extreme particle becomes as follows:

$$
X_{E3} = \arg \max_{X \in_{G1}} d_{gi}(f_{E1}(X_{E1}), f_{E2}(X_{E2}),\ f_{G1j}(X))
$$



a) Extreme guide mechanism

b) Nearest particles belonging to the Pareto front guide

Fig. 1 The behavior of different selection mechanisms

$$f_{E3}(X_{E3}) = \{f_1(X_{E3}),\ f_2(X_{E3}), \ldots f_k(X_{E3})\}^T \tag{9}$$

As a general rule, starting from the 3rd extreme particle and onwards, the evaluation of an extreme particle (e) is determined using $f_{gi} = \prod_{l=1}^{e-1} \| f_{EI}(X_{EI}) - f_{G1i} \|_2$, for $i = 1, \ldots, ng_1 - (e-1)$. In this equation, ng_1 represents the number of particles in the f_{G1} archive. The extreme particle is defined based on this calculation:

$$\begin{aligned} X_{Ee} &= \arg \max_{X \in X_{G1}} d_{gi}, \\ f_{E3}(X_{E3}) &= \{f_1(X_{E3}),\ f_2(X_{E3}), \ldots f_k(X_{E3})\}^T \end{aligned} \tag{10}$$

The algorithm proceeds by selecting the particle located at the furthest extreme. This selection method leads to the behavior illustrated in Fig. 1a, particularly in scenarios involving two objective functions. This behavior results in a broader Pareto front and increased dispersion among particles.

Notably, the introduction of the geometric-based approach was observed to enhance both the convergence and spread of the algorithm during testing with several of the aforementioned benchmark functions. This improvement is a significant feature of a multi-objective mechanism and stands out as one of the innovations introduced in this study.

Figure 1b illustrates the behavior influenced by the guide particle when the selection process is based on the nearest particles within the Pareto front. This mechanism contributes to the algorithm's improved convergence. To maintain diversity, the guide particle is chosen randomly from a list of the n th nearest particles, where n th is assumed to be 15% of the f_{G1} archive (hereinafter referred to as $guid\_perc$). Concerning constraint handling, if an infeasible solution x is created during particle generation, it is removed and a new solution is created as a replacement.

A simplified flowchart depicting the MOQPSO algorithm can be found in Fig. 2.

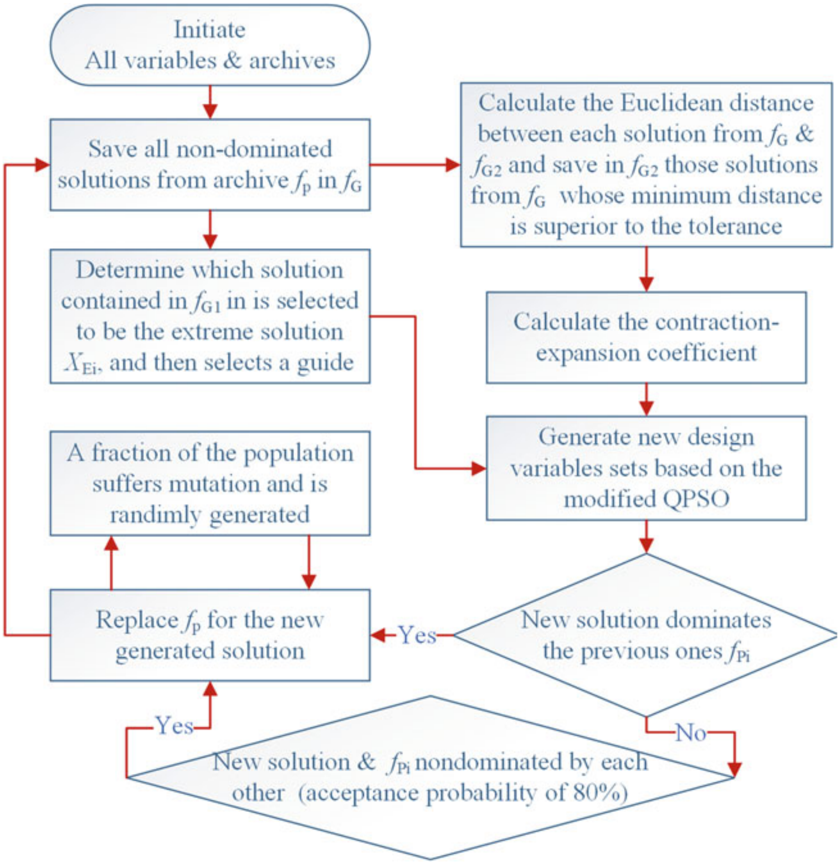


Fig. 2 The procedure diagram of MOQPSO

4 State of the Optimization Problem

This study has been executed in three distinct stages, as outlined in Fig. 3. The initial phase involved the creation of a spatial database. This database encompasses five distinct objective functions and four modes of transportation. The urban public transportation system of Tehran was employed for this purpose. Moving on to the second stage, the multi-modal problem was investigated through the utilization of a multi-modal graph, network analysis, and GIS modeling. The issue was then addressed using MOQPSO. Additionally, the fundamental Genetic algorithm was employed for result evaluation. Finally, the results will be assessed according to factors such as convergence during runs, algorithm execution time, and a comparison between the path achieved by MOQPSO and that derived from the NSGA II and MOPSO. The refined path will also be visualized to enhance comprehension.

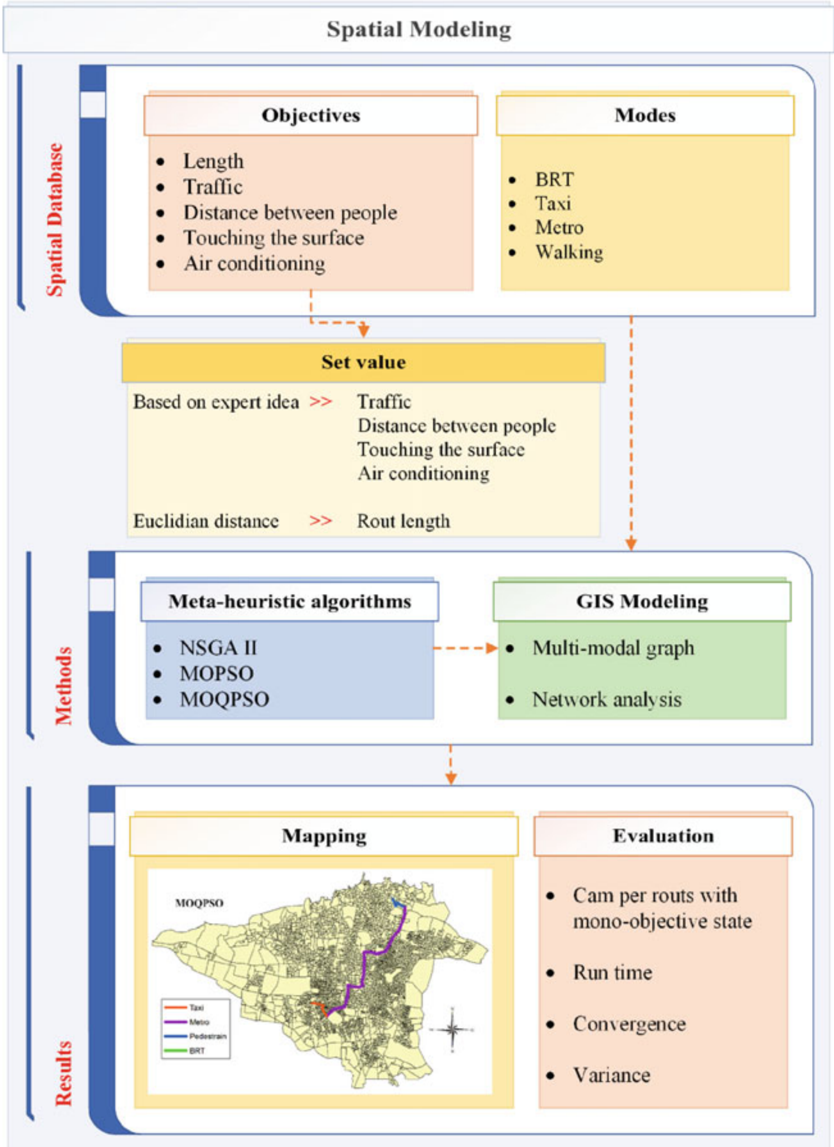


Fig. 3 The framework of the routing process

4.1 *Considered Area for the Study*

The focal area of investigation encompasses the Tehran metropolis, which serves as both the capital and the most densely populated city of Iran. Geographically, Tehran spans 751 square kilometers and is situated between 51° 6′ to 51° 38′ East longitude

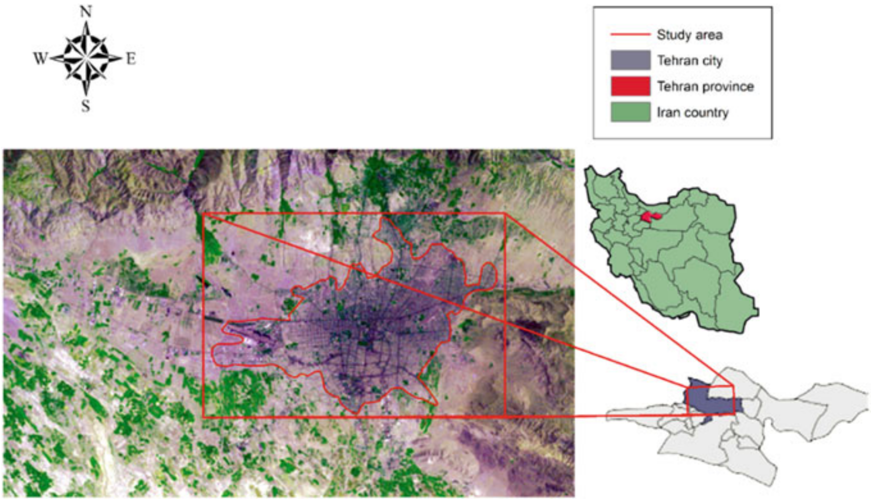


Fig. 4 Adjacency matrix for multi-modal network modeling

and 35° 34′ to 35° 51′ North latitude. The population of Tehran surpasses 8,693,706 individuals.

The city contains the most intricate and robust transportation network of Iran which may have unprecedented challenges and shifts induced heightened traffic levels and a cascade of associated issues. Refer to Fig. 4 for an illustration of the study area's layout.

4.2 Multi-modal Routing

The concept of a multi-modal network involves the integration of various travel modes, such as taxi, subway, walking, etc., in a combined manner for network analysis [23]. Modeling transportation systems, particularly when dealing with an increased number of travel modes, can be challenging. A Multi-modal Transportation System (MTS) encompasses a blend of all possible travel states and transportation systems, functioning through diverse modalities. This implies that commuters employ a mix of transportation methods simultaneously to reach their destinations [34].

In the modeling of a multi-modal network, graphs, label matrices, and adjacency matrices come into play. Nodes symbolize stations, while edges indicate connections between these stations. The adjacency matrix is congruent with the network structure and consists of 0 s and 1 s. It elucidates the presence or absence of connections between stations [23]. The main diagonal entries of the adjacency matrix are set to zero, reflecting the absence of self-loops. The label matrix aligns with the network and allocates modes to edges, essentially assigning labels to connections. In Fig. 5, a demonstration of both adjacency and label matrices is provided. In Fig. 5a, a

"1" signifies the existence of a connection between two nodes, while "0" denotes the absence of a connection. In Fig. 5b, each number designates the type of travel structure between two nodes (stations). For instance, in the second entry, the number "1" specifies the travel structure type between stations numbered 1 and 2.

Figure 6 illustrates a representative example of a multi-modal route. Within this route, four modes of transportation are incorporated: taxis, subways, buses, and walking. Walking is utilized as a mode of transition between the other modes. The route comprises of ten nodes and nine edges.

In the context of public transportation in Tehran, the principal components contain walking, buses, taxis, and subways. Given this emphasis, the study concentrated on these four modes. In the analyzed network, a total of 408 stations were employed, as outlined in Table 1. Notably, the walking mode is characterized by its absence of

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 & 0 \end{bmatrix} \qquad \begin{bmatrix} 0 & 1 & 2 & 0 & 1 & 3 \\ 1 & 0 & 1 & 4 & 3 & 1 \\ 2 & 1 & 0 & 0 & 4 & 1 \\ 0 & 4 & 0 & 0 & 2 & 0 \\ 1 & 3 & 4 & 2 & 0 & 1 \\ 3 & 1 & 1 & 0 & 1 & 0 \end{bmatrix}$$

(a) (b)

Fig. 5 Multi-modal matrix **a** Adjacency matrix, **b** Label matrix

Fig. 6 An example of a multi-modal rout

Table 1 The number of stations in each mode

Mode	Vertex	Edge
Subway	90	93
Taxi	83	82
Bus	235	237
Total	408	412

dedicated stations and its purpose of facilitating transfers between different modes of transportation.

GIS (Geographic Information System) has proven to be highly effective in handling spatial data and is applied across a wide range of domains, including transportation systems. Specifically, GIS-T (GIS in Transportation) leverages GIS functionalities to address transportation-related challenges. Beyond the conventional GIS tools like buffers and overlays, GIS-T is particularly adept at resolving transportation concerns. Its key strengths lie in tasks such as shortest path analysis, network flow modeling, and urban transportation analysis [38]. Notably, network analysis within GIS serves as a robust approach for modeling multi-modal route problems, which has been precisely employed in this research.

4.3 Modeling

The multi-modal problem explored in this research encompasses multiple objectives, including individual distance (empty space), path length, surface contact, air conditioning, and traffic. Furthermore, a constraint is imposed on the maximum allowable stay time in each mode, limited to 15 min. The multi-modal network encompasses subways, buses, pedestrian pathways, and taxis. The initial step involves determining the method for defining responses to model the problem using both the MOQPSO.

A multi-modal path is constructed based on a sequence of stations and the corresponding mode type. Each solution of the algorithm represents a potential route. Solutions within a multi-modal problem consist of station vertices and mode types. Specifically, odd-numbered variables correspond to stations, while even-numbered variables indicate the mode type between the adjacent variables.

4.3.1 Objective Function

Individuals have distinct criteria when selecting a route. For example, during the virus pandemics many prioritize paths that minimize the risk of health-related issues. Therefore, factors like individual distance (spacing), surface contact, air conditioning, and time spent in each mode are influential in virus transmission within public transportation.

Table 2 Goal prioritization according to the modal type

Objective mode	Traffic	Surface contact	Air conditioning	Individual distance
Bus	3	3	2	3
Metro	1	4	4	4
Taxi	4	2	2	2
Walking	2	1	1	1

The objectives of the study encompass path length, individual distance (spacing), surface contact, air conditioning, and traffic. Furthermore, the algorithm incorporates a constraint on modal durations, limiting them to 15 min. An optimal route, in this context, entails a shorter path length, reduced traffic, and minimized surface contact. Moreover, factors like air conditioning and spacing between individuals are taken into account. Path length is computed independently of mode type through Eq. (11) or the equation of Euclidean Distance.

$$Fitness\ Function = Min \sum D_{i,j}$$

$$D_{i,j} = \sqrt{\left(x_j, x_i\right)^2 + \left(y_j, y_i\right)^2} \tag{11}$$

Four distinct objective functions are under consideration, namely individual distance (spacing), surface contact, air conditioning, and traffic. These functions are ranked based on professional judgment. As the optimization problem involves minimization goals, the highest-priority objective function is assigned a value of 1, while the lowest-priority function is assigned a value of 4.

Considering Table 2, it's observed that the traffic function holds the lowest value within the subway mode, indicating higher priority. For the other three objective functions, the subway mode is assigned a lower priority. Conversely, the walking mode takes the highest priority for three functions: individual distance, surface contact, and air conditioning. In simpler terms, the walking mode features the least contact surface and the most effective air conditioning.

5 Result Discussion on the Case Studies

In this section, the value of objective function, convergence trend, and execution time of the algorithms are compared. The runtime of the NSGA II, MOPSO, and MOQPSO, were 1.7215, 0.758, and 0.558s in 100 iterations, respectively. The solution of the algorithms is shown in Fig. 7. The odd cells represent the stations and the even cells indicate the edge mode type ("1" = bus mode, "2" = metro mode, "3" = pedestrian mode, and "4" = taxi mode). Figure 8 shows the route found by algorithms; the green line is related to the bus mode, a blue line for pedestrian, and

1	**1**	**44**	**2**	**42**	**4**	**71**	**3**	**95**	**4**	**196**	**2**	**79**	**4**	**200**

a) NSGA II

1	**1**	**127**	**1**	**175**	**4**	**14**	**2**	**154**	**1**	**104**	**2**	**200**

b) MOPSO

1	**3**	**100**	**2**	**76**	**3**	**72**	**3**	**18**	**3**	**69**	**4**	**200**

c) MOQPSO

Fig. 7 Achieved rout of using different methods

red/pink indicates the subway and taxi mode on the track. According to Fig. 8, the route found by the MOQPSO algorithm is better because the route length is shorter and also more efficient.

The value of the fitness function considering each objective is shown in Fig. 9 based on the best response of each iteration. The value of the fitness function for different methods is presented in Table 3. According to the achieved results, the convergence of the MOQPSO is better than MOPSO and NSGAII. This means that MOQPSO is able to achieve the optimal values faster.

To show the trade-off between different objectives, the Pareto front of pair-wise objectives is depicted in Figs. 10 and 11.

Choosing the optimum solution from these Pareto fronts requires decision-making considerations. For example, in the case of traffic and route length, we expect to have the minimum values for both objectives, and the opposite is true for the case of safety and convenience with the maximum values. For the other objectives such as convenience-length, convenience-traffic, safety-length, and safety-traffic the objectives are conflicting and the optimum solution should satisfy both objectives proportionally.

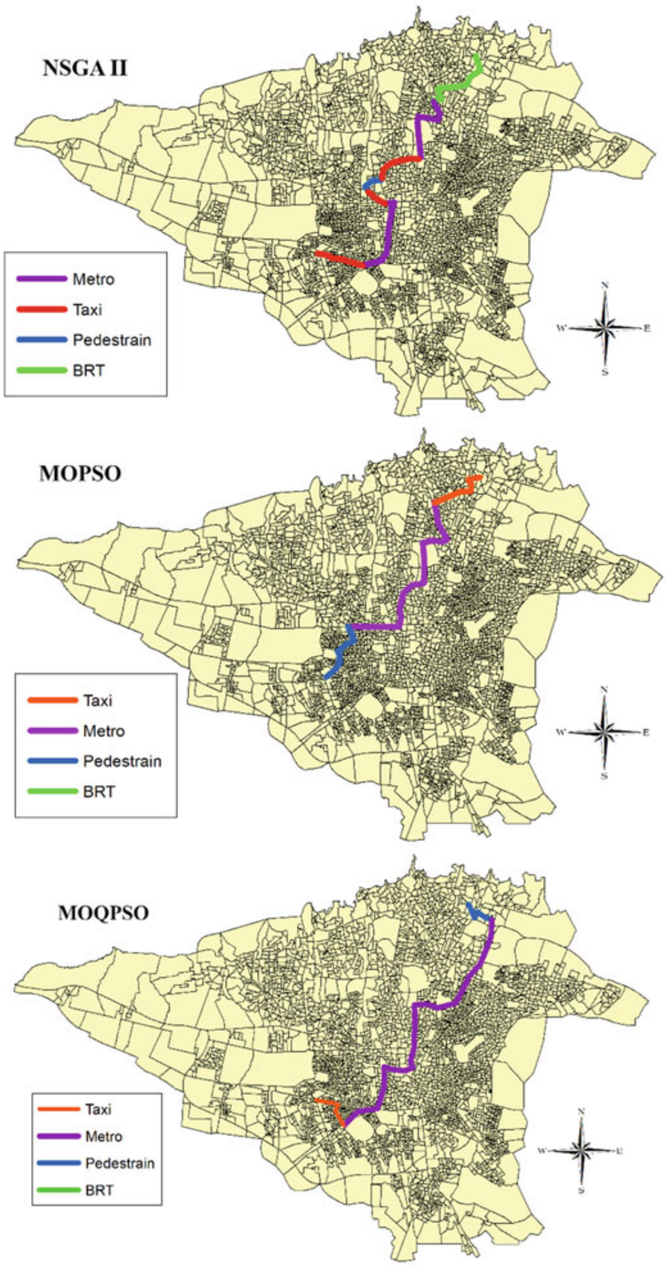


Fig. 8 The multi-modal route found from the MOQPSO and NSGA II algorithms

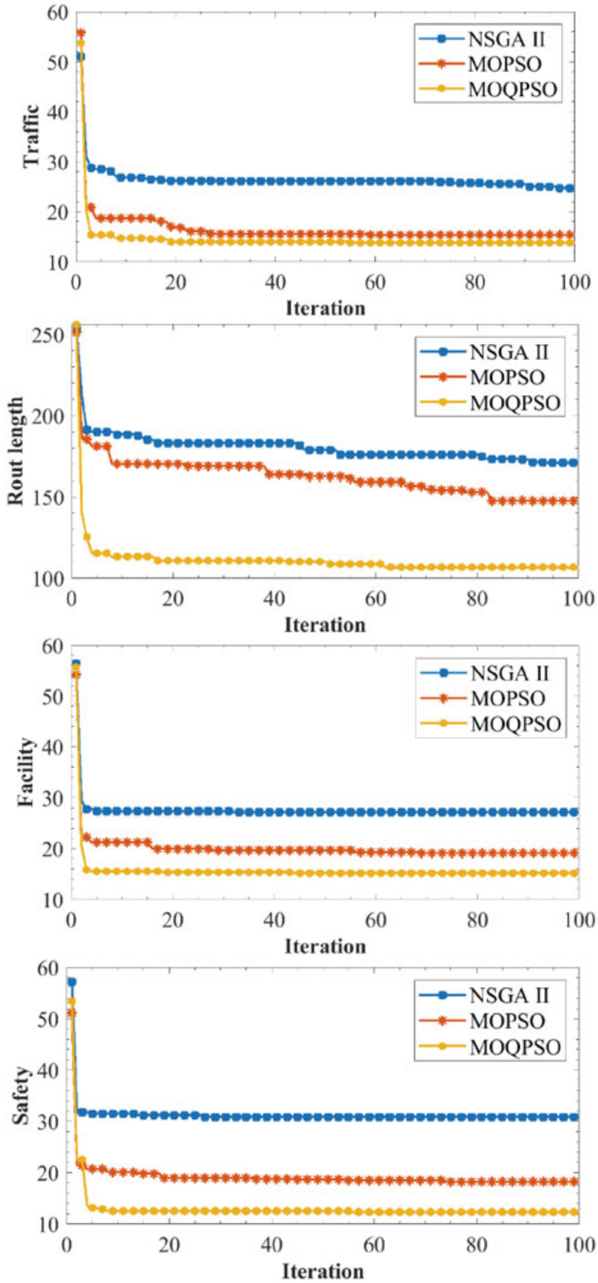


Fig. 9 The convergence of different objectives considering various methods

Table 3 Values of different objectives

Methods/Objective	Traffic	Length	Facility	Safety
NSGAII	24.73	171.00	27.31	30.89
MOPSO	15.40	147.76	19.13	18.27
MOQPSO	13.82	106.65	15.20	18.27

6 Conclusions

Routing in the transport network is a complicated problem, the complexity can be due to the existence of different combinations of the modes, the expansion of the solution space, etc. In this chapter, the multi-modal network is based on the assumption of the bus, metro, and walking modes. A maximum of five mode changes is added, meaning the route that takes more than five mode changes is eliminated. This is an optimization problem that uses meta-heuristic algorithms as good results. In this study, the routing problem is implemented in the multi-modal transport network using MOQPSO. The objective function is according to the minimization of trajectory length. The MOPSO algorithm has been improved using quantum behavior to fit the problem and the results of the MOQPSO are compared with the NSGA II and MOPSO.

The simulated transport network consists of 200 stations in an area of 1600 m^2. To evaluate the proposed algorithm, the convergence criteria, the value of objective function, the route found, and the algorithms' execution time are compared. The developed algorithm has been able to find a more efficient and shorter route than the NSGA II and MOPSO and in fewer generations. The MOQPSO has a simpler computation and parameter tuning. The convergence is also much faster and has a shorter route length as an objective function.

Future research can be used as real data to make the results more realistic. Other algorithms and combinations can also be utilized to solve the multi-modal routing problem. Other modes such as bicycles and scooters can also be used in the multi-mode network.

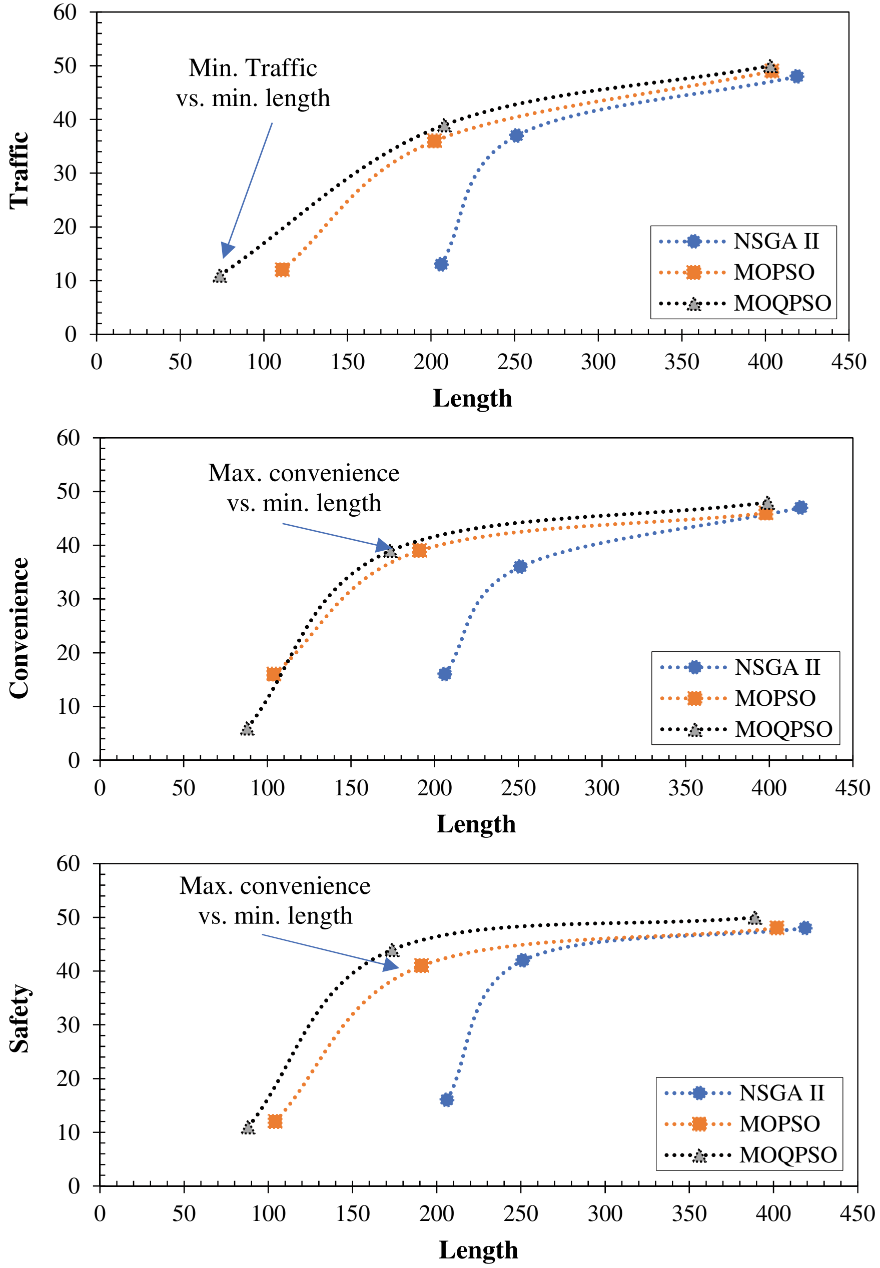


Fig. 10 The Pareto front of length related objectives considering various methods

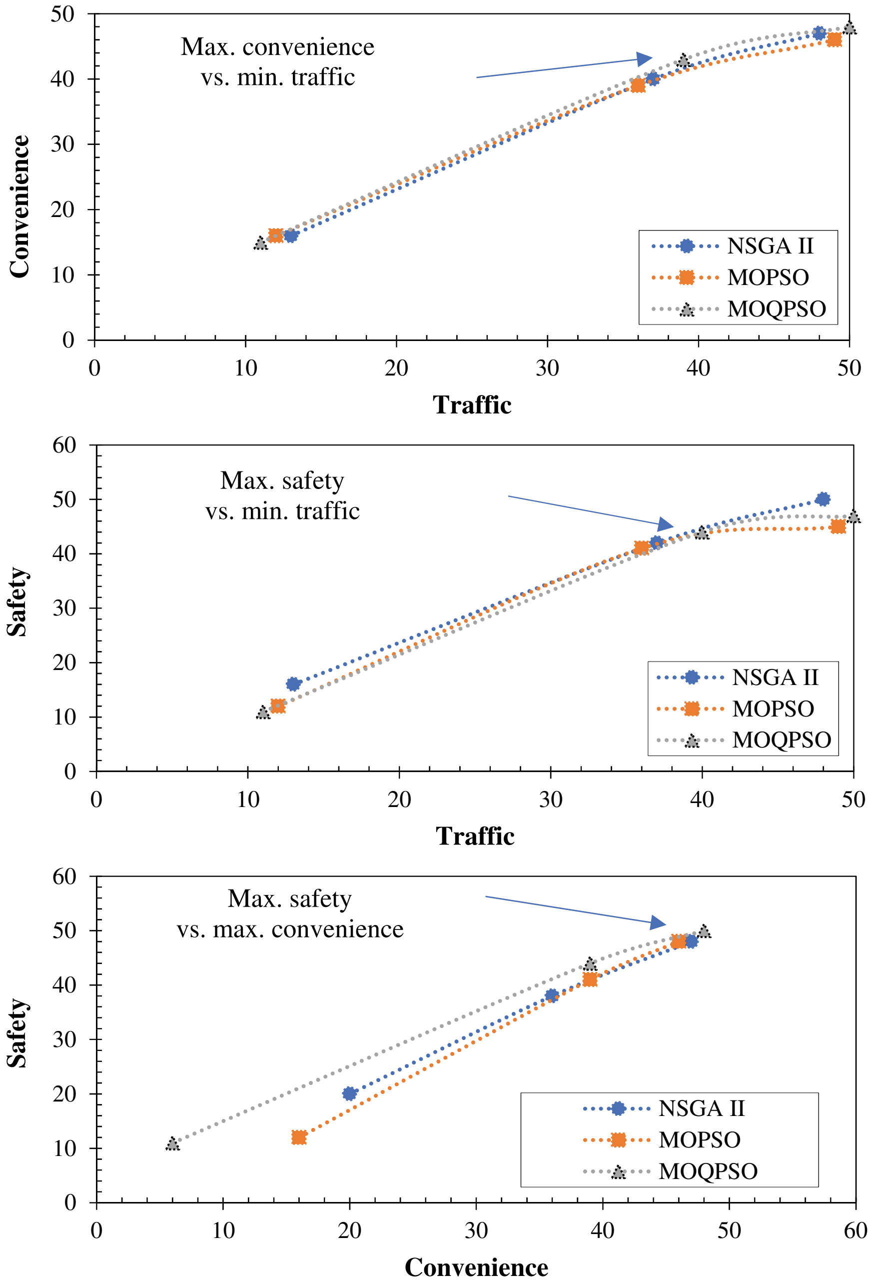


Fig. 11 The Pareto front of different objectives considering various methods

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Chapter 8
Multi-objective Optimization for Feature Selection: A Review

M. B. Dowlatshahi and A. Hashemi

1 Introduction

Methods used by machine learning are often applied to tackle problems containing an extensive list of features. It can consequently be tricky to pick a perfect feature set and neglect redundant ones. In many real-world datasets, some features may be deemed useless due to redundancy and irrelevance within these features. Therefore, taking into account these features is not advantageous and generally results in low classification accuracy. Consequently, the main goal of feature selection is to enhance the efficacy of classification through picking a limited subset of suitable features from the initially extensive feature set. Decreasing the data dimensions would be the natural consequence of removing unnecessary and redundant features. Enhancing the learnt model's simplicity and optimizing its performance can expedite the learning process [1–3].

Since an extensive range of engineering challenges benefit from feature selection, offering efficient methodologies within this domain is imperative. If we denote d as the cardinality of the set of features in a particular dataset, it follows that there exist 2^d unique subsets of features. Due to the exponential growth of the number of potential subsets with increasing feature counts, the feature selection technique is thus identified as an NP-hard task. Therefore, the identification of the most suitable subset of features presents a considerable issue. In recent decades, numerous feature selection methodologies have been introduced, employing many different strategies [4, 5].

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Feature selection involves attempting to accomplish two main objectives that often conflict with each other: improving the accuracy of classification while simultaneously reducing the number of features utilized, aiming to mitigate the curse of dimensionality phenomena. Feature selection is an MOP that balances two opposing goals. Research on MOP-based feature selection techniques to solve these difficulties has advanced in recent years [3].

This chapter aims to critically analyse the most recent studies on this topic, focusing on the research results from 2012 to 2023 and considering the different methods and procedures. A previous study [3] presented feature selection methods based on MOP to review the methods from 2012 to 2019. In addition, these methods were reviewed to determine whether they are filters or wrappers and their application in machine learning tasks. In this chapter, in addition to these methods, we also want to review the works published since 2020. We focus in this chapter on categorizing these methods based on the supervision perspective (supervised, unsupervised, semisupervised) and specifying the nature of each method in accordance with how it communicates with the learning algorithm (wrapper, filter, hybrid, embedded).

Here is the structure of this chapter: Sect. 2 explains the MOP strategy and its formulation. Section 3 explains the feature selection task and its variants based on different perspectives. Section 4 discusses the feature selection algorithms based on MOP, considering the monitoring perspective. Section 5 concludes the chapter.

2 Multi-objective Optimization

In a single-objective problem, objective function values can be compared to determine which response is best. Conversely, in the case of an MOP, dominance is employed to identify and select the optimal solution. A nondominated solution set can be characterized as a collection of solutions not influenced or handled by any individual inside the set. The Pareto optimal system refers to the selection space that is not dominated by any other possible decision space. The Pareto optimum set maps every possible point into a set, and its front is the region that contains that set [3].

Identifying major attributes is crucial in developing an efficient feature selection procedure since the objective is to discover optimal solutions to an optimization problem in which a function must be modified according to the relative relevance of features concerning classes of data. It is possible to determine the importance of feature subsets by maximizing numerous criteria simultaneously. It gets trickier, however, when these goals conflict [6].

Consequently, utilizing MOP techniques provides a viable approach for addressing this challenge. The obstacles that come with multi-objective scenarios emerge whenever it is a requirement for making optimal decisions that require a compromise between two objectives that are commonly in conflict. To minimize or maximize conflicting components, MOP uses objective functions. The challenge of minimizing multi-objective functions is addressed mathematically without adequately losing generality [3, 6]:

$$\textit{Minimize}\, F(x) = [O_1(x), O_2(x), \ldots, O_k(x)] \tag{1}$$

$$\textit{subject to}\, g_i(x) \leq 0, i = 1, 2, \ldots, m \tag{2}$$

$$h_i(x) = 0, i = 1, 2, \ldots, l \tag{3}$$

The function value $O_k(x)$ is contingent upon the selection variables vector y, whereby y denotes a set of decision parameters. The parameter "i" represents the quantity of objective functions that require minimization. The constraint functions can be denoted as $g\_i(x)$ and $h\_i(x)$. The inherent trade-offs between opposing objectives underscore the effectiveness of multi-objective algorithmic solutions. The problem of minimizing the i-objective has been found to have two distinct solutions, which are represented by the variables c and d. If the aforementioned criteria are satisfied, one can assert that c exhibits dominance over d or c possesses a greater quality in comparison to d.

$$\forall i : o_i(c) \leq o_i(d)\, \textit{and}\, \exists j : o_i(c) < o_i(d), i, j \in \{1, 2, \ldots, k\} \tag{4}$$

There are some limitations in MOP, one of which is the curse of dimensionality. This refers to the increase in computational complexity as the number of objectives or decision factors grows. Moreover, the existence of contradictory goals can pose challenges in identifying a singular optimal resolution. Assessing the comparative significance of goals and integrating the preferences of decision-makers might present difficulties as well [6].

3 Feature Selection

In order to gain insight from raw data, preprocessing is a crucial step, and it involves two different methods: feature selection and feature extraction. Feature selection is the procedure by which a system's intended features are chosen. In contrast, the process of extracting features involves representing a mixture of the features that were initially used by making use of transfer methods. In the process of feature extraction, each newly extracted feature can serve as a representation of one or many existing features. Feature selection is the process of reducing a huge set of potential features to a manageable subset by removing redundant or unnecessary features. Feature selection is preferable to feature extraction since it keeps the initial values of the data while still decreasing the number of dimensions. The primary objectives of feature selection encompass enhancing model performance by optimizing predictive accuracy, enhancing model interpretability, and constructing an adaptive model that minimizes overfitting, facilitating improved generalization capabilities [7–9].

The feature selection process can potentially decrease the complexity of models and minimize the time needed for training in machine learning. Consequently, a significant amount of research has been undertaken on utilizing feature selection in activities related to machine learning. These tasks are clustering, regression, and classification techniques that are applied in many real-world scenarios. Not all features in a dataset have equal predictive power, and some features may exhibit redundancy or irrelevance. The rise in the dimension of data leads to the development of a feature space with a vast number of dimensions. The curse of dimensions refers to the exponential expansion of data volume resulting from the progressive increase in the number of dimensions or qualities. Consequently, researchers conducted thorough research to improve the efficiency of feature selection methods for high-dimensional data [10–12].

Different feature selection approaches can be classified into four main groups namely, wrapper, embedded, filter, and hybrid, each of which is determined by how they communicate with learning algorithms. Filter methods evaluate features using statistical metrics rather than learning techniques. Wrapper approaches involve the utilization of a classification algorithm to determine the various potential subsets of features. This implies that many feature subsets are examined during learning to capture the optimal subset that yields the highest classification accuracy. Embedded approaches involve training a model and subsequently assessing the relevance of features using the trained model, such as filter techniques. Embedded approaches involve the utilization of learning algorithms, such as lasso, ridge regression, and autoencoder, to assess the predictive potential of the feature set. Subsequently, the selection of features is determined by these values. Hybrid methods refer to a mixture of previous methodologies [5, 13, 14].

The classification of feature selection methods is determined by the way in which they communicate with the class label they are assigned, resulting in three distinct groups: unsupervised, supervised, and semisupervised. The rating of features is based on the degree to which they have an impact on the classification in supervised feature selection procedures. In unsupervised feature selection methods, the absence of class label information is a notable characteristic. A negligible fraction of the entire dataset has labels while working with semisupervised data [15, 16].

For supervised and semisupervised data, there is not always a class label. In certain datasets, it is observed that there are many class labels. Data that have only one class label are referred to as single-label data, while data that contain many class labels are referred to as multilabel data. Single-label data are common in various applications today. For example, consider a news item. This news item can be categorized as political, economic, and social at the same time. Hence, this communication possesses three distinct classification categories. So, based on the amount of labels in the dataset, two groups of feature selection methods might be formed. When it comes to machine learning, the ideas of single-label and multilabel classification are quite essential. When we talk about single-label classification, we are referring to the process of providing a single class label to each example [17, 18].

Figure 1 illustrates three separate types of feature selection strategies, each representing various points of view.

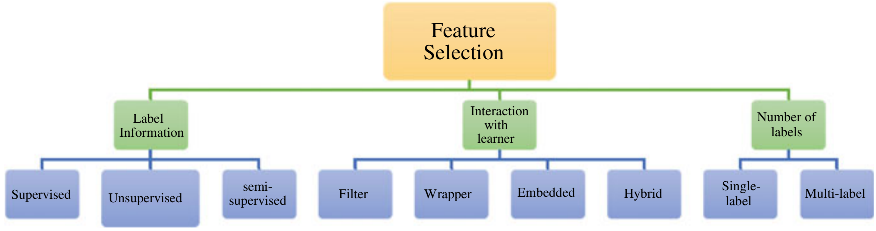

Fig. 1 Classification of feature selection techniques in general

4 Feature Selection Algorithms Based on MOP

Techniques for selecting features using the MOP are discussed in this section from the perspective of interaction with the learning algorithm (filter, wrapper, embedded, hybrid). Moreover, in a separate subsection, we classify these methods based on the supervision perspective (supervised, unsupervised, and semisupervised).

4.1 Filter MOP-Based Feature Selection Methods

We will go over filter-based methods for supervised MOP-based feature selection in this section. In the study cited in reference [19], two feature selection methods based on filter strategy are developed utilizing the NSGA-II and SPEA2 algorithms, both of which are multi-objective in nature. Four multi-objective feature selection procedures are generated by incorporating mutual entropy and information as separate evaluation criteria in both proposed platforms.

In the referenced paper [20], the authors propose two multi-objective feature selection methods. The two approaches are referred to as multi-objective binary particle swarm optimization with nondominated sorting (NSBPSO) and multi-objective binary particle swarm optimization, including crowding, mutation, and dominance (CMDBPSO). The suggested methods incorporate two filter-based evaluation criteria, entropy, and mutual information, as distinct criteria.

The authors [21] introduced a novel approach called FAEMODE in their study. The main breakthrough of this study was the incorporation of Elitism in the algorithm, along with the development of an objective formula that takes into account both nonlinear and linear dependencies among features. The aforementioned formula efficiently removes irrelevant and duplicative features from the dataset under examination.

The MECY-FS method [22] presents a multi-objective strategy that utilizes redundancy and correlation metrics. The MECY-FS method utilizes Pareto optimality as a means of assessing prospective feature subsets and identifying the most concise subset that exhibits the lowest redundancy and highest correlation.

PMFS [23] has used a biobjective optimization technique to address the problem of multilabel feature selection. The degrees of redundancy and relevancy are the two goals that are pursued here. A nondominated sorting algorithm is applied to rank the features. PEFS [24] is an ensemble of feature selection algorithms using the Pareto dominance concept. Three filter rankers are applied as three objectives in this optimization problem. mRMR-EFS [25] has also used the maximum relevancy degree and minimum redundancy of features as two objectives in single-label datasets.

The MMOFS-2S and MMOFS-3S techniques [26] serve as two approaches for feature selection in multilabel classification. The selection of suitable features involves the utilization of a three-phase filter method. The initial stage involves utilizing an evolutionary-based particle swarm optimization method specifically designed for the set of arriving features inside a multi-objective structure. In the next phase, the features selected thus far are compared with those that have already been selected to determine how much overlap there is between the two sets. The final step will find the features in our original set of candidates that lose their importance as we add and remove others.

The MOMFS algorithm, as described in [27], is a feature selection method that utilizes two particle swarms to address multiple objectives and multiple labels. Mutual information is employed as a criterion to assess the importance of features in relation to label sets and the redundancy of features. These two objectives are considered in the analysis.

The MOOSML approach [28] employs a multi-objective search methodology for the selection of streaming characteristics. This is accomplished by utilizing mutual information and Pareto optimal set theories.

The authors suggest I-SFS and G-SFS [29] as online streaming feature selection approaches in multilabel classification, utilizing the cuckoo search algorithm.

The DENCA methodology, proposed by [30], is a feature selection method that incorporates an objective function inspired by the neighborhood component analysis (NCA) technique. This approach applies in singleobjective and multi-objective cases and is integrated into the differential evolution framework.

4.2 Wrapper MOP-Based Feature Selection Methods

In previous research, the authors introduced a novel approach known as the MOFS-BDE method [31]. To optimize how it functions, the following operators are suggested: (1) The concept of probability inequality in binary mutation. (2) The phenomenon of purifying search. The term “one-bit operator” refers to an operator operating on a single bit of information. The proposed approach includes the utilization of a crowding distance-based, nondominated sorting operator.

The publication [32] study presents a multi-objective wrapper strategy that utilizes the NSGA-II algorithm. The approach incorporates two objective functions, with the first objective having the maximization of predicted profit. Furthermore, the process involves the reduction of the subset of features. The following approach is proposed for addressing the issue of credit scoring. In Ref. [33], a novel approach utilizing mating operators is suggested as an extension of NSGA-II for addressing the task of feature selection in motor imaging data. Additionally, an approach for rating features has been offered to ensure the stability of the proposed algorithm.

The MOTLBO is a feature selection algorithm using a multi-objective [34]. Researchers have offered three multi-objective teaching–learning-based optimization (TLBO) methods. The three algorithms considered in this study are (1) MOTLBO-ST, (2) MO-TLBO-NS, and (3) MO-TLBO-MD.

The research article by [35] introduced a multi-objective approach called RSPSOFS, which utilizes a Particle Swarm Optimization (PSO) algorithm. This method prioritizes features based on their frequency within a given set of archives. The previously mentioned steps are employed to optimize record collection and direct the particles. The authors of the paper [36] introduced a novel approach that combines a multi-objective artificial bee colony algorithm with nondominated sorting genetic operators. Two distinct variations of the suggested algorithm, namely, Num-MOABC and Bin-MOABC, have been designed as binary and continuous versions, respectively.

The MOBBA algorithm, as proposed in reference [37], enhances the initial Bat Algorithm by incorporating more effective formulas for random walks, efficient multi-objective operators, and local procedures for research purposes. In this study, a hybrid approach focusing on the Fisher metric has been employed to examine crucial biomarkers that demonstrate the effectiveness of the existing genomic evaluation method across three extensive microarray cancer datasets. The objective was to identify the top discriminative biomarkers for classification purposes.

The feature selection approach, known as MOFSRank, suggested in reference [38], is based on multi-objective optimization and has been utilized in the context of learning to rank. This approach covers three distinct methods. The first step involves proposing a method for the selection of instances, which aims to choose relevant instances from the ranking dataset. This process eliminates irrelevant data and enhances the training process's effectiveness. Furthermore, a novel approach utilizing adaptive mutation according to multi-objective optimization was introduced for the selected subsets. This method aims to obtain optimal feature subsets by selecting features with high rankings and low redundancies. In conclusion, the ranker utilized linear support vector machines (SVMs) to construct an improved feature set by utilizing the feature subsets obtained.

The paper by [39] introduced two distinct techniques for multi-objective optimization using a binary gray wolf optimization (GWO) algorithm. One possible approach to address multi-objective MOGWO is by scalarization. One approach that has been proposed for nondominated sorting is the nondominated sorting gray wolf optimizer (NSGWO). Both methods are employed in the wrapper approach to choose the most suitable feature subset for improved efficiency in detecting cervical lesions.

MODE [40] is a wrapper feature selection technique proposed for multi-objective optimization in facial expression recognition systems. The linear support vector machine (SVM) has been used as a classification model. The proposed technique aimed to achieve two objectives: improving classification accuracy and reducing feature attribute size. The study conducted by the authors [41] employed a binary version of the Particle Swarm Optimization (PSO) algorithm, namely, the multi-objective variant known as MOPSO, to introduce a two-step approach for diagnosing faults in energy transformers. This approach is referred to as 2-ADOPT. This research paper presents a novel approach that combines subset selection and an ensemble classifier to enhance the precision of gas-dissolved examination in power converters, and it is necessary to implement measures that boost the diagnostic accuracy.

The article [42] presents a multi-objective technique to accomplish feature selection in intrusion detection systems. This approach incorporates two distinct strategies for population evolution, specifically a novel dominating method and a predefined multitargeted search strategy. The NSGA-III algorithm is employed for the purpose of obtaining a suitable subset of attributes that yield favorable outcomes. The current research introduces an innovative approach to niche preservation, wherein an improved multi-objective algorithm, referred to as I-NSGA-III, is offered.

The study conducted by [43] employs a multi-objective algorithm called ENORA for feature selection in online advertising sales forecasting. The authors provide a proposed approach that integrates feature selection into regression analysis, model evaluation, and decision-making processes. This approach aims to determine the best possible effective model by utilizing posterior probabilities.

The study conducted in reference [44] introduced a novel approach for feature selection with multiple objectives, aiming to predict warfarin dosage. This research was conducted on a sample size of 553 individuals. Both the NSGA-II and MOPSO algorithms were utilized to select the features to be used. Additionally, artificial neural networks are employed for the assessment of the chosen features.

The study conducted in reference [45] examines the integration of machine learning and multi-objective algorithms in the context of feature subset selection techniques for binary classification problems. The proposed approach has undergone two stages of experimentation. The application of a Genetic Algorithm and Machine Learning techniques for Business Continuity Planning is explored in this study. A genetic algorithm is mostly employed to investigate the algorithm's intelligence because of the challenges associated with completing a full evaluation of all subgroup features. During the second phase, the chosen subset's efficacy is determined by utilizing machine learning techniques.

The development of a biobjective feature selection technique is discussed in the referenced work [46]. The ideal selection for further data classification was determined by selecting a subset of weighted features. The MOEA/D algorithm was introduced, utilizing two evaluation metrics: redundancy and relevancy.

The authors of the work [47] utilized the MOPSO algorithm to tackle the issue of unreliable data feature selection. First, the multi-objective concept contains two primary objectives: the measure of the correctness of a classification model's predictions, commonly referred to as classification accuracy. The issue of reliability. Furthermore, a proposed approach is presented that enhances the efficiency of a bare-bones Particle Swarm Optimization (PSO) algorithm for multi-objective selection. This is achieved by adding two additional operators into the algorithm. One of the two operators is the reinforced memory technique. (2) A hybrid genetic alteration. The study [48] employs NSGA-II to identify the best characteristics. The ID3 classifier is specifically developed to evaluate the suitability of a certain group of characteristics. The test accuracy that has been obtained is thereafter assigned to the fitness value.

The study referenced in [49] presents a method where the process of selecting features, which is dependent on the exchange of details between both sides, is changed into a global optimization endeavor. In this study, we discuss a new method called single-objective multi-input feature selection (SO-MIFS), which integrates both redundancy and correlation. The method exhibits strong global search capabilities and demonstrates excellent computational effectiveness. In addition, this research presents a new multi-objective feature selection methodology known as MO-MIFS, with the intention of enhancing the effectiveness and productivity of the feature selection procedure.

The paper by [50] presents a novel feature selection technique that leverages the shared nearest neighbour (SNN) distance. The technique maintains a resemblance among pairs of examples while selecting a selection of fewer features. The formulation and optimization of a feature assessment criterion are conducted inside a multi-objective framework simultaneously with the cardinality of the feature set with respect to the SNN distance. The NSGA-II optimization approach was employed to explore the search space and identify a collection of nondominated features.

The article referenced as [51] explores the topic of large-scale feature selection through the utilization of separate subgroup evolution, which collaboratively operates after a specified number of distinct generations. This approach contributes to the parallel implementation of multi-objective evolutionary algorithms (MOEA). In addition, the topic of feature selection was additionally approached as an unsupervised multi-objective clustering dilemma.

The MOUFSA algorithm [52], which stands for Multi-Objective Unsupervised Feature Selection Algorithm, is a method that operates without supervision to choose features. It is based on a multi-objective approach. The current goal involves the utilization of relationship coefficients and cardinalities to evaluate feature subsets. This objective measures the occurrence of selected features and assigns target attributes to each particular feature subset size. Moreover, a straightforward chronicle approach based on box-based and negative ε-advantage techniques is designed to safeguard prospective agreements, irrespective of their dominance. Furthermore, three mutation operators with different capabilities are suggested to enhance the method.

The study presented in [53] introduces a novel approach to multi-objective feature selection for intrusion detection. This approach is based on the utilization of the unsupervised clustering procedure of a growth hierarchical self-organizing map (GHSOM). As a result of this study, a novel method for identifying units and an improved system for selecting the winning unit have been developed. These techniques are employed to address the intrusion detection problem. The present study focuses on the subject of network anomaly identification. The proposed multi-objective approach enables the differentiation between normal and abnormal traffic and facilitates the distinction of various types of abnormalities.

The combined optimizer proposed in the study [54] utilizes a multi-objective approach and incorporates a neural network model to choose and classify data effectively. Within the scope of this investigation, the MmGA method is utilized to build a set enhancer. In order to achieve its primary goal, the MmGA-based set enhancer is designed to concurrently decrease the number of input features that are required for classification while simultaneously increasing the efficiency of the neural network model's classification capabilities.

In the publication [55], the authors proposed a genetic-based multi-objective method for feature selection tasks in facial recognition. The suggested approach aims to explore a wide range of potential decisions to reduce the number of elements in the subset while enhancing its ability to distinguish between different elements. The publication's authors [56] have presented a multi-objective feature selection method that utilizes a genetic algorithm. This strategy has been designed to effectively identify miRNA markers from miRNA datasets. Additionally, the algorithm contains an SVM classifier for recognition. The proposed approach effectively enhances multiple performance metrics while concurrently expanding the subset of essential features, specifically miRNAs.

The MOPSO algorithm was proposed in [57]. This study investigates two algorithms focused on MOP, NSPSOFS-A, and CMDPSOFS-B, based on PSO. The initial approach introduces the notion of nondominated sorting within Particle Swarm Optimization (PSO) as a means to address challenges associated with feature selection. The second approach pertains to Particle Swarm Optimization (PSO) and incorporates the principles of crowding, mutation, and dominance to identify alternative solutions on the Pareto front.

The study conducted by the researchers [58] utilized stacking autoencoders to achieve higher-level generalization in representing features. A feature-learning technique was specifically designed to address medical precision concerns, with a particular emphasis on identifying and categorizing risk variables associated with hypertension within a socioeconomically disadvantaged section of the population, namely, African Americans. The publication by [59] presents a deep feature selection method that can find and remove duplicate features in enormous datasets. The primary objective of this approach is to decrease the number of input variables that are considered during the learning phase. The approach employed in this study involved the utilization of deep Boltzmann models, and the utilization of obtained insight during the training step is crucial for the purpose of eliminating specific traits at the first steps in the process of learning.

In the study conducted by [60], a feature selection methodology is proposed that leverages deep learning methodologies. The present methodology conceptualizes the matter of feature selection as a task of feature rebuilding. The deep confidence network, a commonly employed deep learning method, reduces the amount of error that occurs during reconstruction throughout the entire feature collection to achieve feature abstraction. As a result, features that exhibit less reconstruction error are more prone to being impacted.

The binary NSGA-III approach, as described in reference [61], is an optimization technique applied to multi-objective optimization problems. This study aims to determine the best possible features for multilabel classification. This is achieved by addressing two specific goals: achieving the highest possible correlation between features and labels while also reducing the amount of processing resources required for these features.

The SHAPFS-ML [62] approach is developed using the Shapley value concept, while the NSGA3-θ [63] method employs an enhanced version of the NSGA III algorithm that incorporates two archives designed explicitly for multilabel datasets.

4.3 Hybrid MOP-Based Feature Selection Methods

A hybrid method known as C-HMOSHSSA [64] incorporates the salp swarm algorithm with the multi-objective spotted hyena optimizer to address the feature selection problem. The methodology comprises two distinct stages. Initially, the filter technique was employed to reduce extraneous features. Additionally, the hybrid approach described in this study was employed to investigate the best features.

MOSSO, a hybrid filter-wrapper approach, is utilized for gene selection [65]. This approach employs techniques for aggregating and filtering data to identify the best genes. Additionally, a proposed approach involves the utilization of a simplified swarm-based multi-objective algorithm, which is integrated as a wrapper with an SVM classifier. This hybrid approach aims to efficiently explore and identify the optimal subset of genes from the preselected collection.

The study conducted by the authors [66] introduced a fuzzy measure for multi-objective unsupervised feature selection. This was achieved by implementing the hybridized filter-wrapper technique, known as FC-MOFS. These formulas facilitate the selection of features from a given pool and decrease the risk of confusion caused by features that overlap through the use of crisp clustering within a traditional multi-objective optimization process.

The research article [67] introduces a novel interactive filter-wrapper multi-objective evolutionary algorithm called GR-MOEA. This method incorporates guiding and repairing procedures to effectively identify feature subsets of superior quality. The suggested approach involves the simultaneous evolution of wrapper and filter populations.

The unique hybrid filter-wrapper feature selection strategy, FW-GPAWOA [68], incorporates the use of the whale optimization algorithm (WOA). The approach being suggested is a multi-objective algorithm that aims to optimize both the wrapper and filter fitness functions concurrently.

The MO-IBSSA system, proposed by [69], is an innovative approach for identifying the severity of cracks. It utilizes a hybrid filter-wrapper technique combined with a multi-objective optimization feature selection method. The suggested methodology includes two primary elements: (1) the extraction of features by a combination of manual feature engineering and deep feature learning using convolutional neural networks (CNNs) and (2) the selection of features using a hybrid filter-wrapper technique, incorporating a multi-objective improved salp swarm optimization algorithm.

A hybrid feature selection technique called QOMOJaya [70] has been developed specifically for microarray datasets. The technique under consideration consists of two distinct parts. The initial step involves utilizing the PROMETHEE (Preference Rating Organization approach for Enrichment Evaluations) approach to obtain an aggregate rating of five widely recognized filter methods. This ranking is then employed to exclude undesirable features. The features that have been chosen during the initial phase are subsequently transferred to the second phase to undergo additional refinement. The second phase involves the utilization of the quasi oppositional-based multi-objective Jaya algorithm, which has been specifically designed for the purpose of wrapper-based feature selection.

4.4 Classifying Based on Supervision Perspective

In this section, we categorize the methods that were reviewed in the previous sections based on the supervision perspective into four categories: supervised (single-label), supervised (multilabel), semisupervised, and unsupervised. This classification is presented in Table 1.

Table 1 Classifying MOP-based feature selection methods based on the supervision perspective

Category	MOP-based feature selection methods
Supervised (Single-label)	NSGAII [19], SPEA2 [19], NSBPSO [20], CMDBPSO [20], MOFS-BDE [31], NSGA-II [32], NSGA-II [33], C-HMOSHSSA [64], MOTLBO [34], RSPSOFS [35], Num-MOABC [36], Bin-MOABC [36], MOBBA [37], MOFSRank [38], MOGWO [39], NSGWO [39], MODE [40], FAEMODE [21], I-NSGA-III [42], 2-ADOPT [41], ENORA [43], MOPSO [44], MOPSO [45], MOEA/D [46], MOPSO [47], NSGA- II [48], SO-MIFS [49], MO-MIFS [49], MmGA [54], MOGA [55], MOGA [56], CMDPSOFS-B [57], NSPSOFS-A [57], SAFS [58], Deep-FS [59], DBNFS [60], PEFS [24], mRMR-EFS [25], GR-MOEA [67], FW-GPAWOA [68], MO-IBSSA [69], QOMOJaya [70]
Supervised (Multilabel)	PMFS [23], NSGA-III [61], PEFS [24], MOMFS [27], MMOFS-2S and MMOFS-3S [26], mRMR-EFS [25], MOOSML [28], SHAPFS-ML [62], NSGA3-θ [63], I-SFS and G-SFS [29]
Semisupervised	–
Unsupervised	MOEA [51], MOUFSA [52], GHSOM [53], FC-MOFS [66]

5 Conclusions

This chapter provides a summary of feature selection methods based on the MOP. These methods were categorized from two viewpoints of feature selection. Based on this review, we found that most of the techniques presented thus far have been for supervised data, and other areas have received less attention. Only four methods are presented for unsupervised feature selection. Three of them are wrappers, and one is a hybrid. There are no filter methods for unsupervised data. Additionally, methods based on filter, wrapper, embedded, and hybrid methods have not been proposed for semisupervised learning. Several papers are presented for unsupervised data, which are all filters, considering that there is no class label in this type of data. The noteworthy point is the lack of embedded approaches in all kinds of data, which can be an interesting challenge for researchers in this field. Additionally, in supervised multilabel data, all methods are of filter and wrapper types. Therefore, hybrid and embedded approaches can be considered. In addition, more methods can be presented for online streaming data.

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Correction to: Multi-objective Adaptive Guided Differential Evolution for Passively Controlled Structures Equipped with a Tunned Mass Damper

Salar Farahmand-Tabar and Sina Shirgir

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In the original version of the book, a reference in chapter 3 was missed inadvertently. This has now been rectified and the reference [58] has been added. The correct version is given below:

58. Duman S, Akbel M, Kahraman HT (2021) Development of the multi-objective adaptive guided differential evolution and optimization of the MO-ACOPF for wind/PV/tidal energy sources. Appl Soft Comput 112:107814. https://doi.org/10.1016/j.asoc.2021.107814

The correction chapter and the book have been updated with the change.

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